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A Global DSGE Model for Oil-Dependent Economies

Haytem Troug and Francis Vitek

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A Global DSGE Model for Oil-Dependent Economies

Prepared by Haytem Troug and Francis Vitek*

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ABSTRACT: This paper develops a global DSGE model linking oil-dependent economies to the systemic economies, and applies it to the GCC countries. The model features a range of nominal and real rigidities, extensive macrofinancial linkages, and diverse spillover transmission channels. Its modular structure also encompasses common features of oil-dependent economies, including a state-owned oil company, a capital-intensive oil extraction technology, a sovereign wealth fund, an oil consumption subsidy, and membership of an international oil cartel. The calibrated model is used to simulate global risk and domestic reform scenarios for the GCC economies, revealing how their distinct structural characteristics shape their responses to shocks.

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Prepared by Haytem Troug and Francis Vitek[1]

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1. Introduction

Oil-dependent economies tend to exhibit distinct structural characteristics. Typically, a state-owned enterprise extracts oil using a capital-intensive technology subject to an oil production quota set by the augmented Organization of the Petroleum Exporting Countries (OPEC+) cartel, and remits profits to the government. The government in turn subsidizes domestic oil consumption and invests oil profits not needed to finance its budget predominantly abroad via a sovereign wealth fund, often under a fixed exchange rate regime. The members of the Gulf Cooperation Council (GCC) generally fit this description and are among the most oil-dependent economies in the world, as measured by the oil share of output, exports and government revenues. These are Bahrain, Kuwait, Oman, Qatar, Saudi Arabia, and the United Arab Emirates.

This paper develops a global dynamic stochastic general equilibrium (DSGE) model linking a set of oil-dependent economies to the rest of the world, to support quantitative scenario-based policy, risk and spillover analysis. The set of oil-dependent economies is represented by the six members of the GCC, while the rest of the world is disaggregated into four systemically important economies, including China, the Euro Area and the United States. Each economy consists of households, firms, banks, a state-owned enterprise, and a government. Households and firms are subject to a range of nominal and real rigidities, building on Christiano, Eichenbaum and Evans (2005), Smets and Wouters (2003), and Monacelli (2005). Banks are subject to a variety of financial frictions, building on Hülsewig, Mayer and Wollmershäuser (2009), Gerali, Neri, Sessa and Signoretti (2010), and Gabaix and Maggiori (2015). Interactions among households, firms and banks are subject to extensive macrofinancial linkages, including financial accelerator mechanisms connecting the housing and production sectors to the banking sector, following Adrian, Gaspar and Vitek (2024). The state-owned enterprise extracts oil using a capital-intensive technology subject to an oil production quota set by an international oil cartel if the economy is a member, and remits profits to the government. The government in turn encompasses a sovereign wealth fund that invests oil profits not needed to finance the budget abroad, a monetary authority that conducts monetary policy under a flexible, managed or fixed exchange rate regime, a fiscal authority that implements fiscal policy using an extensive set of instruments including an oil consumption subsidy and a non-oil import tariff, and a macroprudential authority that implements macroprudential policy. Economies interact via diverse spillover transmission channels, spanning a variety of international trade, financial and oil price linkages.

Scenario analysis using our calibrated global DSGE model reveals how the distinct structural characteristics of oil-dependent economies shape their responses to shocks. Indeed, simulation results for the GCC economies under global risk and domestic reform scenarios support several conventional views while elucidating the mechanisms at work. First, more oil-dependent members of the GCC experience larger output losses under this global downside risk scenario—which generates a temporary but persistent global growth slowdown associated with tighter global financial conditions and lower oil prices—as the non-oil sector dampens the economic impact of lower oil prices and production through consumption smoothing. This finding supports economic diversification as a strategy for oil-dependent economies to mitigate their exposure to global risks that could impact the oil market. Second, a key determinant of the impact of this global risk scenario on the GCC economies is the response of the international oil cartel, which faces a policy trade-off. Simulating the scenario under a policy counterfactual reveals that while its production quota cuts materially dampen the fall in the price of oil—significantly mitigating the deteriorations in the fiscal and external balances for the aggregate GCC economy—this comes at the cost of a substantially larger output loss. Third, sovereign wealth fund transactions help stabilize

government debt dynamics across the GCC economies under our global risk and domestic reform scenarios, by serving as an alternative source of net financing for the budget. In particular, government debt accumulation to finance higher budgetary fiscal deficits is mitigated by sovereign wealth fund drawdowns under our global risk scenario, and vice versa under our domestic reform scenario.

This paper adds to the emerging literature on modeling oil-dependent economies within a global DSGE framework. Its main contribution is the integration of all of the distinguishing structural characteristics of oil-dependent economies within a unified theoretical framework. Government oil revenues are managed via a sovereign wealth fund within a global DSGE framework in Bergholt, Larsen and Seneca (2019). Like ours, this model treats sovereign wealth fund assets as exclusively external. Domestic oil consumption is subsidized within a global DSGE framework in Balke, Plante and Yücel (2015). Unlike ours, this model lacks nominal and real rigidities, as well as macrofinancial linkages. Oil prices and production are jointly endogenously determined within a global DSGE framework incorporating an international oil cartel in Nakov and Pescatori (2010), Nakov and Nuño (2013), and Bornstein, Krusell and Rebelo (2023). These models determine the optimal level of cartel oil production, whereas ours postulates a flexible feedback rule governing it. However, they also generally lack nominal and real rigidities, as well as macrofinancial linkages. None of these existing global DSGE models combine all of the distinguishing structural characteristics of oil-dependent economies.

The organization of this paper is as follows. The next section develops the theoretical framework, while section three describes the corresponding empirical framework. These sections are necessarily technically complex given the scope of our scenario analysis, and readers content to understand our simulation results at an intuitive level without fully knowing their underpinnings can skip them. Calibration of our empirical framework is the subject of section four, while scenario analysis using it is conducted in section five. Finally, section six concludes.

2. The Theoretical Framework

Consider a finite set of national economies indexed by $i \in \{1,\ldots,N\}$ which constitutes the world economy. These economies are structurally heterogenous but modularly nested within an encompassing theoretical framework. Each of them consists of households, firms, banks, a state-owned enterprise, and a government. The government in turn encompasses a sovereign wealth fund, a monetary authority, a fiscal authority, and a macroprudential authority. Households, firms, banks and state-owned enterprises optimize intertemporally, interacting with governments and an international oil cartel in an uncertain environment to determine equilibrium prices and quantities under rational expectations in globally integrated output and financial markets. Dominant economy i^* issues the quotation currency for transactions in the foreign exchange market.

A. The Household Sector

There exists a continuum of households indexed by $h \in [0,1]$ that are differentiated according to whether they are credit constrained, but are otherwise identical. Credit unconstrained households of type $Z = U$ and measure ϕ^U have access to domestic banks where they accumulate deposits and secure mortgage loans, to the domestic housing market where they trade houses, and to domestic financial markets where they trade government bonds, firm stocks and bank stocks, where $0 < \phi^U < 1$. In contrast, credit constrained households of

type $Z = C$ and measure ϕ^C do not have access to banks, or to the housing and financial markets, where $0 < \phi^C < 1$ and $\phi^U + \phi^C = 1$. Houses are in fixed supply at one per credit unconstrained household. Government bonds are perpetual bonds with coupon payments that decay exponentially at rate ω^G where $0 < \omega^G \leq 1$, following Woodford (2001).

In a reinterpretation of the labor market in the model of nominal wage rigidity proposed by Erceg, Henderson and Levin (2000) to incorporate involuntary unemployment along the lines of Galí (2011), each household consists of a continuum of members represented by the unit square and indexed by $(k,l) \in [0,1] \times [0,1]$. There is full risk sharing among household members, who supply indivisible differentiated intermediate labor services indexed by $k \in [0,1]$, incurring disutility from work determined by $l \in [0,1]$ if they are employed and zero otherwise. Trade-specific intermediate labor services supplied by credit unconstrained and credit constrained households are perfect substitutes.

Consumption, Saving, and Residential Investment

Each infinitely lived household h has preferences defined over consumption $C_{i,h,s}$, housing $S^H_{i,h,s}H_{i,h,s}$, labor supply $\{L_{i,h,k,s}\}^1_{k=0}$, houses $S^H_{i,h,s+1}$, government bonds $\{S^G_{i,h,v,s+1}\}^s_{v=1}$ and firm stocks $\{S^F_{i,h,f,s+1}\}^1_{f=0}$ represented by intertemporal utility function

$$U_{i,h,t} = \mathrm{E}_t \sum_{s=t}^{\infty} \beta^{s-t} u(C_{i,h,s}, S^H_{i,h,s}H_{i,h,s}, \{L_{i,h,k,s}\}^1_{k=0}, S^H_{i,h,s+1}, \{S^G_{i,h,v,s+1}\}^s_{v=1}, \{S^F_{i,h,f,s+1}\}^1_{f=0}), \tag{1}$$

where E_t denotes the expectations operator conditional on information available in period t, and $0 < \beta < 1$. The intratemporal utility function is additively separable and represents external habit formation preferences in consumption and labor supply,

$$\begin{aligned} u(C_{i,h,s}, S^H_{i,h,s}H_{i,h,s}, \{L_{i,h,k,s}\}^1_{k=0}, S^H_{i,h,s+1}, \{S^G_{i,h,v,s+1}\}^s_{v=1}, \{S^F_{i,h,f,s+1}\}^1_{f=0}) = \upsilon^C_{i,s} \Bigg[& \frac{1}{1-1/\sigma} \left(C_{i,h,s} - \alpha^C \frac{C^Z_{i,s-1}}{\phi^Z} \right)^{1-1/\sigma} + \upsilon^D_{i,s} \frac{(S^H_{i,h,s}H_{i,h,s})^{1-1/\varsigma}}{1-1/\varsigma} \\ - \upsilon^N_{i,s} \int_0^1 \int_{\alpha^L \frac{L^Z_{i,k,s-1}}{\phi^Z}}^{L_{i,h,k,s}} & \left(l - \alpha^L \frac{L^Z_{i,k,s-1}}{\phi^Z} \right)^{1/\eta} dl dk - \upsilon^H_{i,s} \frac{V^H_{i,h,s} S^H_{i,h,s+1}}{(1+\tau^C_{i,s}) P^C_{i,s}} - \upsilon^B_{i,s} \sum_{v=1}^{s} \frac{V^G_{i,v,s} S^G_{i,h,v,s+1}}{(1+\tau^C_{i,s}) P^C_{i,s}} - \upsilon^S_{i,s} \int_0^1 \frac{V^F_{i,f,s} S^F_{i,h,f,s+1}}{(1+\tau^C_{i,s}) P^C_{i,s}} df \Bigg], \end{aligned} \tag{2}$$

where $0 \leq \alpha^C < 1$ and $0 \leq \alpha^L < 1$. To introduce exogenous asset risk premia that affect intertemporal substitution, we specify preference shifters $\upsilon^C_{i,s}$, $\upsilon^H_{i,s}$, $\upsilon^B_{i,s}$ and $\upsilon^S_{i,s}$ as functions of aggregate endogenous variables or structural shocks,

$$\begin{aligned} \upsilon^C_{i,t} &= \nu^C_{i,t} - \frac{\phi^C_H}{1-\rho_H} \nu^H_{i,t} - \frac{\phi^C_B}{1-\rho_B} \tilde{\nu}^B_{i,t} - \frac{\phi^C_S}{1-\rho_S} \tilde{\nu}^S_{i,t}, \ \upsilon^H_{i,s} = \frac{\tilde{\nu}^B_{i,s} + \nu^H_{i,s}}{v(C^Z_{i,s}, C^Z_{i,s-1})}, \\ \upsilon^B_{i,s} &= \frac{\tilde{\nu}^B_{i,s}}{v(C^Z_{i,s}, C^Z_{i,s-1})}, \ \upsilon^S_{i,s} = \frac{\tilde{\nu}^B_{i,s} + \tilde{\nu}^S_{i,s}}{v(C^Z_{i,s}, C^Z_{i,s-1})}, \ v(C^Z_{i,s}, C^Z_{i,s-1}) = \left(\frac{C^Z_{i,s}}{\phi^Z} - \alpha^C \frac{C^Z_{i,s-1}}{\phi^Z} \right)^{1/\sigma}, \end{aligned} \tag{3}$$

where $\phi^C_H > 0$, $\phi^C_B > 0$ and $\phi^C_S > 0$. To support the existence of a long run balanced growth path, we also specify preference shifters $\upsilon^D_{i,s}$ and $\upsilon^N_{i,s}$ as functions of aggregate endogenous variables or structural shocks,

$$\upsilon^D_{i,s} = \frac{\nu^D}{v(C^Z_{i,s}, C^Z_{i,s-1})} (C_{i,s})^{1/\varsigma}, \ \upsilon^N_{i,s} = \frac{\mathcal{A}^T_{i,s}}{v(C^Z_{i,s}, C^Z_{i,s-1})} \frac{(L_{i,s} / \nu^N_{i,s})^{\mu/\eta}}{w(L^Z_{i,s}, L^Z_{i,s-1})}, \ w(L^Z_{i,s}, L^Z_{i,s-1}) = \left(\frac{L^Z_{i,s}}{\phi^Z} - \alpha^L \frac{L^Z_{i,s-1}}{\phi^Z} \right)^{1/\eta}, \tag{4}$$

where $\mu > 0$. We assume that the intratemporal utility function is strictly increasing with respect to consumption and housing, and is strictly decreasing with respect to labor supply. This implies restrictions on serially correlated private consumption demand shock $\nu^C_{i,s}$ and labor supply shock $\nu^N_{i,s}$, given mean zero and serially correlated housing risk premium shock $\nu^H_{i,s}$, as well as mean zero and internationally and serially correlated duration risk premium shock $\tilde{\nu}^B_{i,s}$ and equity risk premium shock $\tilde{\nu}^S_{i,s}$. Given these and other restrictions, this intratemporal utility function is strictly concave with respect to consumption, housing and labor supply if $\sigma > 0$, $\varsigma > 0$ and $\eta > 0$.

The household enters period s with previously accumulated wealth, allocated across the values of bank deposits, mortgage loans, houses, government bonds, firm stocks and bank stocks. In particular, it holds bank deposits $B^{H,D}_{i,h,s}$ which pay interest at risk free deposit rate $i^D_{i,s-1}$, secures new borrowing $B^{H,H}_{i,h,s}$ from banks on its $S^H_{i,h,s}$ mortgage loans, owns $S^H_{i,h,s}$ houses worth $V^H_{i,h,s}$, owns $\{S^G_{i,h,v,s}\}^{s-1}_{v=1}$ vintage-specific government bonds worth $\{V^G_{i,v,s}\}^{s-1}_{v=1}$ which pay coupons $\{\Pi^G_{i,v,s}\}^{s-1}_{v=1}$, owns $\{S^F_{i,h,f,s}\}^1_{f=0}$ firm stocks worth $\{V^F_{i,f,s}\}^1_{f=0}$ which pay dividends $\{\Pi^F_{i,f,s}\}^1_{f=0}$, and owns $\{S^B_{i,h,b,s}\}^1_{b=0}$ bank stocks worth $\{V^B_{i,b,s}\}^1_{b=0}$ which pay dividends $\{\Pi^B_{i,b,s}\}^1_{b=0}$. The user cost per house $\Pi^H_{i,h,s}$ satisfies $\Pi^H_{i,h,s} = P^H_{i,s}H_{i,h,s} + (B^{H,H}_{i,h,s} - C^{H,H}_{i,h,s}) - P^{I^H}_{i,s}I^H_{i,h,s}$, where $P^H_{i,s}$ denotes the implicit rental price of housing. The coupon payment per government bond $\Pi^G_{i,v,s}$ satisfies $\Pi^G_{i,v,s} = (1 + i^G_{i,v} - \omega^G)(\omega^G)^{s-v}V^G_{i,v,v}$, where $i^G_{i,v}$ denotes the vintage-specific yield to maturity on government bonds at issuance when $V^G_{i,v,v} = 1$. During period s, the household supplies differentiated intermediate labor services $\{L_{i,h,k,s}\}^1_{k=0}$, earning labor income at trade-specific wages $\{W_{i,k,s}\}^1_{k=0}$. The government levies a tax on household labor income at rate $\tau^L_{i,s}$, and remits household type-specific lump sum transfer payment $T^Z_{i,h,s}$. These sources of funds are summed in household dynamic budget constraint:

$$\begin{aligned} &B^{H,D}_{i,h,s+1} + S^H_{i,h,s}C^{H,H}_{i,h,s} + V^H_{i,h,s}S^H_{i,h,s+1} + \sum_{v=1}^{s} V^G_{i,v,s}S^G_{i,h,v,s+1} + \int_0^1 V^F_{i,f,s}S^F_{i,h,f,s+1}df + \int_0^1 V^B_{i,b,s}S^B_{i,h,b,s+1}db \\ &+(1+\tau^C_{i,s})P^C_{i,s}C_{i,h,s} + P^{I^H}_{i,s}S^H_{i,h,s}I^H_{i,h,s} = (1+i^D_{i,s-1})B^{H,D}_{i,h,s} + S^H_{i,h,s}B^{H,H}_{i,h,s} + V^H_{i,h,s}S^H_{i,h,s} + \sum_{v=1}^{s-1}(\Pi^G_{i,v,s} + V^G_{i,v,s})S^G_{i,h,v,s} \\ &+\int_0^1 (\Pi^F_{i,f,s} + V^F_{i,f,s})S^F_{i,h,f,s}df + \int_0^1 (\Pi^B_{i,b,s} + V^B_{i,b,s})S^B_{i,h,b,s}db + (1-\tau^L_{i,s})\int_0^1 W_{i,k,s}L_{i,h,k,s}dk + T^Z_{i,h,s}. \end{aligned} \quad (5)$$

According to this dynamic budget constraint, at the end of period s the household reallocates its wealth across the values of bank deposits, mortgage loans, houses, government bonds, firm stocks and bank stocks. In particular, it holds bank deposits $B^{H,D}_{i,h,s+1}$, pays debt service cost $C^{H,H}_{i,h,s}$ to banks on its $S^H_{i,h,s}$ mortgage loans, owns $S^H_{i,h,s+1}$ houses worth $V^H_{i,h,s}$, owns $\{S^G_{i,h,v,s+1}\}^s_{v=1}$ vintage-specific government bonds worth $\{V^G_{i,v,s}\}^s_{v=1}$, owns $\{S^F_{i,h,f,s+1}\}^1_{f=0}$ firm stocks worth $\{V^F_{i,f,s}\}^1_{f=0}$, and owns $\{S^B_{i,h,b,s+1}\}^1_{b=0}$ bank stocks worth $\{V^B_{i,b,s}\}^1_{b=0}$. Finally, the household purchases private consumption good $C_{i,h,s}$ at price $P^C_{i,s}$, and residential investment good $I^H_{i,h,s}$ for its $S^H_{i,h,s}$ houses at price $P^{I^H}_{i,s}$. The government levies a tax on household consumption expenditures at rate $\tau^C_{i,s}$, implying total consumption price $P^{C^T}_{i,s} = (1+\tau^C_{i,s})P^C_{i,s}$.

The household enters period s with housing of previously accumulated quality $H_{i,h,s}$, which subsequently evolves according to accumulation function

$$H_{i,h,s+1} = (1-\delta_H)H_{i,h,s} + \mathcal{H}^H(I^H_{i,h,s}, I^H_{i,h,s-1}), \quad (6)$$

where $0 \le \delta_H \le 1$. Effective residential investment function $\mathcal{H}^H(I^H_{i,h,s}, I^H_{i,h,s-1})$ incorporates convex adjustment costs in the gross growth rate of the ratio of nominal residential investment to aggregate nominal output,

$$\mathcal{H}^H(I^H_{i,h,s},I^H_{i,h,s-1}) = \nu^{I^H}_{i,s}\left[1-\frac{\chi^H}{2}\left(\frac{P^{I^H}_{i,s}I^H_{i,h,s}}{P^{I^H}_{i,s-1}I^H_{i,h,s-1}}\frac{P^Y_{i,s-1}Y_{i,s-1}}{P^Y_{i,s}Y_{i,s}}-1\right)^2\right]I^H_{i,h,s}, \tag{7}$$

where serially correlated residential investment demand shock $\nu^{I^H}_{i,s}$ satisfies $\nu^{I^H}_{i,s}>0$, while $\chi^H>0$. In steady state equilibrium, these adjustment costs equal zero, and effective residential investment equals actual residential investment.

The household enters period s with previously accumulated mortgage debt $D^H_{i,h,s}$, which subsequently evolves according to accumulation function

$$D^H_{i,h,s+1} = (1-\alpha^H)D^H_{i,h,s} + B^{H,H}_{i,h,s}, \tag{8}$$

where $0\le\alpha^H\le 1$. Adopting the collateralized borrowing variant of the financial accelerator mechanism due to Kiyotaki and Moore (1997), the household secures new mortgage borrowing from banks to finance a fraction of the installed value of residential investment,

$$B^{H,H}_{i,h,s} = \phi^H Q^H_{i,s}I^H_{i,h,s}, \tag{9}$$

given by regulatory mortgage loan to value ratio limit ϕ^H. Its debt service cost satisfies $C^{H,H}_{i,h,s} = (1-\delta^H_{i,s})(\alpha^H + i^H_{i,s-1})D^H_{i,h,s}$, reflecting an amortization payment at rate α^H and an interest payment at mortgage loan rate $i^H_{i,s-1}$ on the outstanding stock of mortgage debt, net of a writedown at mortgage loan default rate $\delta^H_{i,s}$.

Credit Unconstrained Households

In period t, the credit unconstrained household chooses state contingent sequences for consumption $\{C_{i,h,s}\}^\infty_{s=t}$, residential investment $\{I^H_{i,h,s}\}^\infty_{s=t}$, housing quality $\{H_{i,h,s+1}\}^\infty_{s=t}$, labor force participation $\{\{N_{i,h,k,s}\}^1_{k=0}\}^\infty_{s=t}$, bank deposits $\{A^D_{i,h,s+1}\}^\infty_{s=t}$, mortgage debt $\{D^H_{i,h,s+1}\}^\infty_{s=t}$, houses $\{S^H_{i,h,s+1}\}^\infty_{s=t}$, government bonds $\{\{S^G_{i,h,v,s+1}\}^s_{v=1}\}^\infty_{s=t}$, firm stocks $\{\{S^F_{i,h,f,s+1}\}^1_{f=0}\}^\infty_{s=t}$ and bank stocks $\{\{S^B_{i,h,b,s+1}\}^1_{b=0}\}^\infty_{s=t}$ to maximize intertemporal utility function (1) subject to dynamic budget constraint (5), housing quality accumulation function (6), mortgage debt accumulation function (8), and terminal nonnegativity constraints $H_{i,h,T+1}\ge 0$, $A^D_{i,h,T+1}\ge 0$, $D^H_{i,h,T+1}\ge 0$, $S^H_{i,h,T+1}\ge 0$, $S^G_{i,h,v,T+1}\ge 0$, $S^F_{i,h,f,T+1}\ge 0$ and $S^B_{i,h,b,T+1}\ge 0$ for $T\to\infty$. In equilibrium, the solutions to this utility maximization problem satisfy intertemporal and intratemporal optimality conditions:

$$\mathrm{E}_t\frac{\beta u_C(i,h,t+1)}{u_C(i,h,t)}\frac{1+\tau^C_{i,t}}{1+\tau^C_{i,t+1}}\frac{P^C_{i,t}}{P^C_{i,t+1}}(1+i^D_{i,t}) = 1, \tag{10}$$

$$\frac{Q^H_{i,h,t}}{P^{I^H}_{i,t}}\mathcal{H}^H_1(i,h,t) + \phi^H(1+R^H_{i,h,t})\frac{Q^H_{i,t}}{P^{I^H}_{i,t}} + \mathrm{E}_t\frac{\beta u_C(i,h,t+1)}{u_C(i,h,t)}\frac{1+\tau^C_{i,t}}{1+\tau^C_{i,t+1}}\frac{P^C_{i,t}}{P^C_{i,t+1}}\frac{P^{I^H}_{i,t+1}}{P^{I^H}_{i,t}}\frac{Q^H_{i,h,t+1}}{P^{I^H}_{i,t+1}}\mathcal{H}^H_2(i,h,t+1) = 1, \tag{11}$$

$$\frac{Q^H_{i,h,t}}{P^{I^H}_{i,t}} = \mathrm{E}_t\frac{\beta u_C(i,h,t+1)}{u_C(i,h,t)}\frac{1+\tau^C_{i,t}}{1+\tau^C_{i,t+1}}\frac{P^C_{i,t}}{P^C_{i,t+1}}\frac{P^{I^H}_{i,t+1}}{P^{I^H}_{i,t}}\left[\frac{P^H_{i,t+1}}{P^{I^H}_{i,t+1}} + (1-\delta_H)\frac{Q^H_{i,h,t+1}}{P^{I^H}_{i,t+1}}\right], \tag{12}$$

$$\frac{u_{S^HH}(i,h,t)}{u_C(i,h,t)} = \frac{1}{1+\tau^C_{i,t}}\frac{P^H_{i,t}}{P^C_{i,t}}, \tag{13}$$

$$R^H_{i,h,t} = \mathrm{E}_t\frac{\beta u_C(i,h,t+1)}{u_C(i,h,t)}\frac{1+\tau^C_{i,t}}{1+\tau^C_{i,t+1}}\frac{P^C_{i,t}}{P^C_{i,t+1}}\left[(1-\alpha^H)R^H_{i,h,t+1} - (1-\delta^H_{i,t+1})(\alpha^H + i^H_{i,t})\right], \tag{14}$$

$$-\frac{u_{L_k}(i,h,k,t)}{u_C(i,h,t)}=\frac{1-\tau^L_{i,t}}{1+\tau^C_{i,t}}\frac{W_{i,k,t}}{P^C_{i,t}}, \tag{15}$$

$$\mathrm{E}_t\frac{\beta u_C(i,h,t+1)}{u_C(i,h,t)}\frac{1+\tau^C_{i,t}}{1+\tau^C_{i,t+1}}\frac{P^C_{i,t}}{P^C_{i,t+1}}\left[\frac{\Pi^H_{i,h,t+1}+V^H_{i,h,t+1}}{V^H_{i,h,t}}-(1+i^D_{i,t})\right]=\frac{u_C(i,t)}{u_C(i,h,t)}(\tilde{\nu}^B_{i,t}+\nu^H_{i,t}), \tag{16}$$

$$\mathrm{E}_t\frac{\beta u_C(i,h,t+1)}{u_C(i,h,t)}\frac{1+\tau^C_{i,t}}{1+\tau^C_{i,t+1}}\frac{P^C_{i,t}}{P^C_{i,t+1}}\left[\frac{\Pi^G_{i,v,t+1}+V^G_{i,v,t+1}}{V^G_{i,v,t}}-(1+i^D_{i,t})\right]=\frac{u_C(i,t)}{u_C(i,h,t)}\tilde{\nu}^B_{i,t}, \tag{17}$$

$$\mathrm{E}_t\frac{\beta u_C(i,h,t+1)}{u_C(i,h,t)}\frac{1+\tau^C_{i,t}}{1+\tau^C_{i,t+1}}\frac{P^C_{i,t}}{P^C_{i,t+1}}\left[\frac{\Pi^F_{i,f,t+1}+V^F_{i,f,t+1}}{V^F_{i,f,t}}-(1+i^D_{i,t})\right]=\frac{u_C(i,t)}{u_C(i,h,t)}(\tilde{\nu}^B_{i,t}+\tilde{\nu}^S_{i,t}), \tag{18}$$

$$\mathrm{E}_t\frac{\beta u_C(i,h,t+1)}{u_C(i,h,t)}\frac{1+\tau^C_{i,t}}{1+\tau^C_{i,t+1}}\frac{P^C_{i,t}}{P^C_{i,t+1}}\left[\frac{\Pi^B_{i,b,t+1}+V^B_{i,b,t+1}}{V^B_{i,b,t}}-(1+i^D_{i,t})\right]=0. \tag{19}$$

Here $Q^H_{i,h,s}$ denotes the shadow price of housing, which is proportional to the Lagrange multiplier on the period s housing quality accumulation function, while $R^H_{i,h,s}$ denotes the shadow price of mortgage debt, which is proportional to the Lagrange multiplier on the period s mortgage debt accumulation function.

Credit Constrained Households

In period t, the credit constrained household chooses state contingent sequences for consumption $\{C_{i,h,s}\}^{\infty}_{s=t}$ and labor force participation $\{\{N_{i,h,k,s}\}^1_{k=0}\}^{\infty}_{s=t}$ to maximize intertemporal utility function (1) subject to dynamic budget constraint (5) and the applicable access restrictions. In equilibrium, the solutions to this utility maximization problem satisfy:

$$(1+\tau^C_{i,t})P^C_{i,t}C_{i,h,t}=(1-\tau^L_{i,t})\int_0^1 W_{i,k,t}L_{i,h,k,t}dk+T^C_{i,h,t}, \tag{20}$$

$$-\frac{u_{L_k}(i,h,k,t)}{u_C(i,h,t)}=\frac{1-\tau^L_{i,t}}{1+\tau^C_{i,t}}\frac{W_{i,k,t}}{P^C_{i,t}}. \tag{21}$$

Here household static budget constraint (20) equates after-tax consumption expenditures to disposable labor income plus transfers.

Labor Supply

The unemployment rate $u^L_{i,t}$ measures the share of the labor force $N_{i,t}$ in unemployment $U_{i,t}$, that is $u^L_{i,t}=U_{i,t}/N_{i,t}$, where unemployment equals the labor force less employment $L_{i,t}$, that is $U_{i,t}=N_{i,t}-L_{i,t}$. The labor force satisfies $N_{i,t}=\int_0^1 N_{i,k,t}dk$.

There exist a large number of perfectly competitive firms that combine differentiated intermediate labor services $\{L_{i,k,t}\}^1_{k=0}$ supplied by trade unions of workers to produce final labor service $L_{i,t}$ according to constant elasticity of substitution production function:

$$L_{i,t} = \left[\int_0^1 (L_{i,k,t})^{\frac{\theta_{i,t}^L - 1}{\theta_{i,t}^L}} dk \right]^{\frac{\theta_{i,t}^L}{\theta_{i,t}^L - 1}}. \tag{22}$$

The representative final labor service firm maximizes profits from production of the final labor service with respect to inputs of intermediate labor services, implying:

$$L_{i,k,t} = \left(\frac{W_{i,k,t}}{W_{i,t}} \right)^{-\theta_{i,t}^L} L_{i,t}, \tag{23}$$

$$W_{i,t} = \left[\int_0^1 (W_{i,k,t})^{1-\theta_{i,t}^L} dk \right]^{\frac{1}{1-\theta_{i,t}^L}}. \tag{24}$$

Serially uncorrelated wage markup shock $\vartheta_{i,t}^L$ satisfies $\vartheta_{i,t}^L = \theta_{i,t}^L / (\theta_{i,t}^L - 1)$, where the wage elasticity of demand for intermediate labor services $\theta_{i,t}^L$ satisfies $\theta_{i,t}^L > 1$.

In an extension of the model of nominal wage rigidity proposed by Erceg, Henderson and Levin (2000) along the lines of Smets and Wouters (2003), each period a randomly selected fraction $1-\omega^L$ of trade unions adjust their wage optimally, where $0 \le \omega^L < 1$. The remaining fraction ω^L of trade unions adjust their wage to account for past wage inflation, as well as past consumption price inflation and trend productivity growth, according to partial indexation rule

$$W_{i,k,t} = \left[\left(\frac{W_{i,t-1}}{W_{i,t-2}} \right)^{1-\mu^L} \left(\frac{P_{i,t-1}^C \mathcal{A}_{i,t-1}^T}{P_{i,t-2}^C \mathcal{A}_{i,t-2}^T} \right)^{\mu^L} \right]^{\gamma^L} \left[\left(\frac{\bar{W}_{i,t-1}}{\bar{W}_{i,t-2}} \right)^{1-\mu^L} \left(\frac{\bar{P}_{i,t-1}^C \bar{\mathcal{A}}_{i,t-1}}{\bar{P}_{i,t-2}^C \bar{\mathcal{A}}_{i,t-2}} \right)^{\mu^L} \right]^{1-\gamma^L} W_{i,k,t-1}, \tag{25}$$

where $0 \le \gamma^L \le 1$ and $0 \le \mu^L \le 1$. If trade union k can adjust its wage optimally in period t, then it does so to maximize intertemporal utility function (1) subject to dynamic budget constraint (5), intermediate labor service demand function (23), and the assumed form of nominal wage rigidity. In equilibrium, the solution to this wage setting problem satisfies:

$$\frac{W_{i,t}^*}{W_{i,t}} = -\frac{\mathrm{E}_t \sum_{s=t}^{\infty} (\omega^L)^{s-t} \frac{\beta^{s-t} u_C(i,h,s)}{u_C(i,h,t)} \theta_{i,s}^L \frac{u_{L_k}(i,h,k,s)}{u_C(i,h,s)} \left\{ \left[\left(\frac{W_{i,t-1}}{W_{i,s-1}} \right)^{1-\mu^L} \left(\frac{P_{i,t-1}^C \mathcal{A}_{i,t-1}^T}{P_{i,s-1}^C \mathcal{A}_{i,s-1}^T} \right)^{\mu^L} \right]^{\gamma^L} \left[\left(\frac{\bar{W}_{i,t-1}}{\bar{W}_{i,s-1}} \right)^{1-\mu^L} \left(\frac{\bar{P}_{i,t-1}^C \bar{\mathcal{A}}_{i,t-1}}{\bar{P}_{i,s-1}^C \bar{\mathcal{A}}_{i,s-1}} \right)^{\mu^L} \right]^{1-\gamma^L} \frac{W_{i,s}}{W_{i,t}} \right\}^{\theta_{i,s}^L} \left(\frac{W_{i,t}^*}{W_{i,t}} \right)^{-\theta_{i,s}^L} L_{i,h,s}}{\mathrm{E}_t \sum_{s=t}^{\infty} (\omega^L)^{s-t} \frac{\beta^{s-t} u_C(i,h,s)}{u_C(i,h,t)} (\theta_{i,s}^L - 1) \frac{1-\tau_{i,s}^L}{1+\tau_{i,s}^C} \frac{W_{i,s}}{P_{i,s}^C} \left\{ \left[\left(\frac{W_{i,t-1}}{W_{i,s-1}} \right)^{1-\mu^L} \left(\frac{P_{i,t-1}^C \mathcal{A}_{i,t-1}^T}{P_{i,s-1}^C \mathcal{A}_{i,s-1}^T} \right)^{\mu^L} \right]^{\gamma^L} \left[\left(\frac{\bar{W}_{i,t-1}}{\bar{W}_{i,s-1}} \right)^{1-\mu^L} \left(\frac{\bar{P}_{i,t-1}^C \bar{\mathcal{A}}_{i,t-1}}{\bar{P}_{i,s-1}^C \bar{\mathcal{A}}_{i,s-1}} \right)^{\mu^L} \right]^{1-\gamma^L} \frac{W_{i,s}}{W_{i,t}} \right\}^{\theta_{i,s}^L - 1} \left(\frac{W_{i,t}^*}{W_{i,t}} \right)^{-\theta_{i,s}^L} L_{i,h,s}}, \tag{26}$$

$$W_{i,t} = \left\{ (1-\omega^L)(W_{i,t}^*)^{1-\theta_{i,t}^L} + \omega^L \left\{ \left[\left(\frac{W_{i,t-1}}{W_{i,t-2}} \right)^{1-\mu^L} \left(\frac{P_{i,t-1}^C \mathcal{A}_{i,t-1}^T}{P_{i,t-2}^C \mathcal{A}_{i,t-2}^T} \right)^{\mu^L} \right]^{\gamma^L} \left[\left(\frac{\bar{W}_{i,t-1}}{\bar{W}_{i,t-2}} \right)^{1-\mu^L} \left(\frac{\bar{P}_{i,t-1}^C \bar{\mathcal{A}}_{i,t-1}}{\bar{P}_{i,t-2}^C \bar{\mathcal{A}}_{i,t-2}} \right)^{\mu^L} \right]^{1-\gamma^L} W_{i,t-1} \right\}^{1-\theta_{i,t}^L} \right\}^{\frac{1}{1-\theta_{i,t}^L}}. \tag{27}$$

Here $W_{i,t}^*$ denotes the common wage set by those trade unions that adjust their wage optimally in period t.

B. The Extraction Sector

The extraction sector supplies oil. In doing so, a state-owned enterprise accumulates extraction capital through extraction investment, subject to an oil production quota set by an international oil cartel if the economy is a member.

Extraction Investment

Acting in the interests of the government, the state-owned enterprise maximizes its implicit pre-dividend stock market value

$$\Pi_{i,t}^{O} + V_{i,t}^{O} = \mathrm{E}_t \sum_{s=t}^{\infty} \frac{\beta^{s-t} \lambda_{i,s}^{U}}{\lambda_{i,t}^{U}} \Pi_{i,s}^{O}, \tag{28}$$

where $\lambda_{i,h,s}$ denotes the Lagrange multiplier on the period s household dynamic budget constraint. Ownership of the state-owned enterprise entitles the government to a profit transfer $\Pi_{i,s}^{O}$, defined as revenues from sales of extracted oil $Y_{i,s}^{O}$ at quotation currency denominated global oil price $P_{i^*,s}^{O}$ minus extraction investment expenditures,

$$\Pi_{i,s}^{O} = \mathcal{E}_{i,i^*,s} P_{i^*,s}^{O} Y_{i,s}^{O} - P_{i,s}^{I^O} I_{i,s}^{O}, \tag{29}$$

where $Y_{i,s}^{O} = \mathcal{F}^{O}(u_{i,s}^{O} K_{i,s}^{O})$. The state-owned enterprise utilizes extraction capital $K_{i,s}^{O}$ at rate $u_{i,s}^{O}$ to extract oil $Y_{i,s}^{O}$ according to a production function that exhibits decreasing returns to scale,

$$\mathcal{F}^{O}(u_{i,s}^{O} K_{i,s}^{O}) = (u_{i,s}^{O} K_{i,s}^{O})^{\phi^O}, \tag{30}$$

where $0 < \phi^{O} < 1$. In steady state equilibrium, the extraction capital utilization rate equals one.

The state-owned enterprise enters period s with previously accumulated extraction capital stock $K_{i,s}^{O}$, which subsequently evolves according to accumulation function

$$K_{i,s+1}^{O} = (1 - \delta_O) K_{i,s}^{O} + \mathcal{H}^{O}(I_{i,s}^{O}, I_{i,s-1}^{O}), \tag{31}$$

where $0 \le \delta_O \le 1$. Motivated by Christiano, Eichenbaum and Evans (2005), effective extraction investment function $\mathcal{H}^{O}(I_{i,s}^{O}, I_{i,s-1}^{O})$ incorporates convex adjustment costs in the gross growth rate of the ratio of nominal extraction investment to aggregate nominal output,

$$\mathcal{H}^{O}(I_{i,s}^{O}, I_{i,s-1}^{O}) = \nu_{i,s}^{I^O} \left[1 - \frac{\chi^{O}}{2} \left(\frac{P_{i,s}^{I^O} I_{i,s}^{O}}{P_{i,s-1}^{I^O} I_{i,s-1}^{O}} \frac{P_{i,s-1}^{Y} Y_{i,s-1}}{P_{i,s}^{Y} Y_{i,s}} - 1 \right)^2 \right] I_{i,s}^{O}, \tag{32}$$

where serially correlated extraction investment demand shock $\nu_{i,s}^{I^O}$ satisfies $\nu_{i,s}^{I^O} > 0$, while $\chi^{O} > 0$. In steady state equilibrium, these adjustment costs equal zero, and effective extraction investment equals actual extraction investment.

In period t, the state-owned enterprise chooses state contingent sequences for extraction investment $\{I_{i,s}^{O}\}_{s=t}^{\infty}$ and the extraction capital stock $\{K_{i,s+1}^{O}\}_{s=t}^{\infty}$ to maximize implicit pre-dividend stock market value (28) subject to extraction capital accumulation function (31) and terminal nonnegativity constraint $K_{i,T+1}^{O} \ge 0$ for $T \to \infty$. In equilibrium, the solutions to this value maximization problem satisfy necessary first order conditions:

$$\frac{Q^O_{i,t}}{P^{I^O}_{i,t}}\mathcal{H}^O_1(i,t)+\mathsf{E}_t\frac{\beta\lambda^U_{i,t+1}}{\lambda^U_{i,t}}\frac{P^{I^O}_{i,t+1}}{P^{I^O}_{i,t}}\frac{Q^O_{i,t+1}}{P^{I^O}_{i,t+1}}\mathcal{H}^O_2(i,t+1)=1, \tag{33}$$

$$\frac{Q^O_{i,t}}{P^{I^O}_{i,t}}=\mathsf{E}_t\frac{\beta\lambda^U_{i,t+1}}{\lambda^U_{i,t}}\frac{P^{I^O}_{i,t+1}}{P^{I^O}_{i,t}}\left[u^O_{i,t+1}\mathcal{F}^O_{u^O K^O}(i,t+1)\frac{\mathcal{E}_{i,i^*,t+1}P^O_{i^*,t+1}}{P^{I^O}_{i,t+1}}+(1-\delta_O)\frac{Q^O_{i,t+1}}{P^{I^O}_{i,t+1}}\right]. \tag{34}$$

Here $Q^O_{i,s}$ denotes the shadow price of extraction capital, which is the Lagrange multiplier on the period s extraction capital accumulation function.

Oil Output Supply

For members of the international oil cartel, oil production $Y^O_{i,t}$ satisfies $Y^O_{i,t}=\nu^O_{i,t}Y^{O^Q}_{i,t}$, where $Y^{O^Q}_{i,t}$ denotes an oil production quota, while serially correlated oil supply shock $\nu^O_{i,t}$ captures deviations from quota. For all other economies, oil production satisfies $Y^O_{i,t}=\nu^O_{i,t}\bar{Y}^O_{i,t}$, where serially correlated oil supply shock $\nu^O_{i,t}$ captures deviations from steady state equilibrium.

The international oil cartel sets economy-specific oil production quotas $\{Y^{O^Q}_{i,t}\}^N_{i=1}$ to manage the relative quotation currency denominated global oil price according to an oil supply rule exhibiting partial adjustment dynamics of the form

$$\ln Y^{O^Q}_t-\ln\bar{Y}^{O^Q}_t=\rho^{O^Q}(\ln Y^{O^Q}_{t-1}-\ln\bar{Y}^{O^Q}_{t-1})+(1-\rho^{O^Q})\xi^{O^Q}\left(\ln\frac{P^O_{i^*,t}}{P^Y_{i^*,t}}-\ln\frac{\bar{P}^O_{i^*,t}}{\bar{P}^Y_{i^*,t}}\right)+\nu^{O^Q}_t, \tag{35}$$

where cartel oil production quota $Y^{O^Q}_t$ satisfies $Y^{O^Q}_t=\sum_{i=1}^N Y^{O^Q}_{i,t}$ with $Y^{O^Q}_{i,t}=\omega^O_i Y^{O^Q}_t$, $0\le\omega^O_i\le 1$ and $\sum_{i=1}^N\omega^O_i=1$, while $0\le\rho^{O^Q}<1$ and $\xi^{O^Q}>0$. The deviation of the cartel oil production quota from its steady state equilibrium value is inertially increasing in the deviation of the relative quotation currency denominated global oil price from its target value, and is distributed across economies according to fixed oil production quota shares. This inertial response makes the price elasticity of cartel oil supply lower in the short run than in the long run. Indicator function $\mathcal{I}_i$ activates if and only if the economy is a member of the oil cartel, in which case $\omega^O_i>0$, while $\omega^O_i=0$ otherwise. Deviations from this oil supply rule are captured by mean zero and serially uncorrelated cartel oil supply shock $\nu^{O^Q}_t$.

C. The Production Sector

The production sector supplies a final non-oil output good. In doing so, firms demand the final labor service from households, obtain corporate loans from banks, and accumulate private physical capital through business investment.

Non-Oil Output Demand

There exist a large number of perfectly competitive firms that combine differentiated intermediate non-oil output goods $\{Y^N_{i,f,t}\}^1_{f=0}$ supplied by intermediate non-oil output good firms to produce final non-oil output good $Y^N_{i,t}$ according to constant elasticity of substitution production function:

$$Y_{i,t}^{N} = \left[\int_{0}^{1} (Y_{i,f,t}^{N})^{\frac{\theta_{i,t}^{Y}-1}{\theta_{i,t}^{Y}}} df \right]^{\frac{\theta_{i,t}^{Y}}{\theta_{i,t}^{Y}-1}}. \quad (36)$$

The representative final non-oil output good firm maximizes profits from production of the final non-oil output good with respect to inputs of intermediate non-oil output goods, implying:

$$Y_{i,f,t}^{N} = \left(\frac{P_{i,f,t}^{Y^N}}{P_{i,t}^{Y^N}} \right)^{-\theta_{i,t}^{Y}} Y_{i,t}^{N}, \quad (37)$$

$$P_{i,t}^{Y^N} = \left[\int_{0}^{1} (P_{i,f,t}^{Y^N})^{1-\theta_{i,t}^{Y}} df \right]^{\frac{1}{1-\theta_{i,t}^{Y}}}. \quad (38)$$

Serially uncorrelated non-oil output price markup shock $\vartheta_{i,t}^{Y}$ satisfies $\vartheta_{i,t}^{Y} = \theta_{i,t}^{Y} / (\theta_{i,t}^{Y} - 1)$, where the price elasticity of demand for intermediate non-oil output goods $\theta_{i,t}^{Y}$ satisfies $\theta_{i,t}^{Y} > 1$.

Labor Demand and Business Investment

There exists a continuum of monopolistically competitive intermediate non-oil output good firms indexed by $f \in [0,1]$ that supply differentiated intermediate non-oil output goods, but are otherwise identical. Each intermediate non-oil output good firm f sells shares to domestic credit unconstrained households at price $V_{i,f,t}^{F}$. Acting in the interests of its shareholders, it maximizes its pre-dividend stock market value

$$\Pi_{i,f,t}^{F} + V_{i,f,t}^{F} = \mathrm{E}_t \sum_{s=t}^{\infty} \frac{\beta^{s-t} \lambda_{i,s}^{F}}{\lambda_{i,t}^{F}} \Pi_{i,f,s}^{F}, \quad (39)$$

where $\lambda_{i,s}^{F} = \lambda_{i,s}^{U} \prod_{r=1}^{s-1} (1 + \tilde{v}_{i,r}^{B} + \tilde{v}_{i,r}^{S})^{-1}$. Shares entitle households to dividend payments equal to net profits $\Pi_{i,f,s}^{F}$, defined as after-tax earnings, plus new corporate borrowing $B_{i,f,s}^{F,F}$ less debt service cost $C_{i,f,s}^{F,F}$, minus business investment expenditures,

$$\Pi_{i,f,s}^{F} = (1 - \tau_{i,s}^{K})(P_{i,f,s}^{Y^N} Y_{i,f,s}^{N} - W_{i,s} L_{i,f,s} - \Phi_{i,f,s}^{F}) + (B_{i,f,s}^{F,F} - C_{i,f,s}^{F,F}) - P_{i,s}^{I^K} I_{i,f,s}^{K}, \quad (40)$$

where $Y_{i,f,s}^{N} = \mathcal{F}(u_{i,f,s}^{K} K_{i,f,s}, \mathcal{A}_{i,s} L_{i,f,s})$. Earnings are defined as revenues from sales of differentiated intermediate non-oil output good $Y_{i,f,s}^{N}$ at price $P_{i,f,s}^{Y^N}$ less expenditures on final labor service $L_{i,f,s}$, and other variable costs $\Phi_{i,f,s}^{F}$. The government levies a tax on corporate earnings at rate $\tau_{i,s}^{K}$.

The intermediate non-oil output good firm utilizes private physical capital $K_{i,f,s}$ at rate $u_{i,f,s}^{K}$ and rents final labor service $L_{i,f,s}$ to produce differentiated intermediate non-oil output good $Y_{i,f,s}^{N}$ according to production function:

$$\mathcal{F}(u_{i,f,s}^{K} K_{i,f,s}, \mathcal{A}_{i,s} L_{i,f,s}) = (u_{i,f,s}^{K} K_{i,f,s})^{\phi^Y} (\mathcal{A}_{i,s} L_{i,f,s})^{1-\phi^Y}. \quad (41)$$

This production function exhibits constant returns to scale, with $0 \le \phi^{Y} \le 1$. Productivity $\mathcal{A}_{i,s}$ depends on the ratio of the government capital stock to the aggregate labor force,

$$\mathcal{A}_{i,s} = (\tilde{v}_{i,s}^{\mathcal{A}})^{\phi^{\mathcal{A}}} \left(\frac{K_{i,s}^{G}}{N_{i,s}} \right)^{1-\phi^{\mathcal{A}}}, \quad (42)$$

where internationally and serially correlated productivity shock $\tilde{\nu}_{i,s}^{\mathcal{A}}$ satisfies $\tilde{\nu}_{i,s}^{\mathcal{A}} > 0$, while $0 < \phi^{\mathcal{A}} \leq 1$. Trend productivity $\mathcal{A}_{i,s}^{T}$ exhibits partial adjustment dynamics, that is $\mathcal{A}_{i,s}^{T} = (\mathcal{A}_{i,s-1}^{T})^{\rho^{\mathcal{A}^T}} (\mathcal{A}_{i,s})^{1-\rho^{\mathcal{A}^T}}$ where $0 \leq \rho^{\mathcal{A}^T} < 1$.

In utilizing private physical capital to produce output, the intermediate non-oil output good firm incurs a cost $\mathcal{G}^{K}(u_{i,f,s}^{K}, K_{i,f,s})$ denominated in terms of business investment:

$$\Phi_{i,f,s}^{F} = P_{i,s}^{I^K} \mathcal{G}^{K}(u_{i,f,s}^{K}, K_{i,f,s}) + F_{i,s}^{F}. \tag{43}$$

Following Christiano, Eichenbaum and Evans (2005), this capital utilization cost is increasing in the private physical capital utilization rate at an increasing rate,

$$\mathcal{G}^{K}(u_{i,f,s}^{K}, K_{i,f,s}) = \mu^{K}\left[e^{\eta^{K}(u_{i,f,s}^{K}-1)} - 1\right] K_{i,f,s}, \tag{44}$$

where $\mu^{K} > 0$ and $\eta^{K} > 0$. In steady state equilibrium, the private physical capital utilization rate equals one, and the cost of utilizing private physical capital equals zero. Fixed cost $F_{i,s}^{F}$ ensures that $\int_0^1 \Phi_{i,f,s}^{F} df = 0$.

The intermediate non-oil output good firm enters period s with previously accumulated private physical capital stock $K_{i,f,s}$, which subsequently evolves according to accumulation function

$$K_{i,f,s+1} = (1-\delta_K) K_{i,f,s} + \mathcal{H}^{K}(I_{i,f,s}^{K}, I_{i,f,s-1}^{K}), \tag{45}$$

where $0 \leq \delta_K \leq 1$. Motivated by Christiano, Eichenbaum and Evans (2005), effective business investment function $\mathcal{H}^{K}(I_{i,f,s}^{K}, I_{i,f,s-1}^{K})$ incorporates convex adjustment costs in the gross growth rate of the ratio of nominal business investment to aggregate nominal output,

$$\mathcal{H}^{K}(I_{i,f,s}^{K}, I_{i,f,s-1}^{K}) = \nu_{i,s}^{I^K}\left[1 - \frac{\chi^{K}}{2}\left(\frac{P_{i,s}^{I^K} I_{i,f,s}^{K}}{P_{i,s-1}^{I^K} I_{i,f,s-1}^{K}} \frac{P_{i,s-1}^{Y} Y_{i,s-1}}{P_{i,s}^{Y} Y_{i,s}} - 1\right)^2\right] I_{i,f,s}^{K}, \tag{46}$$

where serially correlated business investment demand shock $\nu_{i,s}^{I^K}$ satisfies $\nu_{i,s}^{I^K} > 0$, while $\chi^{K} > 0$. In steady state equilibrium, these adjustment costs equal zero, and effective business investment equals actual business investment.

The intermediate non-oil output good firm enters period s with previously accumulated corporate debt $D_{i,f,s}^{F}$, which subsequently evolves according to accumulation function

$$D_{i,f,s+1}^{F} = (1-\alpha^{F}) D_{i,f,s}^{F} + B_{i,f,s}^{F,F}, \tag{47}$$

where $0 \leq \alpha^{F} \leq 1$. Adopting the collateralized borrowing variant of the financial accelerator mechanism due to Kiyotaki and Moore (1997), the intermediate non-oil output good firm secures new corporate borrowing from banks to finance a fraction of the installed value of business investment,

$$B_{i,f,s}^{F,F} = \phi^{F} Q_{i,s}^{K} I_{i,f,s}^{K}, \tag{48}$$

given by regulatory corporate loan to value ratio limit ϕ^{F}. Its debt service cost satisfies $C_{i,f,s}^{F,F} = (1-\delta_{i,s}^{F})(\alpha^{F} + i_{i,s-1}^{F}) D_{i,f,s}^{F}$, reflecting an amortization payment at rate α^{F} and an interest payment at corporate loan rate $i_{i,s-1}^{F}$ on the outstanding stock of corporate debt, net of a writedown at corporate loan default rate $\delta_{i,s}^{F}$.

In period t, the intermediate non-oil output good firm chooses state contingent sequences for employment $\{L_{i,f,s}\}_{s=t}^{\infty}$, the private physical capital utilization rate $\{u_{i,f,s}^{K}\}_{s=t}^{\infty}$, business investment $\{I_{i,f,s}^{K}\}_{s=t}^{\infty}$, the private physical capital stock $\{K_{i,f,s+1}\}_{s=t}^{\infty}$ and corporate debt $\{D_{i,f,s+1}^{F}\}_{s=t}^{\infty}$ to maximize pre-dividend stock market value (39) subject to production function (41), private physical capital accumulation function (45), corporate debt accumulation

function (47), and terminal nonnegativity constraints $K_{i,f,T+1} \geq 0$ and $D^F_{i,f,T+1} \geq 0$ for $T \to \infty$. In equilibrium, the solutions to this value maximization problem satisfy necessary first order conditions:

$$\mathcal{F}_{AL}(i,f,t)\Psi^F_{i,f,t} = (1-\tau^K_{i,t})\frac{W_{i,t}}{P^{Y^N}_{i,t}\mathcal{A}_{i,t}}, \tag{49}$$

$$\mathcal{F}_{u^K K}(i,f,t)\frac{P^{Y^N}_{i,t}}{P^{I^K}_{i,t}}\Psi^F_{i,f,t} = (1-\tau^K_{i,t})\frac{\mathcal{G}^K_{u^K}(i,f,t)}{K_{i,f,t}}, \tag{50}$$

$$\frac{Q^K_{i,f,t}}{P^{I^K}_{i,t}}\mathcal{H}^K_1(i,f,t) + \phi^F(1+R^K_{i,f,t})\frac{Q^K_{i,t}}{P^{I^K}_{i,t}} + \mathsf{E}_t\frac{\beta\lambda^F_{i,t+1}}{\lambda^F_{i,t}}\frac{P^{I^K}_{i,t+1}}{P^{I^K}_{i,t}}\frac{Q^K_{i,f,t+1}}{P^{I^K}_{i,t+1}}\mathcal{H}^K_2(i,f,t+1) = 1, \tag{51}$$

$$\frac{Q^K_{i,f,t}}{P^{I^K}_{i,t}} = \mathsf{E}_t\frac{\beta\lambda^F_{i,t+1}}{\lambda^F_{i,t}}\frac{P^{I^K}_{i,t+1}}{P^{I^K}_{i,t}}\left[u^K_{i,f,t+1}\mathcal{F}_{u^K K}(i,f,t+1)\frac{P^{Y^N}_{i,t+1}}{P^{I^K}_{i,t+1}}\Psi^F_{i,f,t+1} - (1-\tau^K_{i,t+1})\mathcal{G}^K_K(i,f,t+1) + (1-\delta_K)\frac{Q^K_{i,f,t+1}}{P^{I^K}_{i,t+1}}\right], \tag{52}$$

$$R^K_{i,f,t} = \mathsf{E}_t\frac{\beta\lambda^F_{i,t+1}}{\lambda^F_{i,t}}\left[(1-\alpha^F)R^K_{i,f,t+1} - (1-\delta^F_{i,t+1})(\alpha^F + i^F_{i,t})\right]. \tag{53}$$

Here $\Psi^F_{i,f,s}$ denotes real marginal cost, which is proportional to the Lagrange multiplier on the period s production technology constraint, while $Q^K_{i,f,s}$ denotes the shadow price of private physical capital, which is the Lagrange multiplier on the period s private physical capital accumulation function, and $R^K_{i,f,s}$ denotes the shadow price of corporate debt, which is proportional to the Lagrange multiplier on the period s corporate debt accumulation function.

Non-Oil Output Supply

In an extension of the model of nominal price rigidity proposed by Calvo (1983) along the lines of Smets and Wouters (2003), each period a randomly selected fraction $1-\omega^Y$ of intermediate non-oil output good firms adjust their price optimally, where $0 \leq \omega^Y < 1$. The remaining fraction ω^Y of intermediate non-oil output good firms adjust their price to account for past non-oil output price inflation according to partial indexation rule

$$P^{Y^N}_{i,f,t} = \left(\frac{P^{Y^N}_{i,t-1}}{P^{Y^N}_{i,t-2}}\right)^{\gamma^Y}\left(\frac{\bar{P}^{Y^N}_{i,t-1}}{\bar{P}^{Y^N}_{i,t-2}}\right)^{1-\gamma^Y} P^{Y^N}_{i,f,t-1}, \tag{54}$$

where $0 \leq \gamma^Y \leq 1$. If the intermediate non-oil output good firm can adjust its price optimally in period t, then it does so to maximize pre-dividend stock market value (39) subject to production function (41), intermediate non-oil output good demand function (37), and the assumed form of nominal price rigidity. In symmetric equilibrium, the solution to this non-oil output price setting problem satisfies:

$$\frac{P^{Y^N,*}_{i,t}}{P^{Y^N}_{i,t}} = \frac{\mathsf{E}_t\sum_{s=t}^{\infty}(\omega^Y)^{s-t}\frac{\beta^{s-t}\lambda^F_{i,s}}{\lambda^F_{i,t}}\theta^Y_{i,s}\Psi^F_{i,f,s}\left[\left(\frac{P^{Y^N}_{i,t-1}}{P^{Y^N}_{i,s-1}}\right)^{\gamma^Y}\left(\frac{\bar{P}^{Y^N}_{i,t-1}}{\bar{P}^{Y^N}_{i,s-1}}\right)^{1-\gamma^Y}\frac{P^{Y^N}_{i,s}}{P^{Y^N}_{i,t}}\right]^{\theta^Y_{i,s}}\left(\frac{P^{Y^N,*}_{i,t}}{P^{Y^N}_{i,t}}\right)^{-\theta^Y_{i,s}} P^{Y^N}_{i,s}Y^N_{i,s}}{\mathsf{E}_t\sum_{s=t}^{\infty}(\omega^Y)^{s-t}\frac{\beta^{s-t}\lambda^F_{i,s}}{\lambda^F_{i,t}}(\theta^Y_{i,s}-1)(1-\tau^K_{i,s})\left[\left(\frac{P^{Y^N}_{i,t-1}}{P^{Y^N}_{i,s-1}}\right)^{\gamma^Y}\left(\frac{\bar{P}^{Y^N}_{i,t-1}}{\bar{P}^{Y^N}_{i,s-1}}\right)^{1-\gamma^Y}\frac{P^{Y^N}_{i,s}}{P^{Y^N}_{i,t}}\right]^{\theta^Y_{i,s}-1}\left(\frac{P^{Y^N,*}_{i,t}}{P^{Y^N}_{i,t}}\right)^{-\theta^Y_{i,s}} P^{Y^N}_{i,s}Y^N_{i,s}}, \tag{55}$$

$$P_{i,t}^{Y^N} = \left\{ (1-\omega^Y)(P_{i,t}^{Y^N,*})^{1-\theta_{i,t}^Y} + \omega^Y \left[\left(\frac{P_{i,t-1}^{Y^N}}{P_{i,t-2}^{Y^N}} \right)^{\gamma^Y} \left(\frac{\bar{P}_{i,t-1}^{Y^N}}{\bar{P}_{i,t-2}^{Y^N}} \right)^{1-\gamma^Y} P_{i,t-1}^{Y^N} \right]^{1-\theta_{i,t}^Y} \right\}^{\frac{1}{1-\theta_{i,t}^Y}}. \quad (56)$$

Here $P_{i,t}^{Y^N,*}$ denotes the common price set by those intermediate non-oil output good firms that adjust their price optimally in period t.

D. The Banking Sector

The banking sector supplies financial intermediation services subject to financial frictions and regulatory constraints. Banks issue risky mortgage and corporate loans to domestic households and firms at infrequently adjusted predetermined mortgage and corporate loan rates, subject to regulatory loan to value ratio limits. They accumulate bank capital from retained earnings given credit losses to satisfy a regulatory capital ratio requirement, obtaining funding from domestic households via deposits and from the global interbank market via loans, subject to a foreign currency liquidity regulation cost.

Credit Demand

There exist a large number of perfectly competitive banks that combine differentiated intermediate mortgage or corporate loans $\{A_{i,b,t+1}^{B,Z}\}_{b=0}^{1}$ supplied by intermediate banks to produce final mortgage or corporate loan $A_{i,t+1}^{B,Z}$ according to constant elasticity of substitution portfolio aggregator

$$A_{i,t+1}^{B,Z} = \left[\int_0^1 (A_{i,b,t+1}^{B,Z})^{\frac{\theta_{i,t+1}^Z - 1}{\theta_{i,t+1}^Z}} db \right]^{\frac{\theta_{i,t+1}^Z}{\theta_{i,t+1}^Z - 1}}, \quad (57)$$

where $Z \in \{H,F\}$. The representative final bank maximizes profits from intermediation of the final mortgage or corporate loan with respect to inputs of intermediate mortgage or corporate loans, implying:

$$A_{i,b,t+1}^{B,Z} = \left(\frac{\alpha^Z + i_{i,b,t}^Z}{\alpha^Z + i_{i,t}^Z} \right)^{-\theta_{i,t+1}^Z} A_{i,t+1}^{B,Z}, \quad (58)$$

$$\alpha^Z + i_{i,t}^Z = \left[\int_0^1 (\alpha^Z + i_{i,b,t}^Z)^{1-\theta_{i,t+1}^Z} db \right]^{\frac{1}{1-\theta_{i,t+1}^Z}}. \quad (59)$$

Serially correlated mortgage or corporate loan markup shock $\vartheta_{i,t+1}^Z$ satisfies $\vartheta_{i,t+1}^Z = \theta_{i,t+1}^Z / (\theta_{i,t+1}^Z - 1)$, where the rate elasticity of demand for intermediate mortgage or corporate loans $\theta_{i,t+1}^Z$ satisfies $\theta_{i,t+1}^Z > 1$.

Funding Demand and Bank Capital Accumulation

There exists a continuum of monopolistically competitive intermediate banks indexed by $b \in [0,1]$ that supply differentiated intermediate mortgage and corporate loans, but are otherwise identical. Each intermediate bank b

sells shares to domestic credit unconstrained households at price $V^B_{i,b,t}$. Acting in the interests of its shareholders, it maximizes its pre-dividend stock market value:

$$\Pi^B_{i,b,t} + V^B_{i,b,t} = \mathsf{E}_t \sum_{s=t}^{\infty} \frac{\beta^{s-t} \lambda^U_{i,s}}{\lambda^U_{i,t}} \Pi^B_{i,b,s}. \tag{60}$$

Shares entitle households to dividend payments $\Pi^B_{i,b,s}$, defined as profits from providing financial intermediation services less retained earnings $I^B_{i,b,s}$:

$$\begin{aligned}\Pi^B_{i,b,s} = {} & (B^{B,D}_{i,b,s+1} - (1+i^D_{i,s-1})B^{B,D}_{i,b,s}) + (B^{B,B^h}_{i,b,s+1} - (1+i^B_{i,s-1})B^{B,B^h}_{i,b,s}) + \mathcal{E}_{i,i^*,s}(B^{B,B^f}_{i,b,s+1} - (1+i^B_{i^*,s-1})B^{B,B^f}_{i,b,s}) \\ & -(B^{B,H}_{i,b,s} - C^{B,H}_{i,b,s}) - (B^{B,F}_{i,b,s} - C^{B,F}_{i,b,s}) - \Phi^B_{i,b,s} - I^B_{i,b,s}.\end{aligned} \tag{61}$$

Profits are defined as the increase in deposits $B^{B,D}_{i,b,s+1}$ from domestic households net of interest payments at the deposit rate $i^D_{i,s-1}$, plus the increase in net loans $B^{B,B^h}_{i,b,s+1}$ from the domestic interbank market net of interest payments at the domestic interbank rate $i^B_{i,s-1}$, plus the domestic currency denominated increase in net loans $B^{B,B^f}_{i,b,s+1}$ from the global interbank market net of interest payments at the global interbank rate $i^B_{i^*,s-1}$, minus new mortgage lending $B^{B,H}_{i,b,s}$ net of mortgage loan income $C^{B,H}_{i,b,s}$, minus new corporate lending $B^{B,F}_{i,b,s}$ net of corporate loan income $C^{B,F}_{i,b,s}$, minus regulation costs $\Phi^B_{i,b,s}$.

The intermediate bank enters period s with previously accumulated mortgage or corporate credit outstanding $A^{B,Z}_{i,b,s}$, which subsequently evolves according to accumulation function

$$A^{B,Z}_{i,b,s+1} = (1-\alpha^Z)A^{B,Z}_{i,b,s} + B^{B,Z}_{i,b,s}, \tag{62}$$

where $Z \in \{H,F\}$, while $0 \le \alpha^Z \le 1$. During period s, the intermediate bank extends new mortgage or corporate credit $B^{B,Z}_{i,b,s}$ to satisfy demand, ultimately from domestic households or firms. Its mortgage or corporate loan income $C^{B,Z}_{i,b,s}$ satisfies

$$C^{B,Z}_{i,b,s} = (\alpha^Z + i^Z_{i,b,s-1})A^{B,Z}_{i,b,s}, \tag{63}$$

reflecting amortization income at rate α^Z and interest income at variable mortgage or corporate loan rate $i^Z_{i,s-1}$ on the outstanding stock of mortgage or corporate credit.

The intermediate bank transforms deposit and money market funding into risky differentiated intermediate mortgage and corporate loans according to balance sheet identity:

$$A^{B,H}_{i,b,s+1} + A^{B,F}_{i,b,s+1} = B^{B,D}_{i,b,s+1} + B^{B,B^h}_{i,b,s+1} + \mathcal{E}_{i,i^*,s}B^{B,B^f}_{i,b,s+1} + K^B_{i,b,s+1}. \tag{64}$$

The bank credit stock $A^B_{i,s+1}$ measures aggregate bank assets, that is $A^B_{i,s+1} = A^{B,H}_{i,s+1} + A^{B,F}_{i,s+1}$, while the money stock $M^S_{i,s+1}$ measures aggregate bank funding, that is $M^S_{i,s+1} = B^{B,D}_{i,s+1} + B^{B,B^h}_{i,s+1} + \mathcal{E}_{i,i^*,s}B^{B,B^f}_{i,s+1}$. The bank capital ratio $\kappa_{i,s+1}$ equals the ratio of aggregate bank capital to assets, that is $\kappa_{i,s+1} = K^B_{i,s+1} / A^B_{i,s+1}$.

In transforming deposit and money market funding into risky mortgage and corporate loans, the intermediate bank incurs costs of satisfying the regulatory capital and foreign currency liquidity requirements:

$$\Phi^B_{i,b,s} = \mathcal{G}^{B,C}(A^{B,H}_{i,b,s}, A^{B,F}_{i,b,s}, K^B_{i,b,s}) + \mathcal{G}^{B,L}(B^{B,B^f}_{i,b,s}) + F^B_{i,s}. \tag{65}$$

Motivated by Gerali, Neri, Sessa and Signoretti (2010), the capital regulation cost is decreasing in the ratio of bank capital to assets at a decreasing rate,

$$\mathcal{G}^{B,C}(A^{B,H}_{i,b,s+1},A^{B,F}_{i,b,s+1},K^{B}_{i,b,s+1})=\mu^{B}\left[e^{(2+\eta^{B})\left(1-\frac{1}{\kappa^{R}_{i,s+1}}\frac{K^{B}_{i,b,s+1}}{A^{B,H}_{i,b,s+1}+A^{B,F}_{i,b,s+1}}\right)}-1\right]K^{B}_{i,b,s+1}, \tag{66}$$

given bank capital requirement $\kappa^{R}_{i,s+1}$, where $\mu^{B}>0$ and $\eta^{B}>0$. In steady state equilibrium, the bank capital ratio equals its required value, and the cost of capital regulation is constant. Motivated by Gabaix and Maggiori (2015), the foreign currency liquidity regulation cost is proportional to net global interbank market funding,

$$\mathcal{G}^{B,L}(B^{B,B^{f}}_{i,b,s+1})=\frac{\gamma^{B}_{i,s}-\gamma^{B}}{\beta}\mathcal{E}_{i,i^{*},s}B^{B,B^{f}}_{i,b,s+1}, \tag{67}$$

given foreign currency liquidity shifter $\gamma^{B}_{i,s}$, where $\gamma^{B}_{i,s}>0$. Fixed cost $F^{B}_{i,s}$ ensures that $\int_{0}^{1}\Phi^{B}_{i,b,s}db=-\Delta K^{B}_{i,s+1}$.

The intermediate bank enters period s with previously accumulated bank capital stock $K^{B}_{i,b,s}$, which subsequently evolves according to accumulation function:

$$K^{B}_{i,b,s+1}=(1-\delta^{B}_{i,b,s})K^{B}_{i,b,s}+\mathcal{H}^{B}(I^{B}_{i,b,s},I^{B}_{i,b,s-1}). \tag{68}$$

The bank capital destruction rate $\delta^{B}_{i,b,s}$ equals the ratio of credit losses on mortgage and corporate loans to capital, given mortgage loan default rate $\delta^{H}_{i,s}$ and corporate loan default rate $\delta^{F}_{i,s}$:

$$\delta^{B}_{i,b,s}K^{B}_{i,b,s}=\delta^{H}_{i,s}(\alpha^{H}+i^{H}_{i,b,s-1})A^{B,H}_{i,b,s}+\delta^{F}_{i,s}(\alpha^{F}+i^{F}_{i,b,s-1})A^{B,F}_{i,b,s}. \tag{69}$$

The intermediate bank smooths retained earnings intertemporally, and effective retained earnings function $\mathcal{H}^{B}(I^{B}_{i,b,s},I^{B}_{i,b,s-1})$ incorporates convex adjustment costs,

$$\mathcal{H}^{B}(I^{B}_{i,b,s},I^{B}_{i,b,s-1})=\left[1-\frac{\chi^{B}}{2}\left(\frac{I^{B}_{i,b,s}}{I^{B}_{i,b,s-1}}-1\right)^{2}\right]I^{B}_{i,b,s}-F^{C}_{i,s}, \tag{70}$$

where $\chi^{B}>0$. In steady state equilibrium, these adjustment costs equal zero, and effective retained earnings equals actual retained earnings. Fixed cost $F^{C}_{i,s}$ ensures that $\int_{0}^{1}\mathcal{H}^{B}(I^{B}_{i,b,s},I^{B}_{i,b,s-1})db=I^{B}_{i,s}$.

In period t, the intermediate bank chooses state contingent sequences for deposit funding $\{B^{B,D}_{i,b,s+1}\}^{\infty}_{s=t}$, net domestic interbank market funding $\{B^{B,B^{h}}_{i,b,s+1}\}^{\infty}_{s=t}$, net global interbank market funding $\{B^{B,B^{f}}_{i,b,s+1}\}^{\infty}_{s=t}$, retained earnings $\{I^{B}_{i,b,s}\}^{\infty}_{s=t}$ and the bank capital stock $\{K^{B}_{i,b,s+1}\}^{\infty}_{s=t}$ to maximize pre-dividend stock market value (60) subject to balance sheet identity (64), bank capital accumulation function (68), mortgage and corporate credit accumulation functions (62), and terminal nonnegativity constraint $K^{B}_{i,b,T+1}\geq 0$ for $T\to\infty$. In equilibrium, the solutions to this value maximization problem satisfy necessary first order conditions:

$$1+i^{B}_{i,t}=1+i^{D}_{i,t}, \tag{71}$$

$$\mathrm{E}_{t}\frac{\beta\lambda^{U}_{i,t+1}}{\lambda^{U}_{i,t}}\left[(1+i^{B}_{i,t})-(1+i^{B}_{i^{*},t})\frac{\mathcal{E}_{i,i^{*},t+1}}{\mathcal{E}_{i,i^{*},t}}\right]=\mathrm{E}_{t}\frac{\beta\lambda^{U}_{i,t+1}}{\lambda^{U}_{i,t}}\frac{\mathcal{G}^{B,L}_{B}(i,b,t+1)}{\mathcal{E}_{i,i^{*},t}}, \tag{72}$$

$$\frac{Q^{B}_{i,b,t}}{P^{Y}_{i,t}}\mathcal{H}^{B}_{1}(i,b,t)+\mathrm{E}_{t}\frac{\beta\lambda^{U}_{i,t+1}}{\lambda^{U}_{i,t}}\frac{Q^{B}_{i,b,t+1}}{P^{Y}_{i,t+1}}\mathcal{H}^{B}_{2}(i,b,t+1)=1, \tag{73}$$

$$\frac{Q^{B}_{i,b,t}}{P^{Y}_{i,t}}=\mathrm{E}_{t}\frac{\beta\lambda^{U}_{i,t+1}}{\lambda^{U}_{i,t}}\left\{\frac{Q^{B}_{i,b,t+1}}{P^{Y}_{i,t+1}}-\left[\frac{\lambda^{U}_{i,t}}{\beta\lambda^{U}_{i,t+1}}-(1+i^{B}_{i,t})\right]-\mathcal{G}^{B,C}_{K^{B}}(i,b,t+1)\right\}. \tag{74}$$

Here $Q^B_{i,b,s}$ denotes the shadow price of bank capital, which is proportional to the Lagrange multiplier on the period s bank capital accumulation function.

Credit Supply

In an adaptation of the model of nominal price rigidity proposed by Calvo (1983) to the banking sector along the lines of Hülsewig, Mayer and Wollmershäuser (2009), each period a randomly selected fraction $1-\omega^B$ of intermediate banks adjust their gross mortgage and corporate loan rates optimally, where $0 \le \omega^B < 1$. The remaining fraction ω^B of intermediate banks do not adjust their loan rates,

$$\alpha^Z + i^Z_{i,b,t} = \alpha^Z + i^Z_{i,b,t-1}, \tag{75}$$

where $Z \in \{H,F\}$. If the intermediate bank can adjust its gross mortgage and corporate loan rates in period t, then it does so to maximize pre-dividend stock market value (60) subject to balance sheet identity (64), intermediate loan demand function (58), and the assumed financial friction. In symmetric equilibrium, the solutions to this mortgage and corporate loan rate setting problem satisfy:

$$\frac{\alpha^Z + i^{Z,*}_{i,t}}{\alpha^Z + i^Z_{i,t}} = \frac{\mathsf{E}_t \sum_{s=t}^{\infty} (\omega^B)^{s-t} \frac{\beta^{s-t}\lambda^U_{i,s}}{\lambda^U_{i,t}} \theta^Z_{i,s} \frac{(1+i^B_{i,s-1}) + \mathcal{G}^{B,C}_{A^{B,Z}}(i,b,s)}{\alpha^Z + i^Z_{i,s-1}} \left(\frac{\alpha^Z + i^Z_{i,s-1}}{\alpha^Z + i^Z_{i,t}}\right)^{\theta^Z_{i,s}} \left(\frac{\alpha^Z + i^{Z,*}_{i,t}}{\alpha^Z + i^Z_{i,t}}\right)^{-\theta^Z_{i,s}} (\alpha^Z + i^Z_{i,s-1}) A^{B,Z}_{i,s}}{\mathsf{E}_t \sum_{s=t}^{\infty} (\omega^B)^{s-t} \frac{\beta^{s-t}\lambda^U_{i,s}}{\lambda^U_{i,t}} \left[\theta^Z_{i,s}\left(\frac{1+i^{Z,*}_{i,t}}{\alpha^Z + i^{Z,*}_{i,t}} - \delta^Z_{i,s}\frac{Q^B_{i,b,s}}{P^Y_{i,s}}\right) - \left(1-\delta^Z_{i,s}\frac{Q^B_{i,b,s}}{P^Y_{i,s}}\right)\right] \left(\frac{\alpha^Z + i^Z_{i,s-1}}{\alpha^Z + i^Z_{i,t}}\right)^{\theta^Z_{i,s}-1} \left(\frac{\alpha^Z + i^{Z,*}_{i,t}}{\alpha^Z + i^Z_{i,t}}\right)^{-\theta^Z_{i,s}} (\alpha^Z + i^Z_{i,s-1}) A^{B,Z}_{i,s}}, \tag{76}$$

$$\alpha^Z + i^Z_{i,t} = \left[(1-\omega^B)(\alpha^Z + i^{Z,*}_{i,t})^{1-\theta^Z_{i,t+1}} + \omega^B (\alpha^Z + i^Z_{i,t-1})^{1-\theta^Z_{i,t+1}}\right]^{\frac{1}{1-\theta^Z_{i,t+1}}}. \tag{77}$$

Here $i^{Z,*}_{i,t}$ denote the common loan rates set by those intermediate banks that adjust their loan rates in period t.

Loan Defaults and Foreign Currency Liquidity

We define the financial gap as a weighted average of the contemporaneous deviations of real nonfinancial private sector debt and the relative prices of housing and equity from their steady state equilibrium values,

$$\ln F_{i,t} - \ln \bar{F}_{i,t} = \phi^F_B \left(\ln \frac{D^P_{i,t+1}}{P^{Y^N}_{i,t}} - \ln \frac{\bar{D}^P_{i,t+1}}{\bar{P}^{Y^N}_{i,t}}\right) + \phi^F_H \left(\ln \frac{V^H_{i,t}}{P^{Y^N}_{i,t}} - \ln \frac{\bar{V}^H_{i,t}}{\bar{P}^{Y^N}_{i,t}}\right) + \phi^F_F \left(\ln \frac{V^F_{i,t}}{P^{Y^N}_{i,t}} - \ln \frac{\bar{V}^F_{i,t}}{\bar{P}^{Y^N}_{i,t}}\right), \tag{78}$$

where $\phi^F_B > 0$, $\phi^F_H > 0$, $\phi^F_F > 0$ and $\phi^F_B + \phi^F_H + \phi^F_F = 1$. Nonfinancial private sector debt $D^P_{i,t+1}$ satisfies $D^P_{i,t+1} = D^H_{i,t+1} + D^F_{i,t+1}$. The change in the financial gap reflects financial conditions, as well as other drivers of the financial cycle, while its level measures financial vulnerability.

The mortgage and corporate loan default rates satisfy loan default relationships exhibiting partial adjustment dynamics of the form

$$\delta^Z_{i,t} - \delta^Z = \rho^{\delta^Z}(\delta^Z_{i,t-1} - \delta^Z) - (1-\rho^{\delta^Z})\zeta^{\delta^Z}\left[(\ln Y^N_{i,t} - \ln Y^{N,P}_{i,t}) + \zeta^F(\ln F_{i,t} - \ln \bar{F}_{i,t})\right] + \nu^{\delta^Z}_{i,t}, \tag{79}$$

where $Z \in \{H,F\}$, while $0 < \delta^Z < 1$, $0 \le \rho^{\delta^Z} < 1$, $\zeta^{\delta^Z} > 0$ and $\zeta^F > 0$. The deviation of the mortgage or corporate loan default rate from its steady state equilibrium value is inertially decreasing in the contemporaneous non-oil output and financial gaps. Deviations from these default rate relationships are captured by mean zero and serially correlated mortgage or corporate loan default shock $\nu^{\delta^Z}_{i,t}$.

The foreign currency liquidity shifter satisfies a foreign currency liquidity relationship that gradually stabilizes the net foreign asset ratio of the form

$$\gamma_{i,t}^{B} - \gamma^{B} = -\zeta^{\mathcal{E}}\left(\frac{A_{i,t+1}}{P_{i,t}^{Y}Y_{i,t}} - \frac{\bar{A}_{i,t+1}}{\bar{P}_{i,t}^{Y}\bar{Y}_{i,t}}\right) + \nu_{i,t}^{\mathcal{E}}, \tag{80}$$

where $\zeta^{\mathcal{E}} > 0$. The deviation of the foreign currency liquidity shifter from its steady state equilibrium value is decreasing in the contemporaneous deviation of the net foreign asset ratio from its steady state equilibrium value. Deviations from this foreign currency liquidity relationship are captured by mean zero and serially correlated foreign currency liquidity shock $\nu_{i,t}^{\mathcal{E}}$.

E. The Trade Sector

The nominal effective exchange rate $\mathcal{E}_{i,t}$ measures the non-oil trade weighted average price of foreign currency in terms of domestic currency, while the real effective exchange rate $\mathcal{Q}_{i,t}$ measures the non-oil trade weighted average price of foreign output in terms of domestic output,

$$\mathcal{E}_{i,t} = \prod_{j=1}^{N}(\mathcal{E}_{i,j,t})^{w_{i,j}^{T^N}}, \quad \mathcal{Q}_{i,t} = \prod_{j=1}^{N}(\mathcal{Q}_{i,j,t})^{w_{i,j}^{T^N}}, \tag{81}$$

where the real bilateral exchange rate $\mathcal{Q}_{i,j,t}$ satisfies $\mathcal{Q}_{i,j,t} = \mathcal{E}_{i,j,t}P_{j,t}^{Y} / P_{i,t}^{Y}$, and bilateral non-oil trade weight $w_{i,j}^{T^N}$ satisfies $w_{i,i}^{T^N} = 0$, $0 \le w_{i,j}^{T^N} \le 1$ and $\sum_{j=1}^{N} w_{i,j}^{T^N} = 1$. Global terms of trade shifter $\upsilon_t^{\mathcal{T}}$ adjusts to ensure multilateral consistency in nominal trade flows, and in steady state equilibrium satisfies $\upsilon^{\mathcal{T}} = 1$.

The Export Sector

The export sector transforms the final non-oil output good into a final non-oil export good under producer currency pricing. All extracted oil is exported at the quotation currency denominated global oil price.

Non-Oil Export Demand

There exist a large number of perfectly competitive non-oil exporters that combine differentiated intermediate non-oil export goods $\{X_{i,x,t}^{N}\}_{x=0}^{1}$ supplied by intermediate non-oil exporters to produce final non-oil export good $X_{i,t}^{N}$ according to constant elasticity of substitution production function:

$$X_{i,t}^{N} = \left[\int_{0}^{1}(X_{i,x,t}^{N})^{\frac{\theta_{i,t}^{X}-1}{\theta_{i,t}^{X}}}dx\right]^{\frac{\theta_{i,t}^{X}}{\theta_{i,t}^{X}-1}}. \tag{82}$$

The representative final non-oil exporter maximizes profits from production of the final non-oil export good with respect to inputs of intermediate non-oil export goods, implying:

$$X_{i,x,t}^{N} = \left(\frac{P_{i,x,t}^{X^N}}{P_{i,t}^{X^N}}\right)^{-\theta_{i,t}^{X}} X_{i,t}^{N}, \tag{83}$$

$$P_{i,t}^{X^N} = \left[\int_0^1 (P_{i,x,t}^{X^N})^{1-\theta_{i,t}^X} dx\right]^{\frac{1}{1-\theta_{i,t}^X}}. \tag{84}$$

Serially uncorrelated non-oil export price markup shock $\vartheta_{i,t}^X$ satisfies $\vartheta_{i,t}^X = \theta_{i,t}^X / (\theta_{i,t}^X - 1)$, where the price elasticity of demand for intermediate non-oil export goods $\theta_{i,t}^X$ satisfies $\theta_{i,t}^X > 1$.

Non-Oil Export Supply

There exists a continuum of monopolistically competitive intermediate non-oil exporters indexed by $x \in [0,1]$ that supply differentiated intermediate non-oil export goods, but are otherwise identical. Each intermediate non-oil exporter x sells shares to domestic credit unconstrained households at price $V_{i,x,t}^X$. Acting in the interests of its shareholders, it maximizes its pre-dividend stock market value:

$$\Pi_{i,x,t}^X + V_{i,x,t}^X = \mathrm{E}_t \sum_{s=t}^{\infty} \frac{\beta^{s-t}\lambda_{i,s}^U}{\lambda_{i,t}^U} \Pi_{i,x,s}^X. \tag{85}$$

Shares entitle households to dividend payments equal to profits $\Pi_{i,x,s}^X$, defined as earnings less fixed cost $F_{i,s}^X$:

$$\Pi_{i,x,s}^X = P_{i,x,s}^{X^N} X_{i,x,s}^N - P_{i,s}^{Y^N} X_{i,x,s}^N - F_{i,s}^X. \tag{86}$$

Earnings are defined as revenues from sales of differentiated intermediate non-oil export good $X_{i,x,s}^N$ at price $P_{i,x,s}^{X^N}$ minus expenditures on final non-oil output good $X_{i,x,s}^N$. The intermediate non-oil exporter purchases the final non-oil output good and differentiates it. Fixed cost $F_{i,s}^X$ ensures that $\int_0^1 \Pi_{i,x,s}^X dx = 0$.

In an adaptation of the model of nominal import price rigidity proposed by Monacelli (2005) to the export sector, each period a randomly selected fraction $1-\omega^X$ of intermediate non-oil exporters adjust their price optimally, where $0 \le \omega^X < 1$. The remaining fraction ω^X of intermediate non-oil exporters adjust their price to account for past non-oil export price inflation according to partial indexation rule

$$P_{i,x,t}^{X^N} = \left(\frac{P_{i,t-1}^{X^N}}{P_{i,t-2}^{X^N}}\right)^{\gamma^X} \left(\frac{\bar{P}_{i,t-1}^{X^N}}{\bar{P}_{i,t-2}^{X^N}}\right)^{1-\gamma^X} P_{i,x,t-1}^{X^N}, \tag{87}$$

where $0 \le \gamma^X \le 1$. If the intermediate non-oil exporter can adjust its price optimally in period t, then it does so to maximize pre-dividend stock market value (85) subject to intermediate non-oil export good demand function (83), and the assumed form of nominal price rigidity. In equilibrium, the solution to this non-oil export price setting problem satisfies:

$$\frac{P_{i,t}^{X^N,*}}{P_{i,t}^{X^N}} = \frac{\mathrm{E}_t \sum_{s=t}^{\infty} (\omega^X)^{s-t} \frac{\beta^{s-t}\lambda_{i,s}^U}{\lambda_{i,t}^U} \theta_{i,s}^X \frac{P_{i,s}^{Y^N}}{P_{i,s}^{X^N}} \left[\left(\frac{P_{i,t-1}^{X^N}}{P_{i,s-1}^{X^N}}\right)^{\gamma^X} \left(\frac{\bar{P}_{i,t-1}^{X^N}}{\bar{P}_{i,s-1}^{X^N}}\right)^{1-\gamma^X} \frac{P_{i,s}^{X^N}}{P_{i,t}^{X^N}}\right]^{\theta_{i,s}^X} \left(\frac{P_{i,t}^{X^N,*}}{P_{i,t}^{X^N}}\right)^{-\theta_{i,s}^X} P_{i,s}^{X^N} X_{i,s}^N}{\mathrm{E}_t \sum_{s=t}^{\infty} (\omega^X)^{s-t} \frac{\beta^{s-t}\lambda_{i,s}^U}{\lambda_{i,t}^U} (\theta_{i,s}^X - 1) \left[\left(\frac{P_{i,t-1}^{X^N}}{P_{i,s-1}^{X^N}}\right)^{\gamma^X} \left(\frac{\bar{P}_{i,t-1}^{X^N}}{\bar{P}_{i,s-1}^{X^N}}\right)^{1-\gamma^X} \frac{P_{i,s}^{X^N}}{P_{i,t}^{X^N}}\right]^{\theta_{i,s}^X - 1} \left(\frac{P_{i,t}^{X^N,*}}{P_{i,t}^{X^N}}\right)^{-\theta_{i,s}^X} P_{i,s}^{X^N} X_{i,s}^N}, \tag{88}$$

$$P_{i,t}^{X^N} = \left\{(1-\omega^X)(P_{i,t}^{X^N,*})^{1-\theta_{i,t}^X} + \omega^X \left[\left(\frac{P_{i,t-1}^{X^N}}{P_{i,t-2}^{X^N}}\right)^{\gamma^X} \left(\frac{\bar{P}_{i,t-1}^{X^N}}{\bar{P}_{i,t-2}^{X^N}}\right)^{1-\gamma^X} P_{i,t-1}^{X^N}\right]^{1-\theta_{i,t}^X}\right\}^{\frac{1}{1-\theta_{i,t}^X}}. \tag{89}$$

Here $P_{i,t}^{X^N,*}$ denotes the common price set by those intermediate non-oil exporters that adjust their price optimally in period t.

The Import Sector

The import sector transforms economy-specific foreign final non-oil export goods into economy-specific final non-oil import goods under local currency pricing. It then aggregates these economy-specific final non-oil import goods across trading partners into a final non-oil import good. Finally, the import sector combines this final non-oil import good with oil imports to produce a final import good, given an oil consumption subsidy and non-oil import tariff. This aggregation mechanism incorporates dominant currency pricing.

Oil and Non-Oil Import Demand

There exist a large number of perfectly competitive importers that combine oil imports $M_{i,t}^O$ with the final non-oil import good $M_{i,t}^N$ to produce final import good $M_{i,t}$ according to constant elasticity of substitution production function

$$M_{i,t} = \left[(\phi_i^O)^{\frac{1}{\psi^O}} (M_{i,t}^O)^{\frac{\psi^O-1}{\psi^O}} + (1-\phi_i^O)^{\frac{1}{\psi^O}} (\vartheta^X \vartheta^M M_{i,t}^N)^{\frac{\psi^O-1}{\psi^O}} \right]^{\frac{\psi^O}{\psi^O-1}}, \tag{90}$$

where $0 \le \phi_i^O < 1$ and $\psi^O > 0$. The government subsidizes oil consumption at rate $\tau_{i,t}^O$ and levies a tariff on non-oil imports at rate $\tau_{i,s}^M$. The representative final importer maximizes profits from production of the final import good with respect to inputs of oil imports and the final non-oil import good, implying:

$$M_{i,t}^O = \phi_i^O \left((1-\tau_{i,t}^O) \frac{\mathcal{E}_{i,i^*,t} P_{i^*,t}^O}{P_{i,t}^M} \right)^{-\psi^O} M_{i,t},\ M_{i,t}^N = (1-\phi_i^O) \left(\frac{1+\tau_{i,t}^M}{\vartheta^X \vartheta^M} \frac{P_{i,t}^{M^N}}{P_{i,t}^M} \right)^{-\psi^O} \frac{M_{i,t}}{\vartheta^X \vartheta^M}, \tag{91}$$

$$P_{i,t}^M = \left[\phi_i^O ((1-\tau_{i,t}^O) \mathcal{E}_{i,i^*,t} P_{i^*,t}^O)^{1-\psi^O} + (1-\phi_i^O) \left(\frac{1+\tau_{i,t}^M}{\vartheta^X \vartheta^M} P_{i,t}^{M^N} \right)^{1-\psi^O} \right]^{\frac{1}{1-\psi^O}}. \tag{92}$$

The aggregate import price index $P_{i,t}^M$ equals the minimum cost of producing one unit of the final import good, given the prices of oil imports and the final non-oil import good, as well as the oil consumption subsidy and non-oil import tariff rates.

There exist a large number of perfectly competitive non-oil importers that combine economy-specific final non-oil import goods $\{M_{i,j,t}^N\}_{j=1}^N$ to produce final non-oil import good $M_{i,t}^N$ according to fixed proportions production function

$$M_{i,t}^N = \min \left\{ \frac{M_{i,j,t}^N}{\phi_{i,j}^M} \right\}_{j=1}^N, \tag{93}$$

where $\phi_{i,i}^M = 0$, $0 \le \phi_{i,j}^M \le 1$ and $\sum_{j=1}^N \phi_{i,j}^M = 1$. The representative final non-oil importer maximizes profits from production of the final non-oil import good with respect to inputs of economy-specific final non-oil import goods, implying:

$$M^{N}_{i,j,t} = \phi^{M}_{i,j} M^{N}_{i,t}, \tag{94}$$

$$P^{M^N}_{i,t} = \sum_{j=1}^{N} \phi^{M}_{i,j} P^{M^N}_{i,j,t}. \tag{95}$$

The aggregate non-oil import price index $P^{M^N}_{i,t}$ equals the minimum cost of producing one unit of the final non-oil import good, given the prices of economy-specific final non-oil import goods.

There exist a large number of perfectly competitive economy-specific non-oil importers that combine economy-specific differentiated intermediate non-oil import goods $\{M^{N}_{i,j,m,t}\}^{1}_{m=0}$ supplied by economy-specific intermediate non-oil importers to produce economy-specific final non-oil import good $M^{N}_{i,j,t}$ according to constant elasticity of substitution production function:

$$M^{N}_{i,j,t} = \left[\int_{0}^{1} (M^{N}_{i,j,m,t})^{\frac{\theta^{M}_{i,t}-1}{\theta^{M}_{i,t}}} dm \right]^{\frac{\theta^{M}_{i,t}}{\theta^{M}_{i,t}-1}}. \tag{96}$$

The representative economy-specific final non-oil importer maximizes profits from production of the economy-specific final non-oil import good with respect to inputs of economy-specific intermediate non-oil import goods, implying:

$$M^{N}_{i,j,m,t} = \left(\frac{P^{M^N}_{i,j,m,t}}{P^{M^N}_{i,j,t}} \right)^{-\theta^{M}_{i,t}} M^{N}_{i,j,t}, \tag{97}$$

$$P^{M^N}_{i,j,t} = \left[\int_{0}^{1} (P^{M^N}_{i,j,m,t})^{1-\theta^{M}_{i,t}} dm \right]^{\frac{1}{1-\theta^{M}_{i,t}}}. \tag{98}$$

Serially uncorrelated non-oil import price markup shock $\vartheta^{M}_{i,t}$ satisfies $\vartheta^{M}_{i,t} = \theta^{M}_{i,t} / (\theta^{M}_{i,t} - 1)$, where the price elasticity of demand for economy-specific intermediate non-oil import goods $\theta^{M}_{i,t}$ satisfies $\theta^{M}_{i,t} > 1$.

Non-Oil Import Supply

There exists a continuum of monopolistically competitive economy-specific intermediate non-oil importers indexed by $m \in [0,1]$ that supply economy-specific differentiated intermediate non-oil import goods, but are otherwise identical. Each economy-specific intermediate non-oil importer m sells shares to domestic credit unconstrained households at price $V^{M}_{i,j,m,t}$. Acting in the interests of its shareholders, it maximizes its pre-dividend stock market value:

$$\Pi^{M}_{i,j,m,t} + V^{M}_{i,j,m,t} = \mathrm{E}_t \sum_{s=t}^{\infty} \frac{\beta^{s-t} \lambda^{U}_{i,s}}{\lambda^{U}_{i,t}} \Pi^{M}_{i,j,m,s}. \tag{99}$$

Shares entitle households to dividend payments equal to profits $\Pi^{M}_{i,j,m,s}$, defined as earnings less fixed cost $F^{M}_{i,j,s}$:

$$\Pi^{M}_{i,j,m,s} = P^{M^N}_{i,j,m,s} M^{N}_{i,j,m,s} - \mathcal{E}_{i,j,s} P^{X^N}_{j,s} M^{N}_{i,j,m,s} - F^{M}_{i,j,s}. \tag{100}$$

Earnings are defined as revenues from sales of economy-specific differentiated intermediate non-oil import good $M^{N}_{i,j,m,s}$ at price $P^{M^N}_{i,j,m,s}$ minus expenditures on economy-specific foreign final non-oil export good $M^{N}_{i,j,m,s}$. The

economy-specific intermediate non-oil importer purchases the economy-specific foreign final non-oil export good and differentiates it. Fixed cost $F^M_{i,j,s}$ ensures that $\int_0^1 \Pi^M_{i,j,m,s} dm = 0$.

In an extension of the model of nominal import price rigidity proposed by Monacelli (2005), each period a randomly selected fraction $1-\omega^M$ of economy-specific intermediate non-oil importers adjust their price optimally, where $0 \le \omega^M < 1$. The remaining fraction ω^M of economy-specific intermediate non-oil importers adjust their price to account for past economy-specific non-oil import price inflation according to partial indexation rule

$$P^{M^N}_{i,j,m,t} = \left(\frac{P^{M^N}_{i,j,t-1}}{P^{M^N}_{i,j,t-2}}\right)^{\gamma^M} \left(\frac{\bar{P}^{M^N}_{i,j,t-1}}{\bar{P}^{M^N}_{i,j,t-2}}\right)^{1-\gamma^M} P^{M^N}_{i,j,m,t-1}, \tag{101}$$

where $0 \le \gamma^M \le 1$. If the economy-specific intermediate non-oil importer can adjust its price optimally in period t, then it does so to maximize pre-dividend stock market value (99) subject to economy-specific intermediate non-oil import good demand function (97), and the assumed form of nominal price rigidity. In equilibrium, the solution to this economy-specific non-oil import price setting problem satisfies:

$$\frac{P^{M^N,*}_{i,j,t}}{P^{M^N}_{i,j,t}} = \frac{\mathrm{E}_t \sum_{s=t}^{\infty} (\omega^M)^{s-t} \frac{\beta^{s-t}\lambda^U_{i,s}}{\lambda^U_{i,t}} \theta^M_{i,s} \frac{\mathcal{E}_{i,j,s} P^{X^N}_{j,s}}{P^{M^N}_{i,j,s}} \left[\left(\frac{P^{M^N}_{i,j,t-1}}{P^{M^N}_{i,j,s-1}}\right)^{\gamma^M} \left(\frac{\bar{P}^{M^N}_{i,j,t-1}}{\bar{P}^{M^N}_{i,j,s-1}}\right)^{1-\gamma^M} \frac{P^{M^N}_{i,j,s}}{P^{M^N}_{i,j,t}}\right]^{\theta^M_{i,s}} \left(\frac{P^{M^N,*}_{i,j,t}}{P^{M^N}_{i,j,t}}\right)^{-\theta^M_{i,s}} P^{M^N}_{i,j,s} M^N_{i,j,s}}{\mathrm{E}_t \sum_{s=t}^{\infty} (\omega^M)^{s-t} \frac{\beta^{s-t}\lambda^U_{i,s}}{\lambda^U_{i,t}} (\theta^M_{i,s}-1) \left[\left(\frac{P^{M^N}_{i,j,t-1}}{P^{M^N}_{i,j,s-1}}\right)^{\gamma^M} \left(\frac{\bar{P}^{M^N}_{i,j,t-1}}{\bar{P}^{M^N}_{i,j,s-1}}\right)^{1-\gamma^M} \frac{P^{M^N}_{i,j,s}}{P^{M^N}_{i,j,t}}\right]^{\theta^M_{i,s}-1} \left(\frac{P^{M^N,*}_{i,j,t}}{P^{M^N}_{i,j,t}}\right)^{-\theta^M_{i,s}} P^{M^N}_{i,j,s} M^N_{i,j,s}}, \tag{102}$$

$$P^{M^N}_{i,j,t} = \left\{(1-\omega^M)(P^{M^N,*}_{i,j,t})^{1-\theta^M_{i,t}} + \omega^M \left[\left(\frac{P^{M^N}_{i,j,t-1}}{P^{M^N}_{i,j,t-2}}\right)^{\gamma^M} \left(\frac{\bar{P}^{M^N}_{i,j,t-1}}{\bar{P}^{M^N}_{i,j,t-2}}\right)^{1-\gamma^M} P^{M^N}_{i,j,t-1}\right]^{1-\theta^M_{i,t}}\right\}^{\frac{1}{1-\theta^M_{i,t}}}. \tag{103}$$

Here $P^{M,*}_{i,j,t}$ denotes the common price set by those economy-specific intermediate non-oil importers that adjust their price optimally in period t.

F. The Distribution Sector

The distribution sector combines the final non-oil output good with the final import good to produce a final good for absorption by households, firms, the state-owned enterprise, and the government. Exchange rate pass-through to the price of this final absorption good is incomplete in the short run but complete in the long run. In parallel, global oil price passthrough to the price of this final absorption good is incomplete in the short run, but complete in the long run in the absence of the oil consumption subsidy.

Domestic Demand

There exist a large number of perfectly competitive distributors that combine differentiated intermediate absorption goods $\{D_{i,a,t}\}^1_{a=0}$ supplied by intermediate distributors to produce final absorption good $D_{i,t}$ according to constant elasticity of substitution production function:

$$D_{i,t} = \left[\int_0^1 (D_{i,a,t})^{\frac{\theta_{i,t}^D - 1}{\theta_{i,t}^D}} da \right]^{\frac{\theta_{i,t}^D}{\theta_{i,t}^D - 1}}. \quad (104)$$

The representative final distributor maximizes profits from production of the final absorption good with respect to inputs of intermediate absorption goods, implying:

$$D_{i,a,t} = \left(\frac{P_{i,a,t}^D}{P_{i,t}^D} \right)^{-\theta_{i,t}^D} D_{i,t}, \quad (105)$$

$$P_{i,t}^D = \left[\int_0^1 (P_{i,a,t}^D)^{1-\theta_{i,t}^D} da \right]^{\frac{1}{1-\theta_{i,t}^D}}. \quad (106)$$

Serially uncorrelated absorption price markup shock $\vartheta_{i,t}^D$ satisfies $\vartheta_{i,t}^D = \theta_{i,t}^D / (\theta_{i,t}^D - 1)$, where the price elasticity of demand for intermediate absorption goods $\theta_{i,t}^D$ satisfies $\theta_{i,t}^D > 1$.

Import Demand

There exists a continuum of monopolistically competitive intermediate distributors indexed by $a \in [0,1]$ that supply differentiated intermediate absorption goods, but are otherwise identical. Each intermediate distributor a sells shares to domestic credit unconstrained households at price $V_{i,a,t}^D$. Acting in the interests of its shareholders, it maximizes its pre-dividend stock market value:

$$\Pi_{i,a,t}^D + V_{i,a,t}^D = \mathsf{E}_t \sum_{s=t}^{\infty} \frac{\beta^{s-t} \lambda_{i,s}^U}{\lambda_{i,t}^U} \Pi_{i,a,s}^D. \quad (107)$$

Shares entitle households to dividend payments equal to profits $\Pi_{i,a,s}^D$, defined as earnings less fixed cost $F_{i,s}^D$,

$$\Pi_{i,a,s}^D = P_{i,a,s}^D D_{i,a,s} - P_{i,s}^{Y^N} D_{i,a,s}^h - P_{i,s}^M D_{i,a,s}^f - F_{i,s}^D, \quad (108)$$

where $D_{i,a,s} = \mathcal{F}^D(D_{i,a,s}^h, \upsilon_{i,s}^M D_{i,a,s}^f)$. Earnings are defined as revenues from sales of differentiated intermediate absorption good $D_{i,a,s}$ at price $P_{i,a,s}^D$, minus expenditures on final non-oil output good $D_{i,a,s}^h$ and final import good $D_{i,a,s}^f$. Fixed cost $F_{i,s}^D$ ensures that $\int_0^1 \Pi_{i,a,s}^D da = 0$.

The intermediate distributor combines the final non-oil output good $D_{i,a,s}^h$ with the final import good $D_{i,a,s}^f$ to produce differentiated intermediate absorption good $D_{i,a,s}$ according to constant elasticity of substitution production function:

$$\mathcal{F}^D(D_{i,a,s}^h, \upsilon_{i,s}^M D_{i,a,s}^f) = \vartheta^D \left[(1-\phi_i^M)^{\frac{1}{\psi^M}} (D_{i,a,s}^h)^{\frac{\psi^M - 1}{\psi^M}} + (\phi_i^M)^{\frac{1}{\psi^M}} (\nu_{i,s}^M D_{i,a,s}^f)^{\frac{\psi^M - 1}{\psi^M}} \right]^{\frac{\psi^M}{\psi^M - 1}}. \quad (109)$$

This production function exhibits constant returns to scale, with $0 \le \phi_i^M \le 1$ and $\psi^M > 0$, while serially correlated import demand shock $\nu_{i,s}^M$ satisfies $\nu_{i,s}^M > 0$.

In period t, the intermediate distributor chooses state contingent sequences for final non-oil output good demand $\{D_{i,a,s}^h\}_{s=t}^{\infty}$ and final import good demand $\{D_{i,a,s}^f\}_{s=t}^{\infty}$ to maximize pre-dividend stock market value (107) subject to production function (109). In equilibrium, the solutions to this value maximization problem satisfy necessary first order conditions:

$$\mathcal{F}^D_{Y^N}(i,a,t)\Psi^D_{i,a,t} = \frac{P^{Y^N}_{i,t}}{P^D_{i,t}}, \tag{110}$$

$$\mathcal{F}^D_{\upsilon^M M}(i,a,t)\Psi^D_{i,a,t} = \frac{1}{\upsilon^M_{i,t}}\frac{P^M_{i,t}}{P^D_{i,t}}. \tag{111}$$

Here $\Psi^D_{i,a,s}$ denotes real marginal cost, which is proportional to the Lagrange multiplier on the period s production technology constraint.

Domestic Supply

In an adaptation of the model of nominal price rigidity proposed by Calvo (1983) to the distribution sector, each period a randomly selected fraction $1-\omega^D$ of intermediate distributors adjust their price optimally, where $0 \le \omega^D < 1$. The remaining fraction ω^D of intermediate distributors adjust their price to account for past absorption price inflation according to partial indexation rule

$$P^D_{i,a,t} = \left(\frac{P^D_{i,t-1}}{P^D_{i,t-2}}\right)^{\gamma^D}\left(\frac{\bar{P}^D_{i,t-1}}{\bar{P}^D_{i,t-2}}\right)^{1-\gamma^D} P^D_{i,a,t-1}, \tag{112}$$

where $0 \le \gamma^D \le 1$. If the intermediate distributor can adjust its price optimally in period t, then it does so to maximize pre-dividend stock market value (107) subject to production function (109), intermediate absorption good demand function (105), and the assumed form of nominal price rigidity. In equilibrium, the solution to this absorption price setting problem satisfies:

$$\frac{P^{D,*}_{i,t}}{P^D_{i,t}} = \frac{\mathrm{E}_t\sum_{s=t}^{\infty}(\omega^D)^{s-t}\frac{\beta^{s-t}\lambda^U_{i,s}}{\lambda^U_{i,t}}\theta^D_{i,s}\Psi^D_{i,a,s}\left[\left(\frac{P^D_{i,t-1}}{P^D_{i,s-1}}\right)^{\gamma^D}\left(\frac{\bar{P}^D_{i,t-1}}{\bar{P}^D_{i,s-1}}\right)^{1-\gamma^D}\frac{P^D_{i,s}}{P^D_{i,t}}\right]^{\theta^D_{i,s}}\left(\frac{P^{D,*}_{i,t}}{P^D_{i,t}}\right)^{-\theta^D_{i,s}}P^D_{i,s}D_{i,s}}{\mathrm{E}_t\sum_{s=t}^{\infty}(\omega^D)^{s-t}\frac{\beta^{s-t}\lambda^U_{i,s}}{\lambda^U_{i,t}}(\theta^D_{i,s}-1)\left[\left(\frac{P^D_{i,t-1}}{P^D_{i,s-1}}\right)^{\gamma^D}\left(\frac{\bar{P}^D_{i,t-1}}{\bar{P}^D_{i,s-1}}\right)^{1-\gamma^D}\frac{P^D_{i,s}}{P^D_{i,t}}\right]^{\theta^D_{i,s}-1}\left(\frac{P^{D,*}_{i,t}}{P^D_{i,t}}\right)^{-\theta^D_{i,s}}P^D_{i,s}D_{i,s}}, \tag{113}$$

$$P^D_{i,t} = \left\{(1-\omega^D)(P^{D,*}_{i,t})^{1-\theta^D_{i,t}} + \omega^D\left[\left(\frac{P^D_{i,t-1}}{P^D_{i,t-2}}\right)^{\gamma^D}\left(\frac{\bar{P}^D_{i,t-1}}{\bar{P}^D_{i,t-2}}\right)^{1-\gamma^D}P^D_{i,t-1}\right]^{1-\theta^D_{i,t}}\right\}^{\frac{1}{1-\theta^D_{i,t}}}. \tag{114}$$

Here $P^{D,*}_{i,t}$ denotes the common price set by those intermediate distributors that adjust their price optimally in period t.

G. The Government

The government encompasses a sovereign wealth fund, a monetary authority, a fiscal authority, and a macroprudential authority. The sovereign wealth fund manages government oil revenues, the monetary authority conducts monetary policy, the fiscal authority conducts fiscal policy, and the macroprudential authority conducts macroprudential policy. By investing oil profits not needed to finance the budget abroad, the sovereign wealth fund supports fiscal stabilization and intergenerational equity but not economic development.

The Sovereign Wealth Fund

The sovereign wealth fund invests oil profits not needed to finance the budget of the government in quotation currency denominated assets issued by foreign banks, generating profits $\Pi_{i,t}^{W}$ that it transfers to the fiscal authority:

$$\Pi_{i,t}^{W} = \Pi_{i,t}^{O} - \mathcal{E}_{i,i^*,t}(B_{i,t+1}^{W,B^f} - (1+i_{i^*,t-1}^{B})B_{i,t}^{W,B^f}). \tag{115}$$

Profits are defined as the profit transfer from the state-owned enterprise $\Pi_{i,t}^{O}$, minus the domestic currency denominated increase in foreign asset holdings $B_{i,t+1}^{W,B^f}$ net of interest payments at the global interbank rate $i_{i^*,t-1}^{B}$. Indicator function $\mathcal{J}_i$ activates if and only if the economy has a sovereign wealth fund.

The Monetary Authority

The monetary authority implements monetary policy through control of the interbank rate. We differentiate between flexible inflation targeting, managed exchange rate, and fixed exchange rate regimes.

Under a flexible inflation targeting regime, the interbank rate satisfies a monetary policy rule exhibiting partial adjustment dynamics of the form

$$i_{i,t}^{B} - \bar{i}_{i,t}^{B} = \rho^{i^B}(i_{i,t-1}^{B} - \bar{i}_{i,t-1}^{B}) + (1-\rho^{i^B})\left[\xi^{\pi^C}\mathsf{E}_t(\pi_{i,t+1}^{C} - \bar{\pi}_{i,t+1}^{C}) + \xi^{Y^G}(\ln Y_{i,t}^{N} - \ln Y_{i,t}^{N,P})\right] + \nu_{i,t}^{i^B}, \tag{116}$$

where $0 \le \rho^{i^B} < 1$, $\xi^{\pi^C} > 1$ and $\xi^{Y^G} \ge 0$. The deviation of the interbank rate from its steady state equilibrium value is inertially increasing in the expected future deviation of consumption price inflation from its target value, as well as the contemporaneous non-oil output gap. We define the non-oil output gap as the deviation of non-oil output from its potential value, which we define as that non-oil output level consistent with full utilization of private physical capital and effective labor, given the private physical capital stock and effective labor force. Deviations from this monetary policy rule are captured by mean zero and serially uncorrelated monetary policy shock $\nu_{i,t}^{i^B}$.

Under a managed exchange rate regime, the interbank rate satisfies a monetary policy rule exhibiting partial adjustment dynamics of the form

$$\begin{aligned} i_{i,t}^{B} - \bar{i}_{i,t}^{B} &= \rho^{i^B}(i_{i,t-1}^{B} - \bar{i}_{i,t-1}^{B}) \\ &+(1-\rho^{i^B})\left[\xi^{\pi^C}\mathsf{E}_t(\pi_{i,t+1}^{C} - \bar{\pi}_{i,t+1}^{C}) + \xi^{Y^G}(\ln Y_{i,t}^{N} - \ln Y_{i,t}^{N,P}) + \xi^{\Delta\mathcal{E}}(\Delta\ln\mathcal{E}_{i,t} - \Delta\ln\bar{\mathcal{E}}_{i,t})\right] + \nu_{i,t}^{i^B}, \end{aligned} \tag{117}$$

where $\xi^{\Delta\mathcal{E}} \ge 0$. The deviation of the interbank rate from its steady state equilibrium value is inertially increasing in the expected future deviation of consumption price inflation from its target value, as well as the contemporaneous non-oil output gap, and the contemporaneous deviation of the change in the nominal effective exchange rate from its steady state equilibrium value. Deviations from this monetary policy rule are captured by mean zero and serially uncorrelated monetary policy shock $\nu_{i,t}^{i^B}$.

Under a fixed exchange rate regime, the interbank rate satisfies a monetary policy rule exhibiting feedback of the form

$$i_{i,t}^{B} - \bar{i}_{i,t}^{B} = (i_{i^*,t}^{B} - \bar{i}_{i^*,t}^{B}) + (\gamma_{i,t}^{B} - \gamma^{B}) + \xi^{\Delta\mathcal{E}_{i^*}}(\Delta\ln\mathcal{E}_{i,i^*,t} - \Delta\ln\bar{\mathcal{E}}_{i,i^*,t}), \tag{118}$$

where $\xi^{\Delta\mathcal{E}_{i^*}} > 1$. The deviation of the interbank rate from its steady state equilibrium value tracks the sum of the contemporaneous deviations of the global interbank rate and foreign currency liquidity shifter from their steady

state equilibrium values one for one, and is increasing in the contemporaneous deviation of the change in the corresponding nominal bilateral exchange rate from its target value.

The Fiscal Authority

The fiscal authority issues government bonds to domestic households, securing new government borrowing $B_{i,t}^{G,G}$. It also levies taxes on corporate earnings at rate $\tau_{i,t}^{K}$, on household labor income at rate $\tau_{i,t}^{L}$ and on household consumption expenditures at rate $\tau_{i,t}^{C}$, as well as a tariff on non-oil imports at rate $\tau_{i,t}^{M}$. In addition, the fiscal authority receives profit transfer $\Pi_{i,t}^{W}$ from the sovereign wealth fund. These sources of funds are summed in government dynamic budget constraint:

$$\begin{aligned} C_{i,t}^{G,G} + P_{i,t}^{G^C} G_{i,t}^{C} + P_{i,t}^{G^I} G_{i,t}^{I} + \tau_{i,t}^{O} \mathcal{E}_{i,i^*,t} P_{i^*,t}^{O} M_{i,t}^{O} + \int_0^1 T_{i,h,t}^{Z} dh = B_{i,t}^{G,G} + \int_0^1 \tau_{i,t}^{K} (P_{i,f,t}^{Y^N} Y_{i,f,t}^{N} - W_{i,t} L_{i,f,t} - \Phi_{i,f,t}^{F}) df \\ + \int_0^1 \tau_{i,t}^{L} \int_0^1 W_{i,k,t} L_{i,h,k,t} dk dh + \int_0^1 \tau_{i,t}^{C} P_{i,t}^{C} C_{i,h,t} dh + \tau_{i,t}^{M} P_{i,t}^{M^N} M_{i,t}^{N} + \Pi_{i,t}^{W}. \end{aligned} \tag{119}$$

According to this dynamic budget constraint, the fiscal authority services its debt at cost $C_{i,t}^{G,G}$. It also purchases government consumption good $G_{i,t}^{C}$ at price $P_{i,t}^{G^C}$, and government investment good $G_{i,t}^{I}$ at price $P_{i,t}^{G^I}$, accumulating the government capital stock $K_{i,t+1}^{G}$ according to $K_{i,t+1}^{G} = (1-\delta_G) K_{i,t}^{G} + G_{i,t}^{I}$ where $0 \le \delta_G \le 1$. In addition, the fiscal authority subsidizes oil consumption at rate $\tau_{i,t}^{O}$. Finally, it remits household type-specific lump sum transfer payments $\{T_{i,h,t}^{Z}\}_{h=0}^{1}$, which it allocates across nondiscretionary transfers $\{T_{i,h,t}^{Z,N}\}_{h=0}^{1}$ and discretionary transfers $\{T_{i,h,t}^{Z,D}\}_{h=0}^{1}$, that is $T_{i,h,t}^{Z} = T_{i,h,t}^{Z,N} + T_{i,h,t}^{Z,D}$. Nondiscretionary transfers sum to zero across households, that is $\int_0^1 T_{i,h,t}^{Z,N} dh = 0$, while discretionary transfers are targeted at credit constrained households, that is $T_{i,h,t}^{U,D} = 0$. In steady state equilibrium, nondiscretionary transfers equate consumption across credit unconstrained and credit constrained households.

The fiscal authority enters period t with previously accumulated government debt $D_{i,t}^{G}$, which subsequently evolves according to accumulation function

$$D_{i,t+1}^{G} = (1-\alpha^{G}) D_{i,t}^{G} + B_{i,t}^{G,G}, \tag{120}$$

where $\alpha^{G} = 1-\omega^{G}$, which implies that $0 \le \alpha^{G} < 1$. Its debt service cost satisfies $C_{i,t}^{G,G} = (\alpha^{G} + i_{i,t-1}^{G^E}) D_{i,t}^{G}$, reflecting an amortization payment at rate α^{G} and an interest payment at effective rate $i_{i,t-1}^{G^E}$ on the outstanding stock of government debt. This effective government rate evolves according to

$$i_{i,t}^{G^E} = \left(1 - \frac{B_{i,t-1}^{G,G}}{D_{i,t}^{G}}\right) i_{i,t-1}^{G^E} + \frac{B_{i,t-1}^{G,G}}{D_{i,t}^{G}} i_{i,t-1}^{G}, \tag{121}$$

as a weighted average of the interest rates applicable to outstanding and new government debt, the latter given by the past government bond yield.

Government consumption and investment satisfy countercyclical fiscal expenditure rules exhibiting partial adjustment dynamics of the form

$$\ln \frac{G_{i,t}^{Z}}{\overline{G}_{i,t}^{Z}} = \rho^{G^Z} \ln \frac{G_{i,t-1}^{Z}}{\overline{G}_{i,t-1}^{Z}} - (1-\rho^{G^Z}) \xi^{G^Z} (\ln Y_{i,t}^{N} - \ln Y_{i,t}^{N,P}) + \nu_{i,t}^{G^Z}, \tag{122}$$

where $Z \in \{C,I\}$, while $0 \le \rho^{G^Z} < 1$ and $\xi^{G^Z} \ge 0$. The deviation of government consumption or investment from its steady state equilibrium value is inertially decreasing in the contemporaneous non-oil output gap. Deviations from these fiscal expenditure rules are captured by mean zero and serially correlated government consumption or investment demand shock $\nu_{i,t}^{G^Z}$.

The subsidy rate applicable to oil consumption by the domestic economy satisfies a fiscal expenditure rule of the form

$$\tau_{i,t}^{O} = \xi_i^{\tau^O}\left(\ln\frac{P_{i^*,t}^{O}}{P_{i^*,t}^{Y}} - \ln\frac{\bar{P}_{i^*,t}^{O}}{\bar{P}_{i^*,t}^{Y}}\right) + \nu_{i,t}^{\tau^O}, \tag{123}$$

where $\tau_i^O = 0$ and $\xi_i^{\tau^O} \ge 0$. The oil consumption subsidy rate is increasing in the contemporaneous deviation of the quotation currency denominated global oil price from its target value to an economy-specific degree. Deviations from this fiscal expenditure rule are captured by mean zero and serially correlated oil consumption subsidy rate shock $\nu_{i,t}^{\tau^O}$.

The tax rates on corporate earnings, household labor income and household consumption expenditures, as well as the tariff rate on non-oil imports, satisfy acyclical fiscal revenue rules of the form

$$\tau_{i,t}^{Z} - \tau_i^Z = \rho^{\tau^Z}(\tau_{i,t-1}^{Z} - \tau_i^Z) + \nu_{i,t}^{\tau^Z}, \tag{124}$$

where $Z \in \{K,L,C,M\}$, while $0 \le \tau_i^Z < 1$ and $0 \le \rho^{\tau^Z} < 1$, with $\tau_i^M = 0$. Deviations from these fiscal revenue rules are captured by mean zero and serially correlated corporate, labor income or consumption tax, or non-oil import tariff, rate shock $\nu_{i,t}^{\tau^Z}$.

Nondiscretionary transfers to credit constrained households satisfy a transfer payment rule that helps gradually stabilize the net foreign asset ratio of the form

$$\frac{T_{i,t}^{C,N}}{P_{i,t}^{Y}Y_{i,t}} - \frac{\bar{T}_{i,t}^{C,N}}{\bar{P}_{i,t}^{Y}\bar{Y}_{i,t}} = \xi^{T^N}\left(\frac{A_{i,t+1}}{P_{i,t}^{Y}Y_{i,t}} - \frac{\bar{A}_{i,t+1}}{\bar{P}_{i,t}^{Y}\bar{Y}_{i,t}}\right), \tag{125}$$

where $\xi^{T^N} > 0$. The deviation of the nondiscretionary transfer payment ratio from its steady state equilibrium value is increasing in the contemporaneous deviation of the net foreign asset ratio from its steady state equilibrium value. Discretionary transfers to credit constrained households satisfy a transfer payment rule that gradually stabilizes the government debt ratio of the form

$$\frac{T_{i,t}^{C,D}}{P_{i,t}^{Y}Y_{i,t}} - \frac{\bar{T}_{i,t}^{C,D}}{\bar{P}_{i,t}^{Y}\bar{Y}_{i,t}} = -\xi^{T^D}\left(\frac{D_{i,t+1}^{G}}{P_{i,t}^{Y}Y_{i,t}} - \frac{\bar{D}_{i,t+1}^{G}}{\bar{P}_{i,t}^{Y}\bar{Y}_{i,t}}\right) + \nu_{i,t}^{T}, \tag{126}$$

where $\xi^{T^D} > 0$. The deviation of the discretionary transfer payment ratio from its steady state equilibrium value is decreasing in the contemporaneous deviation of the government debt ratio from its target value. Deviations from this transfer payment rule are captured by mean zero and serially correlated transfer payment shock $\nu_{i,t}^{T}$.

The Macroprudential Authority

The regulatory bank capital ratio requirement satisfies a countercyclical capital buffer rule exhibiting partial adjustment dynamics of the form

$$\kappa^R_{i,t+1} - \kappa^R = \rho^{\kappa^R}(\kappa^R_{i,t} - \kappa^R) + (1-\rho^{\kappa^R})\xi^{\kappa^R}\left(\ln\frac{A^B_{i,t+1}}{P^{Y^N}_{i,t}} - \ln\frac{\bar{A}^B_{i,t+1}}{\bar{P}^{Y^N}_{i,t}}\right) + \nu^{\kappa^R}_{i,t}, \quad (127)$$

where $0 < \kappa^R < 1$, $0 \le \rho^{\kappa^R} < 1$ and $\xi^{\kappa^R} \ge 0$. The deviation of the bank capital requirement from its steady state equilibrium value is inertially increasing in the contemporaneous deviation of real bank credit from its steady state equilibrium value. Deviations from this countercyclical capital buffer rule are captured by mean zero and serially correlated bank capital requirement shock $\nu^{\kappa^R}_{i,t}$.

H. Market Clearing Conditions

A rational expectations equilibrium in this DSGE model of the world economy consists of state contingent sequences of allocations for domestic and foreign households, firms, banks and state-owned enterprises that solve their constrained optimization problems given prices and policies, together with state contingent sequences of allocations for domestic and foreign governments and the international oil cartel that satisfy their policy rules and constraints given prices, with supporting prices such that all markets clear.

Clearing of the global oil market requires that world oil supply equal world oil demand, that is $\sum_{i=1}^N X^O_{i,t} = \sum_{i=1}^N M^O_{i,t}$ where $X^O_{i,t} = Y^O_{i,t}$. Clearing of the final non-oil output good market requires that the value of non-oil exports $X^N_{i,t}$ equal the value of final non-oil output good supply minus demand, ultimately from households, firms, the state-owned enterprise and the government,

$$P^{X^N}_{i,t} X^N_{i,t} = P^{Y^N}_{i,t}(Y^N_{i,t} - D^h_{i,t}) - F^D_{i,t}, \quad (128)$$

where $X^N_{i,t} = \sum_{j=1}^N X^N_{i,j,t}$ with $X^N_{i,j,t} = M^N_{j,i,t}$. Clearing of the final import good market requires that the volume of imports $M_{i,t}$ equal the volume of demand, ultimately from households, firms, the state-owned enterprise and the government:

$$M_{i,t} = D^f_{i,t}. \quad (129)$$

In equilibrium, combination of these final output and import good market clearing conditions yields output expenditure decomposition,

$$P^Y_{i,t} Y_{i,t} = P^C_{i,t} C_{i,t} + P^I_{i,t} I_{i,t} + P^G_{i,t} G_{i,t} + P^X_{i,t} X_{i,t} - P^M_{i,t} M_{i,t}, \quad (130)$$

where the price of non-government investment satisfies $P^I_{i,t} = P^{I^H}_{i,t} = P^{I^O}_{i,t} = P^{I^K}_{i,t}$ while non-government investment satisfies $I_{i,t} = I^H_{i,t} + I^O_{i,t} + I^K_{i,t}$, and the price of public demand satisfies $P^G_{i,t} = P^{G^C}_{i,t} = P^{G^I}_{i,t}$ while public demand satisfies $G_{i,t} = G^C_{i,t} + G^I_{i,t}$, with $P^D_{i,t} = P^C_{i,t} = P^I_{i,t} = P^G_{i,t}$ and $D_{i,t} = C_{i,t} + I_{i,t} + G_{i,t}$. Finally, the price of output is defined by $P^Y_{i,t} Y_{i,t} = \mathcal{E}_{i,i^*,t} P^O_{i^*,t} Y^O_{i,t} + P^{Y^N}_{i,t} Y^N_{i,t}$ where output satisfies $Y_{i,t} = Y^O_{i,t} + Y^N_{i,t}$, and the price of exports is defined by $P^X_{i,t} X_{i,t} = \mathcal{E}_{i,i^*,t} P^O_{i^*,t} X^O_{i,t} + P^{X^N}_{i,t} X^N_{i,t}$ where exports satisfy $X_{i,t} = X^O_{i,t} + X^N_{i,t}$.

Let $A_{i,t+1}$ denote the net foreign asset position, which equals the sum of the assets less liabilities of households, firms, banks, the state-owned enterprise and the government,

$$A_{i,t+1} = \mathcal{E}_{i,i^*,t}(B^{W,B^f}_{i,t+1} - B^{B,B^f}_{i,t+1}) + \sum_Z \left(\int_0^1 A^{B,Z}_{i,b,t+1} db - A^{B,Z}_{i,t+1}\right), \quad (131)$$

where $Z \in \{H, F\}$. Imposing equilibrium conditions on government dynamic budget constraint (119) reveals that the fiscal deficit $FD_{i,t} = D^G_{i,t+1} - D^G_{i,t}$ equals the sum of interest expenditures and the non-oil primary fiscal deficit

$PD_{i,t} = P^G_{i,t}G_{i,t} + \tau^O_{i,t}\mathcal{E}_{i,i^*,t}P^O_{i^*,t}M^O_{i,t} + T^{C,D}_{i,t} - (\mathcal{E}_{i,i^*,t}P^O_{i^*,t}Y^O_{i,t} - P^{I^O}_{i,t}I^O_{i,t}) - T_{i,t}$ net of oil profits, plus net investment in the sovereign wealth fund,

$$\begin{aligned} D^G_{i,t+1} - D^G_{i,t} = {} & i^{G^E}_{i,t-1}D^G_{i,t} + P^G_{i,t}G_{i,t} + \tau^O_{i,t}\mathcal{E}_{i,i^*,t}P^O_{i^*,t}M^O_{i,t} + T^{C,D}_{i,t} - (\mathcal{E}_{i,i^*,t}P^O_{i^*,t}Y^O_{i,t} - P^{I^O}_{i,t}I^O_{i,t}) \\ & -\tau^K_{i,t}(P^{Y^N}_{i,t}Y^N_{i,t} - W_{i,t}L_{i,t}) - \tau^L_{i,t}W_{i,t}L_{i,t} - \tau^C_{i,t}P^C_{i,t}C_{i,t} - \tau^M_{i,t}P^{M^N}_{i,t}M^N_{i,t} + \mathcal{E}_{i,i^*,t}(B^{W,B^f}_{i,t+1} - (1+i^B_{i^*,t-1})B^{W,B^f}_{i,t}), \end{aligned} \tag{132}$$

where tax revenues $T_{i,t} = \tau^K_{i,t}(P^{Y^N}_{i,t}Y^N_{i,t} - W_{i,t}L_{i,t}) + \tau^L_{i,t}W_{i,t}L_{i,t} + \tau^C_{i,t}P^C_{i,t}C_{i,t} + \tau^M_{i,t}P^{M^N}_{i,t}M^N_{i,t}$. The budgetary fiscal deficit $BD_{i,t} = FD_{i,t} - \mathcal{E}_{i,i^*,t}(B^{W,B^f}_{i,t+1} - (1+i^B_{i^*,t-1})B^{W,B^f}_{i,t})$ equals the fiscal deficit minus net investment in the sovereign wealth fund. Imposing equilibrium conditions on household dynamic budget constraint (5), then combining it with government dynamic budget constraint (132), firm dividend payment definition (40), bank dividend payment definition (61) and output expenditure decomposition (130) reveals that the current account balance $CA_{i,t} = A_{i,t+1} - A_{i,t}$ equals the sum of net international investment income and the trade balance $TB_{i,t} = P^X_{i,t}X_{i,t} - P^M_{i,t}M_{i,t} - \tau^O_{i,t}\mathcal{E}_{i,i^*,t}P^O_{i^*,t}M^O_{i,t} + \tau^M_{i,t}P^{M^N}_{i,t}M^N_{i,t}$:

$$\begin{aligned} A_{i,t+1} - A_{i,t} = {} & \left((1+i^B_{i^*,t-1})\frac{\mathcal{E}_{i,i^*,t}}{\mathcal{E}_{i,i^*,t-1}} - 1\right)\mathcal{E}_{i,i^*,t-1}(B^{W,B^f}_{i,t} - B^{B,B^f}_{i,t}) - \sum_Z \alpha^Z \left(\int_0^1 A^{B,Z}_{i,b,t}db - A^{B,Z}_{i,t}\right) \\ & + P^X_{i,t}X_{i,t} - P^M_{i,t}M_{i,t} - \tau^O_{i,t}\mathcal{E}_{i,i^*,t}P^O_{i^*,t}M^O_{i,t} + \tau^M_{i,t}P^{M^N}_{i,t}M^N_{i,t}. \end{aligned} \tag{133}$$

The derivation of this result imposes deposit market clearing condition $B^{H,D}_{i,t+1} = B^{B,D}_{i,t+1}$, mortgage and corporate debt market clearing conditions $A^{B,Z}_{i,t+1} = D^Z_{i,t+1}$ for $Z \in \{H,F\}$, government bond market clearing conditions $D^G_{i,t+1} = \sum_{v=1}^{t} V^G_{i,v,t}S^G_{i,v,t+1}$ and $(1+i^{G^E}_{i,t-1})D^G_{i,t} = \sum_{v=1}^{t-1}(\Pi^G_{i,v,t} + V^G_{i,v,t})S^G_{i,v,t}$, stock market clearing conditions $S^F_{i,f,t+1} = 1$ for all $f \in [0,1]$ and $S^B_{i,b,t+1} = 1$ for all $b \in [0,1]$, and domestic interbank market clearing condition $B^{B,B^h}_{i,t+1} = 0$. To close the model, we assume that the economy has net exposure to the global interbank market via at most one sector, with $B^{B,B^f}_{i,t+1} = 0$ if $\mathcal{J}_i = 1$ and $B^{W,B^f}_{i,t+1} = 0$ otherwise. Clearing of the global interbank market requires that the world net foreign asset position equal zero, that is $\sum_{i=1}^{N}\mathcal{E}_{i^*,i,t}A_{i,t+1} = 0$.

3. The Empirical Framework

Scenario analysis is based on a linear state space representation of an approximate multivariate linear rational expectations representation of this DSGE model of the world economy. This multivariate linear rational expectations representation is derived by analytically linearizing the equilibrium conditions of the DSGE model around a stationary deterministic steady state equilibrium that abstracts from long run balanced growth, and consolidating them by substituting out intermediate variables.[1] The response coefficients of these consolidated approximate linear equilibrium conditions are functions of behavioral parameters that have generally been restricted to coincide across economies—occasionally within groups sharing a structural characteristic—and economy-specific structural characteristics implied by steady state equilibrium relationships.

In what follows, $\hat{x}_{i,t}$ denotes the deviation of variable $x_{i,t}$ from its steady state equilibrium value x_i, while $\mathrm{E}_t x_{i,t+s}$ denotes the rational expectation of variable $x_{i,t+s}$ conditional on information available in period t. Bilateral weights $w^{Z^N}_{i,j}$ for evaluating the non-oil trade weighted average of variable $x_{i,t}$ across the trading partners of

[1] In steady state equilibrium $\nu^O_i = \nu^C_i = \nu^{I^H}_i = \nu^{I^O}_i = \nu^{I^K}_i = \nu^M_i = 1$, $\nu^B_i = \nu^H_i = \nu^S_i = \nu^{\delta^H}_i = \nu^{\delta^F}_i = \nu^{\mathcal{E}}_i = \nu^{I^B}_i = \nu^{G^C}_i = \nu^{G^I}_i = \nu^{\tau^K}_i = \nu^{\tau^L}_i = \nu^{\tau^C}_i = \nu^{\tau^M}_i = \nu^{\tau^O}_i = \nu^T_i = \nu^{\kappa^R}_i = \nu^{O^Q} = 0$, and $\tilde{\nu}^Z_i = \nu^Z_i$ for $Z \in \{A,B,S\}$.

economy i are based on non-oil exports for $Z = X$, non-oil imports for $Z = M$, and their average for $Z = T$. World weights w_i^Z for evaluating the weighted average of variable $x_{i,t}$ across all economies are based on output for $Z = Y$ and capital market capitalization for $Z = A$.

A. Endogenous Variables

The deviation of consumption price inflation from its partially indexed value depends on its expected future value, and is driven by a weighted average of the relative prices of non-oil output and imports, determining the price of consumption $\ln\hat{P}_{i,t}^C$ according to Phillips curve:

$$\begin{aligned}\hat{\pi}_{i,t}^C - \gamma^D\hat{\pi}_{i,t-1}^C &= \beta\mathsf{E}_t(\hat{\pi}_{i,t+1}^C - \gamma^D\hat{\pi}_{i,t}^C)\\ &+\frac{(1-\omega^D)(1-\omega^D\beta)}{\omega^D}\left[\left(1-\vartheta^D\frac{M_i}{Y_i}\right)\ln\frac{\hat{P}_{i,t}^{Y^N}}{\hat{P}_{i,t}^C}+\vartheta^D\frac{M_i}{Y_i}\ln\frac{\hat{P}_{i,t}^M}{\hat{v}_{i,t}^M\hat{P}_{i,t}^C}+\ln\hat{\vartheta}_{i,t}^D\right].\end{aligned} \tag{134}$$

The partially indexed value of consumption price inflation depends on its past value. Consumption price inflation $\hat{\pi}_{i,t}^C$ satisfies $\hat{\pi}_{i,t}^C = \ln\hat{P}_{i,t}^C - \ln\hat{P}_{i,t-1}^C$. The total price of consumption $\ln\hat{P}_{i,t}^{C^T}$ satisfies $\ln\hat{P}_{i,t}^{C^T} = \frac{1}{1+\tau_i^C}\hat{\tau}_{i,t}^C + \ln\hat{P}_{i,t}^C$, while total consumption price inflation $\hat{\pi}_{i,t}^{C^T}$ satisfies $\hat{\pi}_{i,t}^{C^T} = \ln\hat{P}_{i,t}^{C^T} - \ln\hat{P}_{i,t-1}^{C^T}$.

Credit unconstrained consumption $\ln\hat{C}_{i,t}^U$ depends on its past and expected future values, and is driven by the real interbank rate adjusted by the housing, duration and equity risk premia, according to Euler equation:

$$\begin{aligned}\ln\hat{C}_{i,t}^U &= \frac{\alpha^C}{1+\alpha^C}\ln\hat{C}_{i,t-1}^U+\frac{1}{1+\alpha^C}\mathsf{E}_t\ln\hat{C}_{i,t+1}^U\\ &-\sigma\frac{1-\alpha^C}{1+\alpha^C}\mathsf{E}_t\left[(\hat{i}_{i,t}^B+\phi_H^C\hat{v}_{i,t}^H+\phi_B^C\hat{\hat{v}}_{i,t}^B+\phi_S^C\hat{\hat{v}}_{i,t}^S-\hat{\pi}_{i,t+1}^C)-\frac{1}{1+\tau_i^C}\Delta\hat{\tau}_{i,t+1}^C+\Delta\ln\hat{v}_{i,t+1}^C\right].\end{aligned} \tag{135}$$

Credit unconstrained consumption is also driven by the change in the consumption tax rate. Credit constrained consumption $\ln\hat{C}_{i,t}^C$ satisfies static budget constraint

$$(1+\tau_i^C)\frac{C_i}{Y_i}\left(\ln\frac{\hat{P}_{i,t}^C\hat{C}_{i,t}^C}{\hat{P}_{i,t}^Y\hat{Y}_{i,t}}+\frac{1}{1+\tau_i^C}\hat{\tau}_{i,t}^C\right)=(1-\tau_i^L)\frac{W_iL_i}{P_i^YY_i}\left(\ln\frac{\hat{W}_{i,t}\hat{L}_{i,t}}{\hat{P}_{i,t}^Y\hat{Y}_{i,t}}-\frac{1}{1-\tau_i^L}\hat{\tau}_{i,t}^L\right)+\frac{1}{\phi^C}\frac{\hat{T}_{i,t}^C}{P_{i,t}^YY_{i,t}}, \tag{136}$$

which equates after-tax credit constrained consumption expenditures to disposable labor income plus transfers.

The deviation of wage inflation from its partially indexed value depends on its expected future value, and is driven by the contemporaneous and past unemployment rate, determining the wage $\ln\hat{W}_{i,t}$ according to Phillips curve:

$$\begin{aligned}\hat{\pi}_{i,t}^W-\gamma^L((1-\mu^L)\hat{\pi}_{i,t-1}^W+\mu^L(\hat{\pi}_{i,t-1}^C+g_{i,t-1}^{A^T}))&=\beta\mathsf{E}_t\left[\hat{\pi}_{i,t+1}^W-\gamma^L((1-\mu^L)\hat{\pi}_{i,t}^W+\mu^L(\hat{\pi}_{i,t}^C+g_{i,t}^{A^T}))\right]\\ &-\frac{(1-\omega^L)(1-\omega^L\beta)}{\omega^L}\left[\frac{1}{\eta}\frac{1}{1-\alpha^L}(\hat{u}_{i,t}^L-\alpha^L\hat{u}_{i,t-1}^L)-\ln\hat{\vartheta}_{i,t}^L\right].\end{aligned} \tag{137}$$

The partially indexed value of wage inflation depends on its past value, as well as past consumption price inflation and trend productivity growth. Wage inflation $\hat{\pi}_{i,t}^W$ satisfies $\hat{\pi}_{i,t}^W = \ln\hat{W}_{i,t} - \ln\hat{W}_{i,t-1}$.

The unemployment rate depends on its past value, and is driven by employment and the real effective wage, determining the labor force $\ln\hat{N}_{i,t}$ according to supply function:

$$\hat{u}^L_{i,t} = \alpha^L \hat{u}^L_{i,t-1} - (1-\alpha^L)\left\{\mu \ln \frac{\hat{L}_{i,t}}{\hat{v}^N_{i,t}} - \eta\left[\ln \frac{\hat{W}_{i,t}}{\hat{P}^C_{i,t}\hat{A}^T_{i,t}} - \frac{1}{1-\tau^L_i}\hat{\tau}^L_{i,t} - \frac{1}{1+\tau^C_i}\hat{\tau}^C_{i,t}\right]\right\}. \quad (138)$$

The unemployment rate also depends on the labor income and consumption tax rates. The unemployment rate $\hat{u}^L_{i,t}$ satisfies $\hat{u}^L_{i,t} = \ln \hat{N}_{i,t} - \ln \hat{L}_{i,t}$.

The price of housing $\ln \hat{V}^H_{i,t}$ depends on its expected future value, and is driven by the user cost of housing, according to asset pricing relationship:

$$\ln \hat{V}^H_{i,t} = \beta \mathrm{E}_t \ln \hat{V}^H_{i,t+1} + (1-\beta)\mathrm{E}_t \ln \hat{\Pi}^H_{i,t+1} - (\hat{i}^B_{i,t} + \hat{\tilde{v}}^B_{i,t} + \hat{v}^H_{i,t}). \quad (139)$$

The price of housing is also driven by the interbank rate adjusted by the duration and housing risk premia. The user cost of housing $\ln \hat{\Pi}^H_{i,t}$ satisfies

$$\frac{\Pi^H_i}{P^Y_i Y_i}\ln \hat{\Pi}^H_{i,t} = \ln(\hat{P}^H_{i,t}\hat{H}_{i,t}) + \frac{A^{B,H}_i}{P^Y_i Y_i}\left[\alpha^H \ln \hat{B}^{H,H}_{i,t} - (1-\delta^H)(\alpha^H + i^H)\ln \hat{C}^{H,H}_{i,t}\right] - \frac{I^H_i}{Y_i}\ln(\hat{P}^C_{i,t}\hat{I}^H_{i,t}), \quad (140)$$

where $\frac{\Pi^H_i}{P^Y_i Y_i} = 1 - \left[1 - \frac{\phi^H}{1+\phi^H(1+R^H)}\left(1-(1-\delta^H)\frac{\alpha^H+i^H}{\alpha^H}\right)\right]\frac{I^H_i}{Y_i}$. New mortgage borrowing $\ln \hat{B}^{H,H}_{i,t}$ satisfies $\ln \hat{B}^{H,H}_{i,t} = \ln(\hat{Q}^H_{i,t}\hat{I}^H_{i,t})$, while mortgage debt service cost $\ln \hat{C}^{H,H}_{i,t}$ satisfies:

$$\ln \hat{C}^{H,H}_{i,t} = \frac{1+i^H}{\alpha^H + i^H}\hat{i}^H_{i,t-1} - \hat{\delta}^H_{i,t} + \ln \hat{A}^{B,H}_{i,t}. \quad (141)$$

Mortgage credit $\ln \hat{A}^{B,H}_{i,t+1}$ is accumulated from new mortgage borrowing according to $\ln \hat{A}^{B,H}_{i,t+1} = (1-\alpha^H)\ln \hat{A}^{B,H}_{i,t} + \alpha^H \ln \hat{B}^{H,H}_{i,t}$.

The deviation of residential investment expenditures from nominal output depends on its past and expected future values, and is driven by the relative shadow price of housing and the shadow price of mortgage debt, determining residential investment $\ln \hat{I}^H_{i,t}$ according to demand function:

$$\begin{aligned}\ln \frac{\hat{P}^C_{i,t}\hat{I}^H_{i,t}}{\hat{P}^Y_{i,t}\hat{Y}_{i,t}} = {} & \frac{1}{1+\beta}\ln \frac{\hat{P}^C_{i,t-1}\hat{I}^H_{i,t-1}}{\hat{P}^Y_{i,t-1}\hat{Y}_{i,t-1}} + \frac{\beta}{1+\beta}\mathrm{E}_t \ln \frac{\hat{P}^C_{i,t+1}\hat{I}^H_{i,t+1}}{\hat{P}^Y_{i,t+1}\hat{Y}_{i,t+1}} \\ & + \frac{1}{\chi^H(1+\beta)}\left[(1+\phi^H(1+R^H))\ln \frac{\hat{Q}^H_{i,t}}{\hat{P}^C_{i,t}} + \phi^H \hat{R}^H_{i,t} + \ln \hat{v}^{I^H}_{i,t}\right].\end{aligned} \quad (142)$$

The relative shadow price of housing depends on its expected future value, and is driven by the real interbank rate adjusted by the duration and housing risk premia, determining the shadow price of housing $\ln \hat{Q}^H_{i,t}$ according to Euler equation:

$$\ln \frac{\hat{Q}^H_{i,t}}{\hat{P}^C_{i,t}} = \mathrm{E}_t\left[\beta(1-\delta_H)\ln \frac{\hat{Q}^H_{i,t+1}}{\hat{P}^C_{i,t+1}} - (\hat{i}^B_{i,t} + \hat{\tilde{v}}^B_t + \hat{v}^H_t - \hat{\pi}^C_{i,t+1}) + (1-\beta(1-\delta_H))\ln \frac{\hat{P}^H_{i,t+1}}{\hat{P}^C_{i,t+1}}\right]. \quad (143)$$

The relative shadow price of housing is also driven by the relative implicit rental price of housing. The implicit rental price of housing $\ln \hat{P}^H_{i,t}$ satisfies:

$$\ln \frac{\hat{P}^H_{i,t}}{\hat{P}^C_{i,t}} - \frac{1}{1+\tau^C_i}\hat{\tau}^C_{i,t} = -\frac{1}{\varsigma}\ln \frac{\hat{H}_{i,t}}{\hat{C}_{i,t}}. \quad (144)$$

The shadow price of mortgage debt $\hat{R}^H_t$ depends on its expected future value, and is driven by the interbank rate adjusted by the duration and housing risk premia, according to Euler equation

$$\frac{\hat{R}^H_{i,t}}{R^H} = \mathsf{E}_t\left[\beta(1-\alpha^H)\frac{\hat{R}^H_{i,t+1}}{R^H} - (\hat{i}^B_{i,t} + \hat{\tilde{v}}^B_t + \hat{v}^H_t) + (1-\beta(1-\alpha^H))\frac{1+i^H}{\alpha^H + i^H}\hat{i}^H_{i,t}\right], \tag{145}$$

where $R^H = -\frac{\beta(1-\delta^H)(\alpha^H+i^H)}{1-\beta(1-\alpha^H)}$. The shadow price of mortgage debt is also driven by the mortgage loan rate. The housing stock $\ln\hat{H}_{i,t+1}$ is accumulated from residential investment according to $\ln\hat{H}_{i,t+1} = (1-\delta_H)\ln\hat{H}_{i,t} + \delta_H \ln(\hat{v}^{I^H}_{i,t}\hat{I}^H_{i,t})$.[2]

The government bond yield $\hat{i}^G_{i,t}$ depends on its expected future value, and is driven by the interbank rate adjusted by the duration risk premium, according to asset pricing relationship:

$$\hat{i}^G_{i,t} = \omega^G\beta\mathsf{E}_t\hat{i}^G_{i,t+1} + \frac{1-\omega^G\beta}{1-\omega^G(1-\omega^G)\beta}(\hat{i}^B_{i,t} + \hat{\tilde{v}}^B_{i,t}). \tag{146}$$

The term premium $\hat{\mu}^B_{i,t}$ depends on its expected future value, and is driven by the duration risk premium, according to:

$$\hat{\mu}^B_{i,t} = \omega^G\beta\mathsf{E}_t\hat{\mu}^B_{i,t+1} + \frac{1-\omega^G\beta}{1-\omega^G(1-\omega^G)\beta}\hat{\tilde{v}}^B_{i,t}. \tag{147}$$

The duration risk premium $\hat{\tilde{v}}^B_{i,t}$ may be internationally correlated given $\hat{\tilde{v}}^B_{i,t} = \frac{\lambda^B}{1-w^A_i}\sum_{j=1}^N w^A_j\hat{v}^B_{j,t} + \left(1-\frac{\lambda^B}{1-w^A_i}\right)\hat{v}^B_{i,t}$ where $0 \le \lambda^B \le 1$, capturing international bond market contagion.

The deviation of extraction investment expenditures from nominal output depends on its past and expected future values, and is driven by the relative shadow price of extraction capital, determining extraction investment $\ln\hat{I}^O_{i,t}$ according to demand function:

$$\ln\frac{\hat{P}^C_{i,t}\hat{I}^O_{i,t}}{\hat{P}^Y_{i,t}\hat{Y}_{i,t}} = \frac{1}{1+\beta}\ln\frac{\hat{P}^C_{i,t-1}\hat{I}^O_{i,t-1}}{\hat{P}^Y_{i,t-1}\hat{Y}_{i,t-1}} + \frac{\beta}{1+\beta}\mathsf{E}_t\ln\frac{\hat{P}^C_{i,t+1}\hat{I}^O_{i,t+1}}{\hat{P}^Y_{i,t+1}\hat{Y}_{i,t+1}} + \frac{1}{\chi^O(1+\beta)}\ln\left(\hat{v}^{I^O}_{i,t}\frac{\hat{Q}^O_{i,t}}{\hat{P}^C_{i,t}}\right). \tag{148}$$

The relative shadow price of extraction capital depends on its expected future value, and is driven by the real interbank rate, determining the shadow price of extraction capital $\ln\hat{Q}^O_{i,t}$ according to Euler equation:

$$\ln\frac{\hat{Q}^O_{i,t}}{\hat{P}^C_{i,t}} = \mathsf{E}_t\left[\beta(1-\delta_O)\ln\frac{\hat{Q}^O_{i,t+1}}{\hat{P}^C_{i,t+1}} - (\hat{i}^B_{i,t} - \hat{\pi}^C_{i,t+1}) + (1-\beta(1-\delta_O))\left(\ln\frac{\hat{\mathcal{E}}_{i,i^*,t+1}\hat{P}^O_{i^*,t+1}}{\hat{P}^C_{i,t+1}} - \ln\frac{\hat{K}^O_{i,t+1}}{\hat{Y}^O_{i,t+1}}\right)\right]. \tag{149}$$

The relative shadow price of extraction capital is also driven by the relative price of oil and the deviation of extraction capital from oil output. The extraction capital utilization rate $\ln\hat{u}^O_{i,t}$ satisfies production function:

$$\ln\hat{Y}^O_{i,t} = \phi^O\ln(\hat{u}^O_{i,t}\hat{K}^O_{i,t}). \tag{150}$$

The extraction capital stock $\ln\hat{K}^O_{i,t+1}$ is accumulated from extraction investment according to $\ln\hat{K}^O_{i,t+1} = (1-\delta_O)\ln\hat{K}^O_{i,t} + \delta_O\ln(\hat{v}^{I^O}_{i,t}\hat{I}^O_{i,t})$.

The deviation of non-oil output price inflation from its partially indexed value depends on its expected future value, and is driven by the labor income share of non-oil output, determining the price of non-oil output $\ln\hat{P}^{Y^N}_{i,t}$ according to Phillips curve:

[2] The inclusion of asset risk premia and exclusion of the mortgage loan default rate in these equations is not microfounded.

$$\hat{\pi}^{Y^N}_{i,t} - \gamma^Y \hat{\pi}^{Y^N}_{i,t-1} = \beta \mathsf{E}_t(\hat{\pi}^{Y^N}_{i,t+1} - \gamma^Y \hat{\pi}^{Y^N}_{i,t}) + \frac{(1-\omega^Y)(1-\omega^Y\beta)}{\omega^Y}\left(\ln\frac{\hat{W}_{i,t}\hat{L}_{i,t}}{\hat{P}^{Y^N}_{i,t}\hat{Y}^N_{i,t}} + \ln\hat{g}^Y_{i,t}\right). \tag{151}$$

The partially indexed value of non-oil output price inflation depends on its past value. Non-oil output price inflation $\hat{\pi}^{Y^N}_{i,t}$ satisfies $\hat{\pi}^{Y^N}_{i,t} = \ln\hat{P}^{Y^N}_{i,t} - \ln\hat{P}^{Y^N}_{i,t-1}$.

Non-oil output depends on utilized private physical capital and effective employment, determining employment $\ln\hat{L}_{i,t}$ according to production function:

$$\ln\hat{Y}^N_{i,t} = \left(1 - \vartheta^Y \frac{W_i L_i}{P^Y_i Y_i}\left(\frac{Y^N_i}{Y_i}\right)^{-1}\right)\ln(\hat{u}^K_{i,t}\hat{K}_{i,t}) + \vartheta^Y \frac{W_i L_i}{P^Y_i Y_i}\left(\frac{Y^N_i}{Y_i}\right)^{-1}\ln(\hat{\mathcal{A}}_{i,t}\hat{L}_{i,t}). \tag{152}$$

Non-oil potential output $\ln\hat{Y}^{N,P}_{i,t}$ depends on the private physical capital stock and effective labor force according to:

$$\ln\hat{Y}^{N,P}_{i,t} = \left(1 - \vartheta^Y \frac{W_i L_i}{P^Y_i Y_i}\left(\frac{Y^N_i}{Y_i}\right)^{-1}\right)\ln\hat{K}_{i,t} + \vartheta^Y \frac{W_i L_i}{P^Y_i Y_i}\left(\frac{Y^N_i}{Y_i}\right)^{-1}\ln(\hat{\mathcal{A}}_{i,t}\hat{N}_{i,t}). \tag{153}$$

The non-oil output gap $\ln\hat{Y}^{N,G}_{i,t}$ satisfies $\ln\hat{Y}^{N,G}_{i,t} = \ln\hat{Y}^N_{i,t} - \ln\hat{Y}^{N,P}_{i,t}$. Productivity $\ln\hat{\mathcal{A}}_{i,t}$ depends on the productivity shifter and government capital intensity according to:

$$\ln\hat{\mathcal{A}}_{i,t} = \phi^{\mathcal{A}}\ln\hat{\tilde{\nu}}^{\mathcal{A}}_{i,t} + (1-\phi^{\mathcal{A}})\ln\frac{\hat{K}^G_{i,t}}{\hat{N}_{i,t}}. \tag{154}$$

The productivity shifter $\ln\hat{\tilde{\nu}}^{\mathcal{A}}_{i,t}$ may be internationally correlated given $\ln\hat{\tilde{\nu}}^{\mathcal{A}}_{i,t} = \frac{\lambda^{\mathcal{A}}}{1-w^Y_i}\sum_{j=1}^N w^Y_j \ln\hat{\nu}^{\mathcal{A}}_{j,t} + \left(1 - \frac{\lambda^{\mathcal{A}}}{1-w^Y_i}\right)\ln\hat{\nu}^{\mathcal{A}}_{i,t}$, where $0 \le \lambda^{\mathcal{A}} \le 1$. Trend productivity $\ln\hat{\mathcal{A}}^T_{i,t}$ satisfies $\ln\hat{\mathcal{A}}^T_{i,t} = \rho^{\mathcal{A}^T}\ln\hat{\mathcal{A}}^T_{i,t-1} + (1-\rho^{\mathcal{A}^T})\ln\hat{\mathcal{A}}_{i,t}$, while trend productivity growth $\hat{g}^{\mathcal{A}^T}_{i,t}$ satisfies $\hat{g}^{\mathcal{A}^T}_{i,t} = \ln\hat{\mathcal{A}}^T_{i,t} - \ln\hat{\mathcal{A}}^T_{i,t-1}$.

The price of equity $\ln\hat{V}^F_{i,t}$ depends on its expected future value, and is driven by corporate profit, according to asset pricing relationship:

$$\ln\hat{V}^F_{i,t} = \beta\mathsf{E}_t\ln\hat{V}^F_{i,t+1} + (1-\beta)\mathsf{E}_t\ln\hat{\Pi}^F_{i,t+1} - (\hat{i}^B_{i,t} + \hat{\tilde{\nu}}^B_{i,t} + \hat{\tilde{\nu}}^S_{i,t}). \tag{155}$$

The price of equity is also driven by the interbank rate adjusted by the duration and equity risk premia. The equity risk premium $\hat{\tilde{\nu}}^S_{i,t}$ may be internationally correlated given $\hat{\tilde{\nu}}^S_{i,t} = \frac{\lambda^S}{1-w^A_i}\sum_{j=1}^N w^A_j\hat{\nu}^S_{j,t} + \left(1 - \frac{\lambda^S}{1-w^A_i}\right)\hat{\nu}^S_{i,t}$ where $0 \le \lambda^S \le 1$, capturing international stock market contagion. Corporate profit $\ln\hat{\Pi}^F_{i,t}$ satisfies

$$\begin{aligned}\frac{\Pi^F_i}{P^Y_i Y_i}\ln\hat{\Pi}^F_{i,t} = (1-\tau^K_i)\left[\frac{Y^N_i}{Y_i}\ln(\hat{P}^{Y^N}_{i,t}\hat{Y}^N_{i,t}) - \frac{W_i L_i}{P^Y_i Y_i}\ln(\hat{W}_{i,t}\hat{L}_{i,t})\right] - \left(\frac{Y^N_i}{Y_i} - \frac{W_i L_i}{P^Y_i Y_i}\right)\hat{\tau}^K_{i,t} \\ + \frac{A^{B,F}_i}{P^Y_i Y_i}\left[\alpha^F\ln\hat{B}^{F,F}_{i,t} - (1-\delta^F)(\alpha^F + i^F)\ln\hat{C}^{F,F}_{i,t}\right] - \frac{I^K_i}{Y_i}\ln(\hat{P}^C_{i,t}\hat{I}^K_{i,t}),\end{aligned} \tag{156}$$

where $\frac{\Pi^F_i}{P^Y_i Y_i} = (1-\tau^K_i)\left(\frac{Y^N_i}{Y_i} - \frac{W_i L_i}{P^Y_i Y_i}\right) - \left[1 - \frac{\phi^F}{1+\phi^F(1+R^K)}\left(1-(1-\delta^F)\frac{\alpha^F+i^F}{\alpha^F}\right)\right]\frac{I^K_i}{Y_i}$. New corporate borrowing $\ln\hat{B}^{F,F}_{i,t}$ satisfies $\ln\hat{B}^{F,F}_{i,t} = \ln(\hat{Q}^K_{i,t}\hat{I}^K_{i,t})$, while corporate debt service cost $\ln\hat{C}^{F,F}_{i,t}$ satisfies:

$$\ln\hat{C}^{F,F}_{i,t} = \frac{1+i^F}{\alpha^F + i^F}\hat{i}^F_{i,t-1} - \hat{\delta}^F_{i,t} + \ln\hat{A}^{B,F}_{i,t}. \tag{157}$$

Corporate credit $\ln\hat{A}^{B,F}_{i,t+1}$ is accumulated from new corporate borrowing according to $\ln\hat{A}^{B,F}_{i,t+1} = (1-\alpha^F)\ln\hat{A}^{B,F}_{i,t} + \alpha^F\ln\hat{B}^{F,F}_{i,t}$.

The deviation of business investment expenditures from nominal output depends on its past and expected future values, and is driven by the relative shadow price of private physical capital and the shadow price of corporate debt, determining business investment $\ln \hat{I}^K_{i,t}$ according to demand function:

$$\ln \frac{\hat{P}^C_{i,t}\hat{I}^K_{i,t}}{\hat{P}^Y_{i,t}\hat{Y}_{i,t}} = \frac{1}{1+\beta}\ln \frac{\hat{P}^C_{i,t-1}\hat{I}^K_{i,t-1}}{\hat{P}^Y_{i,t-1}\hat{Y}_{i,t-1}} + \frac{\beta}{1+\beta}\mathrm{E}_t \ln \frac{\hat{P}^C_{i,t+1}\hat{I}^K_{i,t+1}}{\hat{P}^Y_{i,t+1}\hat{Y}_{i,t+1}} + \frac{1}{\chi^K(1+\beta)}\left[(1+\phi^F(1+R^K))\ln \frac{\hat{Q}^K_{i,t}}{\hat{P}^C_{i,t}} + \phi^F \hat{R}^K_{i,t} + \ln \hat{\nu}^{I^K}_{i,t}\right]. \quad (158)$$

The relative shadow price of private physical capital depends on its expected future value, and is driven by the real interbank rate adjusted by the duration and equity risk premia, determining the shadow price of private physical capital $\ln \hat{Q}^K_{i,t}$ according to Euler equation:

$$\ln \frac{\hat{Q}^K_{i,t}}{\hat{P}^C_{i,t}} = \mathrm{E}_t\left[\beta(1-\delta_K)\ln \frac{\hat{Q}^K_{i,t+1}}{\hat{P}^C_{i,t+1}} - (\hat{i}^B_{i,t} + \hat{\tilde{\nu}}^B_{i,t} + \hat{\tilde{\nu}}^S_{i,t} - \hat{\pi}^C_{i,t+1}) + (1-\beta(1-\delta_K))\left(\eta^K \ln \hat{u}^K_{i,t+1} - \frac{1}{1-\tau^K_i}\hat{\tau}^K_{i,t+1}\right)\right]. \quad (159)$$

The relative shadow price of private physical capital is also driven by the private physical capital utilization and corporate tax rates. The private physical capital utilization rate $\ln \hat{u}^K_{i,t}$ satisfies:

$$\ln \hat{u}^K_{i,t} = \frac{1}{1+\eta^K}\ln \frac{\hat{W}_{i,t}\hat{L}_{i,t}}{\hat{P}^C_{i,t}\hat{K}_{i,t}}. \quad (160)$$

The shadow price of corporate debt $\hat{R}^K_{i,t}$ depends on its expected future value, and is driven by the interbank rate adjusted by the duration and equity risk premia, according to Euler equation

$$\frac{\hat{R}^K_{i,t}}{R^K} = \mathrm{E}_t\left[\beta(1-\alpha^F)\frac{\hat{R}^K_{i,t+1}}{R^K} - (\hat{i}^B_{i,t} + \hat{\tilde{\nu}}^B_{i,t} + \hat{\tilde{\nu}}^S_{i,t}) + (1-\beta(1-\alpha^F))\frac{1+i^F}{\alpha^F + i^F}\hat{i}^F_{i,t}\right], \quad (161)$$

where $R^K = -\frac{\beta(1-\delta^F)(\alpha^F+i^F)}{1-\beta(1-\alpha^F)}$. The shadow price of corporate debt is also driven by the corporate loan rate. The private physical capital stock $\ln \hat{K}_{i,t+1}$ is accumulated from business investment according to $\ln \hat{K}_{i,t+1} = (1-\delta_K)\ln \hat{K}_{i,t} + \delta_K \ln(\hat{\nu}^{I^K}_{i,t}\hat{I}^K_{i,t})$.[3]

The change in the mortgage or corporate loan rate $\hat{i}^Z_{i,t}$ depends on its expected future value, and is driven by the interbank rate and mortgage or corporate loan rate, according to Phillips curves

$$\Delta \hat{i}^Z_{i,t} = \beta \mathrm{E}_t \Delta \hat{i}^Z_{i,t+1} + \frac{(1-\omega^B)(1-\omega^B\beta)}{\omega^B}\left\{\frac{\vartheta^Z}{\beta(1-\delta^Z)(1+i^Z)}\left[\hat{i}^B_{i,t-1} - \eta^B(1-\beta)\left(\hat{\kappa}_{i,t} - \frac{1+\eta^B}{\eta^B}\hat{\kappa}^R_{i,t}\right)\right] + \frac{\alpha^Z + i^Z}{1+i^Z}\left[\hat{\delta}^Z_{i,t} + \frac{\delta^Z}{1-\delta^Z}\ln \frac{\hat{Q}^B_{i,t}}{\hat{P}^Y_{i,t}} + \ln \hat{\vartheta}^Z_{i,t}\right] - \hat{i}^Z_{i,t}\right\}, \quad (162)$$

where $Z \in \{H, F\}$, while $\vartheta^Z = \frac{\beta(1-\delta^Z)(\alpha^Z+i^Z)}{1+\kappa^R(1-\beta)-\beta(1-\alpha^Z)}$. The change in the mortgage or corporate loan rate is also driven by the bank capital ratio and its required value, as well as the mortgage or corporate loan default rate and relative shadow price of bank capital.

[3] The exclusion of the corporate loan default rate in these equations is not microfounded.

The money stock $\ln\hat{M}^S_{i,t+1}$ satisfies $\ln\hat{A}^B_{i,t+1} = (1-\kappa^R)\ln\hat{M}^S_{i,t+1} + \kappa^R \ln\hat{K}^B_{i,t+1}$, while bank credit $\ln\hat{A}^B_{i,t+1}$ depends on mortgage and corporate credit according to

$$\frac{A^B_i}{P^Y_i Y_i}\ln\hat{A}^B_{i,t+1} = \frac{A^{B,H}_i}{P^Y_i Y_i}\ln\hat{A}^{B,H}_{i,t+1} + \frac{A^{B,F}_i}{P^Y_i Y_i}\ln\hat{A}^{B,F}_{i,t+1}, \tag{163}$$

where $\frac{A^B_i}{P^Y_i Y_i} = \frac{A^{B,H}_i}{P^Y_i Y_i} + \frac{A^{B,F}_i}{P^Y_i Y_i}$ with $\frac{A^{B,H}_i}{P^Y_i Y_i} = \frac{1}{\alpha^H}\frac{\phi^H}{1+\phi^H(1+R^H)}\frac{I^H_i}{Y_i}$ and $\frac{A^{B,F}_i}{P^Y_i Y_i} = \frac{1}{\alpha^F}\frac{\phi^F}{1+\phi^F(1+R^K)}\frac{I^K_i}{Y_i}$. Bank retained earnings $\ln\hat{I}^B_{i,t}$ depends on its past and expected future values, and is driven by the relative shadow price of bank capital, according to:

$$\ln\hat{I}^B_{i,t} = \frac{1}{1+\beta}\ln\hat{I}^B_{i,t-1} + \frac{\beta}{1+\beta}\mathrm{E}_t\ln\hat{I}^B_{i,t+1} + \frac{1}{\chi^B(1+\beta)}\ln\frac{\hat{Q}^B_{i,t}}{\hat{P}^Y_{i,t}}. \tag{164}$$

The relative shadow price of bank capital depends on its expected future value, and is driven by the interbank rate, determining the shadow price of bank capital $\ln\hat{Q}^B_{i,t}$ according to Euler equation:

$$\ln\frac{\hat{Q}^B_{i,t}}{\hat{P}^Y_{i,t}} = \beta\mathrm{E}_t\ln\frac{\hat{Q}^B_{i,t+1}}{\hat{P}^Y_{i,t+1}} - \hat{i}^B_{i,t} - (1-\beta)\frac{\eta^B}{\kappa^R}(\hat{\kappa}_{i,t+1} - \hat{\kappa}^R_{i,t+1}). \tag{165}$$

The relative shadow price of bank capital is also driven by the deviation of the bank capital ratio from its required value. The bank capital ratio $\hat{\kappa}_{i,t+1}$ satisfies $\hat{\kappa}_{i,t+1} = \kappa^R(\ln\hat{K}^B_{i,t+1} - \ln\hat{A}^B_{i,t+1})$. The bank capital stock $\ln\hat{K}^B_{i,t+1}$ is accumulated from retained earnings given credit losses according to $\ln\hat{K}^B_{i,t+1} = (1-\delta^B_i)(\ln\hat{K}^B_{i,t} - \hat{\delta}^B_{i,t}) + \delta^B_i\ln\hat{I}^B_{i,t}$.

The financial gap $\ln\hat{F}_{i,t}$ depends on real nonfinancial private sector credit, as well as the relative prices of housing and equity, according to:

$$\ln\hat{F}_{i,t} = \phi^F_B\ln\frac{\hat{A}^B_{i,t+1}}{\hat{P}^{Y^N}_{i,t}} + \phi^F_H\ln\frac{\hat{V}^H_{i,t}}{\hat{P}^{Y^N}_{i,t}} + \phi^F_F\ln\frac{\hat{V}^F_{i,t}}{\hat{P}^{Y^N}_{i,t}}. \tag{166}$$

The mortgage or corporate loan default rate $\hat{\delta}^Z_{i,t}$ responds inertially to the non-oil output and financial gaps according to

$$\hat{\delta}^Z_{i,t} = \rho^{\delta^Z}\hat{\delta}^Z_{i,t-1} - (1-\rho^{\delta^Z})\zeta^{\delta^Z}(\ln\hat{Y}^{N,G}_{i,t} + \zeta^F\ln\hat{F}_{i,t}) + \hat{v}^{\delta^Z}_{i,t}, \tag{167}$$

where $Z \in \{H,F\}$. The bank capital destruction rate $\hat{\delta}^B_{i,t}$ depends on the mortgage and corporate loan default rates according to

$$\delta^B_i\kappa^R\frac{A^B_i}{P^Y_i Y_i}\left[\frac{1-\delta^B_i}{\delta^B_i}\hat{\delta}^B_{i,t} + \ln\hat{K}^B_{i,t}\right] = \sum_Z \delta^Z(\alpha^Z + i^Z)\frac{A^{B,Z}_i}{P^Y_i Y_i}\left[\frac{1+i^Z}{\alpha^Z+i^Z}\hat{i}^Z_{i,t-1} + \frac{1-\delta^Z}{\delta^Z}\hat{\delta}^Z_{i,t} + \ln\hat{A}^{B,Z}_{i,t}\right], \tag{168}$$

where $\delta^B_i\kappa^R\frac{A^B_i}{P^Y_i Y_i} = \delta^H(\alpha^H + i^H)\frac{A^{B,H}_i}{P^Y_i Y_i} + \delta^F(\alpha^F + i^F)\frac{A^{B,F}_i}{P^Y_i Y_i}$. The bank capital destruction rate also depends on the bank capital buffer and credit risk exposures.

Under a flexible inflation targeting regime, the interbank rate $\hat{i}^B_{i,t}$ responds inertially to consumption price inflation and the non-oil output gap according to monetary policy rule:

$$\hat{i}^B_{i,t} = \rho^{i^B}\hat{i}^B_{i,t-1} + (1-\rho^{i^B})(\xi^{\pi^C}\mathrm{E}_t\hat{\pi}^C_{i,t+1} + \xi^{Y^G}\ln\hat{Y}^{N,G}_{i,t}) + \hat{v}^{i^B}_{i,t}. \tag{169}$$

Under a managed exchange rate regime, the interbank rate also responds inertially to the change in the nominal effective exchange rate according to monetary policy rule:

$$\hat{i}^B_{i,t} = \rho^{i^B}\hat{i}^B_{i,t-1} + (1-\rho^{i^B})(\xi^{\pi^C}\mathrm{E}_t\hat{\pi}^C_{i,t+1} + \xi^{Y^G}\ln\hat{Y}^{N,G}_{i,t} + \xi^{\Delta\mathcal{E}}\Delta\ln\hat{\mathcal{E}}_{i,t}) + \hat{v}^{i^B}_{i,t}. \tag{170}$$

Under a fixed exchange rate regime, the interbank rate instead tracks the global interbank rate one for one, while responding to the change in the corresponding nominal bilateral exchange rate, according to monetary policy rule:

$$\hat{i}^{B}_{i,t} = \hat{i}^{B}_{i^*,t} - \zeta^{\mathcal{E}} \frac{\hat{A}_{i,t+1}}{P^{Y}_{i,t} Y_{i,t}} + \xi^{\Delta \mathcal{E}_{i^*}} \Delta \ln \hat{\mathcal{E}}_{i,i^*,t} + \hat{v}^{\mathcal{E}}_{i,t}. \tag{171}$$

The interbank rate also responds to the net foreign asset ratio. It follows that under a fixed exchange rate regime, $\ln \hat{\mathcal{E}}_{i,i^*,t} = 0$ in equilibrium.

Government consumption or investment $\ln \hat{G}^{Z}_{i,t}$ responds inertially to the non-oil output gap according to fiscal expenditure rule

$$\ln \hat{G}^{Z}_{i,t} = \rho^{G^Z} \ln \hat{G}^{Z}_{i,t-1} - (1-\rho^{G^Z}) \xi^{G^Z} \ln \hat{Y}^{N,G}_{i,t} + \hat{v}^{G^Z}_{i,t}, \tag{172}$$

where $Z \in \{C, I\}$. The government capital stock $\ln \hat{K}^{G}_{i,t+1}$ is accumulated from government investment according to $\ln \hat{K}^{G}_{i,t+1} = (1-\delta_G) \ln \hat{K}^{G}_{i,t} + \delta_G \ln \hat{G}^{I}_{i,t}$. The oil consumption subsidy rate $\hat{\tau}^{O}_{i,t}$ responds to the relative quotation currency denominated global oil price according to fiscal expenditure rule:

$$\hat{\tau}^{O}_{i,t} = \xi^{\tau^O}_{I} \ln \frac{\hat{P}^{O}_{i^*,t}}{\hat{P}^{Y}_{i^*,t}} + \hat{v}^{\tau^O}_{i,t}. \tag{173}$$

The corporate, labor income or consumption tax, or non-oil import tariff, rate $\hat{\tau}^{Z}_{t}$ satisfies fiscal revenue rule $\hat{\tau}^{Z}_{i,t} = \rho^{\tau^Z} \hat{\tau}^{Z}_{i,t-1} + \hat{v}^{\tau^Z}_{i,t}$, where $Z \in \{K, L, C, M\}$. The nondiscretionary transfer payment ratio $\frac{\hat{T}^{C,N}_{i,t}}{P^{Y}_{i,t} Y_{i,t}}$ satisfies $\frac{\hat{T}^{C,N}_{i,t}}{P^{Y}_{i,t} Y_{i,t}} = \xi^{T^N} \frac{\hat{A}_{i,t+1}}{P^{Y}_{i,t} Y_{i,t}}$, while the discretionary transfer payment ratio $\frac{\hat{T}^{C,D}_{i,t}}{P^{Y}_{i,t} Y_{i,t}}$ satisfies $\frac{\hat{T}^{C,D}_{i,t}}{P^{Y}_{i,t} Y_{i,t}} = -\xi^{T^D} \frac{\hat{D}^{G}_{i,t+1}}{P^{Y}_{i,t} Y_{i,t}} + \hat{V}^{T}_{i,t}$.

The regulatory bank capital ratio requirement $\hat{\kappa}^{R}_{i,t+1}$ responds inertially to real bank credit according to countercyclical capital buffer rule:

$$\hat{\kappa}^{R}_{i,t+1} = \rho^{\kappa^R} \hat{\kappa}^{R}_{i,t} + (1-\rho^{\kappa^R}) \xi^{\kappa^R} \ln \frac{\hat{A}^{B}_{i,t+1}}{\hat{P}^{Y^N}_{i,t}} + \hat{v}^{\kappa^R}_{i,t}. \tag{174}$$

The regulatory mortgage and corporate loan to value ratio limits are constant.

The nominal bilateral exchange rate $\ln \hat{\mathcal{E}}_{i,i^*,t}$ depends on its expected future value, and is driven by the international interbank rate differential, according to uncovered interest parity condition:

$$\ln \hat{\mathcal{E}}_{i,i^*,t} = \mathrm{E}_t \ln \hat{\mathcal{E}}_{i,i^*,t+1} - (\hat{i}^{B}_{i,t} - \hat{i}^{B}_{i^*,t}) - \zeta^{\mathcal{E}} \frac{\hat{A}_{i,t+1}}{P^{Y}_{i,t} Y_{i,t}} + \hat{v}^{\mathcal{E}}_{i,t}. \tag{175}$$

The nominal bilateral exchange rate is also driven by the net foreign asset ratio. The real bilateral exchange rate $\ln \hat{\mathcal{Q}}_{i,i^*,t}$ satisfies $\ln \hat{\mathcal{Q}}_{i,i^*,t} = \ln \hat{\mathcal{E}}_{i,i^*,t} + \ln \hat{P}^{Y}_{i^*,t} - \ln \hat{P}^{Y}_{i,t}$. The nominal effective exchange rate $\ln \hat{\mathcal{E}}_{i,t}$ satisfies $\ln \hat{\mathcal{E}}_{i,t} = \ln \hat{\mathcal{E}}_{i,i^*,t} - \sum_{j=1}^{N} w^{T^N}_{i,j} \ln \hat{\mathcal{E}}_{j,i^*,t}$, while the real effective exchange rate $\ln \hat{\mathcal{Q}}_{i,t}$ satisfies $\ln \hat{\mathcal{Q}}_{i,t} = \ln \hat{\mathcal{Q}}_{i,i^*,t} - \sum_{j=1}^{N} w^{T^N}_{i,j} \ln \hat{\mathcal{Q}}_{j,i^*,t}$.

Nominal exports depend on nominal oil and non-oil exports, determining the price of exports $\ln \hat{P}^{X}_{i,t}$ according to market clearing condition:

$$\frac{M_i}{Y_i} \ln(\hat{P}^{X}_{i,t} \hat{X}_{i,t}) = \frac{Y^{O}_{i}}{Y_i} \ln(\hat{\mathcal{E}}_{i,i^*,t} \hat{P}^{O}_{i^*,t} \hat{X}^{O}_{i,t}) + \left(\frac{M_i}{Y_i} - \frac{Y^{O}_{i}}{Y_i} \right) \ln(\hat{P}^{X^N}_{i,t} \hat{X}^{N}_{i,t}). \tag{176}$$

Export price inflation $\hat{\pi}^{X}_{i,t}$ satisfies $\hat{\pi}^{X}_{i,t} = \ln \hat{P}^{X}_{i,t} - \ln \hat{P}^{X}_{i,t-1}$. Exports $\ln \hat{X}_{i,t}$ depend on oil and non-oil exports according to

$$\left(\frac{M_i}{Y_i}+(\vartheta^X-1)\frac{Y_i^O}{Y_i}\right)\ln\hat{X}_{i,t}=\vartheta^X\frac{Y_i^O}{Y_i}\ln\frac{\hat{X}_{i,t}^O}{\hat{\nu}_{i,t}^X}+\left(\frac{M_i}{Y_i}-\frac{Y_i^O}{Y_i}\right)\ln\frac{\hat{X}_{i,t}^N}{\hat{\nu}_{i,t}^X}, \tag{177}$$

where oil exports $\ln\hat{X}_{i,t}^O$ satisfy $\ln\hat{X}_{i,t}^O=\ln\hat{Y}_{i,t}^O$, while non-oil exports $\ln\hat{X}_{i,t}^N$ satisfy $w_i^Y\left(\frac{M_i}{Y_i}-\frac{Y_i^O}{Y_i}\right)\ln\hat{X}_{i,t}^N=\sum_{j=1}^N\frac{w_j^Y w_{j,i}^{M^N}}{\vartheta^M}\left(\frac{M_j}{Y_j}-\frac{M_j^O}{Y_j}\right)\ln\hat{M}_{j,t}^N$.[4]

The deviation of non-oil export price inflation from its partially indexed value depends on its expected future value, and is driven by the relative price of non-oil output, determining the price of non-oil exports $\ln\hat{P}_{i,t}^{X^N}$ according to Phillips curve:

$$\hat{\pi}_{i,t}^{X^N}-\gamma^X\hat{\pi}_{i,t-1}^{X^N}=\beta\mathsf{E}_t(\hat{\pi}_{i,t+1}^{X^N}-\gamma^X\hat{\pi}_{i,t}^{X^N})+\frac{(1-\omega^X)(1-\omega^X\beta)}{\omega^X}\left(\ln\frac{\hat{P}_{i,t}^{Y^N}}{\hat{P}_{i,t}^{X^N}}+\ln\hat{\vartheta}_{i,t}^X\right). \tag{178}$$

The partially indexed value of non-oil export price inflation depends on its past value. Non-oil export price inflation $\hat{\pi}_{i,t}^{X^N}$ satisfies $\hat{\pi}_{i,t}^{X^N}=\ln\hat{P}_{i,t}^{X^N}-\ln\hat{P}_{i,t-1}^{X^N}$.

The price of imports $\ln\hat{P}_{i,t}^M$ depends on the domestic currency denominated price of oil, as well as the price of non-oil imports, according to:

$$\ln\hat{P}_{i,t}^M=\frac{M_i^O}{Y_i}\left(\frac{M_i}{Y_i}\right)^{-1}(\ln(\hat{\mathcal{E}}_{i,i^*,t}\hat{P}_{i^*,t}^O)-\hat{\tau}_{i,t}^O)+\left(1-\frac{M_i^O}{Y_i}\left(\frac{M_i}{Y_i}\right)^{-1}\right)(\ln\hat{P}_{i,t}^{M^N}+\hat{\tau}_{i,t}^M). \tag{179}$$

The price of imports also depends on the oil consumption subsidy and non-oil import tariff rates. Import price inflation $\hat{\pi}_{i,t}^M$ satisfies $\hat{\pi}_{i,t}^M=\ln\hat{P}_{i,t}^M-\ln\hat{P}_{i,t-1}^M$. Imports $\ln\hat{M}_{i,t}$ depend on private consumption, non-government investment and government demand according to demand function:

$$\ln\hat{M}_{i,t}=\frac{C_i}{Y_i}\ln\frac{\hat{C}_{i,t}}{\hat{\nu}_{i,t}^M}+\frac{I_i}{Y_i}\ln\frac{\hat{I}_{i,t}}{\hat{\nu}_{i,t}^M}+\frac{G_i}{Y_i}\ln\frac{\hat{G}_{i,t}}{\hat{\nu}_{i,t}^M}-\psi^M\left(1-\vartheta^D\frac{M_i}{Y_i}\right)\ln\frac{\hat{P}_{i,t}^M}{\hat{\nu}_{i,t}^M\hat{P}_{i,t}^{Y^N}}. \tag{180}$$

Imports also depend on the relative price of imports. Oil imports $\ln\hat{M}_{i,t}^O$ satisfy $\ln\hat{M}_{i,t}^O=\ln\hat{M}_{i,t}-\psi^O\left(\ln\frac{\hat{\mathcal{E}}_{i,i^*,t}\hat{P}_{i^*,t}^O}{\hat{P}_{i,t}^M}-\hat{\tau}_{i,t}^O\right)$, while non-oil imports $\ln\hat{M}_{i,t}^N$ satisfy $\ln\hat{M}_{i,t}^N=\ln\hat{M}_{i,t}-\psi^O\left(\ln\frac{\hat{P}_{i,t}^{M^N}}{\hat{P}_{i,t}^M}+\hat{\tau}_{i,t}^M\right)$.

The deviation of non-oil import price inflation from its partially indexed value depends on its expected future value, and is driven by the relative domestic currency denominated foreign price of non-oil exports, determining the price of non-oil imports $\ln\hat{P}_{i,t}^{M^N}$ according to Phillips curve:

$$\hat{\pi}_{i,t}^{M^N}-\gamma^M\hat{\pi}_{i,t-1}^{M^N}=\beta\mathsf{E}_t(\hat{\pi}_{i,t+1}^{M^N}-\gamma^M\hat{\pi}_{i,t}^{M^N})+\frac{(1-\omega^M)(1-\omega^M\beta)}{\omega^M}\left(\sum_{j=1}^N w_{i,j}^{M^N}\ln\frac{\hat{\mathcal{E}}_{i,i^*,t}\hat{P}_{j,t}^{X^N}}{\hat{\mathcal{E}}_{j,i^*,t}\hat{P}_{i,t}^{M^N}}+\ln\hat{\vartheta}_{i,t}^M\right). \tag{181}$$

The partially indexed value of non-oil import price inflation depends on its past value. Non-oil import price inflation $\hat{\pi}_{i,t}^{M^N}$ satisfies $\hat{\pi}_{i,t}^{M^N}=\ln\hat{P}_{i,t}^{M^N}-\ln\hat{P}_{i,t-1}^{M^N}$.

Nominal output depends on nominal oil and non-oil output, determining the price of output $\ln\hat{P}_{i,t}^Y$ according to market clearing condition:

$$\ln(\hat{P}_{i,t}^Y\hat{Y}_{i,t})=\frac{Y_i^O}{Y_i}\ln(\hat{\mathcal{E}}_{i,i^*,t}\hat{P}_{i^*,t}^O\hat{Y}_{i,t}^O)+\frac{Y_i^N}{Y_i}\ln(\hat{P}_{i,t}^{Y^N}\hat{Y}_{i,t}^N). \tag{182}$$

[4] The inclusion of shocks in these equations is not microfounded.

Output price inflation $\hat{\pi}^Y_{i,t}$ satisfies $\hat{\pi}^Y_{i,t} = \ln\hat{P}^Y_{i,t} - \ln\hat{P}^Y_{i,t-1}$. Oil output $\ln\hat{Y}^O_{i,t}$ satisfies $\ln\hat{Y}^O_{i,t} = \mathcal{I}_i \ln\hat{Y}^{O^Q}_t + \ln\hat{\nu}^O_{i,t}$, while non-oil output $\ln\hat{Y}^N_{i,t}$ satisfies $\ln\hat{Y}_{i,t} = \frac{Y^O_i}{Y_i}\ln\hat{Y}^O_{i,t} + \frac{Y^N_i}{Y_i}\ln\hat{Y}^N_{i,t}$ with $\frac{Y^O_i}{Y_i} + \frac{Y^N_i}{Y_i} = 1$. Nominal output also depends on private consumption expenditures, non-government investment expenditures, government expenditures, export revenues and import expenditures, determining output $\ln\hat{Y}_{i,t}$ according to market clearing condition

$$\ln(\hat{P}^Y_{i,t}\hat{Y}_{i,t}) = \frac{C_i}{Y_i}\ln(\hat{P}^C_{i,t}\hat{C}_{i,t}) + \frac{I_i}{Y_i}\ln(\hat{P}^C_{i,t}\hat{I}_{i,t}) + \frac{G_i}{Y_i}\ln(\hat{P}^C_{i,t}\hat{G}_{i,t}) + \frac{M_i}{Y_i}\ln\frac{\hat{P}^X_{i,t}\hat{X}_{i,t}}{\hat{P}^M_{i,t}\hat{M}_{i,t}}, \tag{183}$$

where $\frac{C_i}{Y_i} + \frac{I_i}{Y_i} + \frac{G_i}{Y_i} = 1$. Private consumption $\ln\hat{C}_{i,t}$ satisfies $\ln\hat{C}_{i,t} = (1-\phi^C)\ln\hat{C}^U_{i,t} + \phi^C\ln\hat{C}^C_{i,t}$, while non-government investment $\ln\hat{I}_{i,t}$ satisfies $\frac{I_i}{Y_i}\ln\hat{I}_{i,t} = \frac{I^H_i}{Y_i}\ln\hat{I}^H_{i,t} + \frac{I^O_i}{Y_i}\ln\hat{I}^O_{i,t} + \frac{I^K_i}{Y_i}\ln\hat{I}^K_{i,t}$ with $\frac{I_i}{Y_i} = \frac{I^H_i}{Y_i} + \frac{I^O_i}{Y_i} + \frac{I^K_i}{Y_i}$, and government demand $\ln\hat{G}_{i,t}$ satisfies $\frac{G_i}{Y_i}\ln\hat{G}_{i,t} = \frac{G^C_i}{Y_i}\ln\hat{G}^C_{i,t} + \frac{G^I_i}{Y_i}\ln\hat{G}^I_{i,t}$ with $\frac{G_i}{Y_i} = \frac{G^C_i}{Y_i} + \frac{G^I_i}{Y_i}$.

The fiscal deficit ratio $\frac{\widehat{FD}_{i,t}}{P^Y_{i,t}Y_{i,t}}$ depends on government debt service cost, the non-oil primary fiscal deficit ratio and oil profits, as well as net investment in the sovereign wealth fund if $\mathcal{J}_i = 1$, according to dynamic budget constraint:

$$\begin{aligned}\frac{\widehat{FD}_{i,t}}{P^Y_{i,t}Y_{i,t}} = {} & \frac{1-\omega^G(1-\omega^G)\beta}{\omega^G\beta}\frac{D^G_i}{P^Y_iY_i}\hat{i}^{G^E}_{i,t-1} + \left(\frac{1-\omega^G(1-\omega^G)\beta}{\omega^G\beta} - 1\right)\left(\frac{\hat{D}^G_{i,t}}{P^Y_{i,t-1}Y_{i,t-1}} - \frac{D^G_i}{P^Y_iY_i}\ln\frac{\hat{P}^Y_{i,t}\hat{Y}_{i,t}}{\hat{P}^Y_{i,t-1}\hat{Y}_{i,t-1}}\right) \\ & + \frac{\widehat{PD}_{i,t}}{P^Y_{i,t}Y_{i,t}} - \frac{Y^O_i}{Y_i}\ln\left(\hat{\mathcal{Q}}_{i,i^*,t}\frac{\hat{P}^O_{i^*,t}}{\hat{P}^Y_{i^*,t}}\frac{\hat{Y}^O_{i,t}}{\hat{Y}_{i,t}}\right) + \frac{I^O_i}{Y_i}\ln\frac{\hat{P}^C_{i,t}\hat{I}^O_{i,t}}{\hat{P}^Y_{i,t}\hat{Y}_{i,t}} + \mathcal{J}_i\left(\frac{\hat{A}_{i,t+1}}{P^Y_{i,t}Y_{i,t}} - \frac{1}{\beta}\frac{\hat{A}_{i,t}}{P^Y_{i,t-1}Y_{i,t-1}}\right).\end{aligned} \tag{184}$$

The effective government rate $\hat{i}^{G^E}_{i,t}$ satisfies $\hat{i}^{G^E}_{i,t} = \omega^G\hat{i}^{G^E}_{i,t-1} + (1-\omega^G)\hat{i}^G_{i,t-1}$, while the budgetary fiscal deficit ratio $\frac{\widehat{BD}_{i,t}}{P^Y_{i,t}Y_{i,t}}$ satisfies $\frac{\widehat{BD}_{i,t}}{P^Y_{i,t}Y_{i,t}} = \frac{\widehat{FD}_{i,t}}{P^Y_{i,t}Y_{i,t}} - \mathcal{J}_i\left(\frac{\hat{A}_{i,t+1}}{P^Y_{i,t}Y_{i,t}} - \frac{1}{\beta}\frac{\hat{A}_{i,t}}{P^Y_{i,t-1}Y_{i,t-1}}\right)$. The non-oil primary fiscal deficit ratio $\frac{\widehat{PD}_{i,t}}{P^Y_{i,t}Y_{i,t}}$ depends on the government expenditure and tax revenue ratios according to:

$$\frac{\widehat{PD}_{i,t}}{P^Y_{i,t}Y_{i,t}} = \frac{G_i}{Y_i}\ln\frac{\hat{P}^C_{i,t}\hat{G}_{i,t}}{\hat{P}^Y_{i,t}\hat{Y}_{i,t}} + \frac{M^O_i}{Y_i}\hat{\tau}^O_{i,t} + \frac{\hat{T}^{C,D}_{i,t}}{P^Y_{i,t}Y_{i,t}} - \frac{\hat{T}_{i,t}}{P^Y_{i,t}Y_{i,t}}. \tag{185}$$

The transfer payment ratio $\frac{\hat{T}^C_t}{P^Y_tY_t}$ satisfies $\frac{\hat{T}^C_{i,t}}{P^Y_{i,t}Y_{i,t}} = \frac{\hat{T}^{C,N}_{i,t}}{P^Y_{i,t}Y_{i,t}} + \frac{\hat{T}^{C,D}_{i,t}}{P^Y_{i,t}Y_{i,t}}$. The tax revenue ratio $\frac{\hat{T}_{i,t}}{P^Y_{i,t}Y_{i,t}}$ depends on the corporate, labor income and consumption tax rates and bases, as well as the non-oil import tariff rate, according to:

$$\begin{aligned}\frac{\hat{T}_{i,t}}{P^Y_{i,t}Y_{i,t}} = {} & \left(\frac{Y^N_i}{Y_i} - \frac{W_iL_i}{P^Y_iY_i}\right)\hat{\tau}^K_{i,t} + \tau^K_i\left(\frac{Y^N_i}{Y_i}\ln\frac{\hat{P}^{Y^N}_{i,t}\hat{Y}^N_{i,t}}{\hat{P}^Y_{i,t}\hat{Y}_{i,t}} - \frac{W_iL_i}{P^Y_iY_i}\ln\frac{\hat{W}_{i,t}\hat{L}_{i,t}}{\hat{P}^Y_{i,t}\hat{Y}_{i,t}}\right) \\ & + \frac{W_iL_i}{P^Y_iY_i}\left(\hat{\tau}^L_{i,t} + \tau^L_i\ln\frac{\hat{W}_{i,t}\hat{L}_{i,t}}{\hat{P}^Y_{i,t}\hat{Y}_{i,t}}\right) + \frac{C_i}{Y_i}\left(\hat{\tau}^C_{i,t} + \tau^C_i\ln\frac{\hat{P}^C_{i,t}\hat{C}_{i,t}}{\hat{P}^Y_{i,t}\hat{Y}_{i,t}}\right) + \vartheta^X\vartheta^M\frac{M^N_i}{Y_i}\hat{\tau}^M_{i,t}.\end{aligned} \tag{186}$$

The government debt ratio $\frac{\hat{D}^G_{i,t+1}}{P^Y_{i,t}Y_{i,t}}$ satisfies $\frac{\widehat{FD}_{i,t}}{P^Y_{i,t}Y_{i,t}} = \frac{\hat{D}^G_{i,t+1}}{P^Y_{i,t}Y_{i,t}} - \left(\frac{\hat{D}^G_{i,t}}{P^Y_{i,t-1}Y_{i,t-1}} - \frac{D^G_i}{P^Y_iY_i}\ln\frac{\hat{P}^Y_{i,t}\hat{Y}_{i,t}}{\hat{P}^Y_{i,t-1}\hat{Y}_{i,t-1}}\right)$.

The current account balance ratio $\frac{\widehat{CA}^{i,t}}{P^Y_{i,t}Y_{i,t}}$ depends on net international investment income and the trade balance ratio according to dynamic budget constraint:

$$\frac{\widehat{CA}_{i,t}}{P^Y_{i,t}Y_{i,t}} = \frac{1-\beta}{\beta}\frac{\hat{A}_{i,t}}{P^Y_{i,t-1}Y_{i,t-1}} + \frac{\widehat{TB}_{i,t}}{P^Y_{i,t}Y_{i,t}}. \tag{187}$$

The trade balance ratio $\frac{\widehat{TB}_{i,t}}{P^Y_{i,t}Y_{i,t}}$ depends on export revenues and import expenditures, as well as the oil consumption subsidy and non-oil import tariff rates, according to:

$$\frac{\widehat{TB}_{i,t}}{P^Y_{i,t}Y_{i,t}} = \frac{M_i}{Y_i}\ln\left(\hat{\upsilon}^T_t \frac{\hat{P}^X_{i,t}\hat{X}_{i,t}}{\hat{P}^M_{i,t}\hat{M}_{i,t}}\right) - \frac{M^O_i}{Y_i}\hat{\tau}^O_{i,t} + \left(\frac{M_i}{Y_i} - \frac{M^O_i}{Y_i}\right)\hat{\tau}^M_{i,t}. \tag{188}$$

The net foreign asset ratio $\frac{\hat{A}_{i,t+1}}{P^Y_t Y_t}$ satisfies $\frac{\widehat{CA}_{i,t}}{P^Y_{i,t}Y_{i,t}} = \frac{\hat{A}_{i,t+1}}{P^Y_{i,t}Y_{i,t}} - \frac{\hat{A}_{i,t}}{P^Y_{i,t-1}Y_{i,t-1}}$. Clearing of the global interbank market requires that the world output-weighted average net foreign asset ratio equal zero, that is $\sum_{i=1}^N w^Y_i \frac{\hat{A}_{i,t+1}}{P^Y_{i,t}Y_{i,t}} = 0$, which determines the global terms of trade shifter $\ln\hat{\upsilon}^T_t$. It follows that the world output-weighted average current account and trade balance ratios also equal zero.[5]

The quotation currency denominated global oil price $\ln\hat{P}^O_{i^*,t}$ adjusts to equate global oil supply with global oil demand according to:

$$\sum_{i=1}^N w^Y_i \frac{Y^O_i}{Y_i}\ln\hat{X}^O_{i,t} = \sum_{i=1}^N w^Y_i \frac{M^O_i}{Y_i}\ln\hat{M}^O_{i,t}. \tag{189}$$

The cartel oil production quota $\ln\hat{Y}^{O^Q}_t$ responds inertially to the relative quotation currency denominated global oil price according to oil supply rule:

$$\ln\hat{Y}^{O^Q}_t = \rho^{O^Q}\ln\hat{Y}^{O^Q}_{t-1} + (1-\rho^{O^Q})\xi^{O^Q}\ln\frac{\hat{P}^O_{i^*,t}}{\hat{P}^Y_{i^*,t}} + \hat{\nu}^{O^Q}_t. \tag{190}$$

Quotation currency denominated global oil price inflation $\hat{\pi}^O_{i^*,t}$ satisfies $\hat{\pi}^O_{i^*,t} = \ln\hat{P}^O_{i^*,t} - \ln\hat{P}^O_{i^*,t-1}$.

B. Exogenous Variables

The oil supply $\ln\hat{\nu}^O_{i,t}$, productivity $\ln\hat{\nu}^{\mathcal{A}}_{i,t}$, labor supply $\ln\hat{\nu}^N_{i,t}$, private consumption demand $\ln\hat{\nu}^C_{i,t}$, residential investment demand $\ln\hat{\nu}^{I^H}_{i,t}$, extraction investment demand $\ln\hat{\nu}^{I^O}_{i,t}$, business investment demand $\ln\hat{\nu}^{I^K}_{i,t}$, export demand $\ln\hat{\nu}^X_{i,t}$, import demand $\ln\hat{\nu}^M_{i,t}$, duration risk premium $\hat{\nu}^B_{i,t}$, housing risk premium $\hat{\nu}^H_{i,t}$, equity risk premium $\hat{\nu}^S_{i,t}$ and foreign currency liquidity $\hat{\nu}^{\varepsilon}_{i,t}$ shocks follow stationary first order autoregressive processes driven by normally distributed innovations

$$\hat{\nu}^Z_{i,t} = \rho_Z\hat{\nu}^Z_{i,t-1} + \varepsilon^Z_{i,t},\ \varepsilon^Z_{i,t} \sim \mathcal{N}(0,\sigma^2_Z), \tag{191}$$

where $\hat{\nu}^Z_{i,t} \in \{\ln\hat{\nu}^O_{i,t}, \ln\hat{\nu}^{\mathcal{A}}_{i,t}, \ln\hat{\nu}^N_{i,t}, \ln\hat{\nu}^C_{i,t}, \ln\hat{\nu}^{I^H}_{i,t}, \ln\hat{\nu}^{I^O}_{i,t}, \ln\hat{\nu}^{I^K}_{i,t}, \ln\hat{\nu}^X_{i,t}, \ln\hat{\nu}^M_{i,t}, \hat{\nu}^B_{i,t}, \hat{\nu}^H_{i,t}, \hat{\nu}^S_{i,t}, \hat{\nu}^{\varepsilon}_{i,t}\}$, with corresponding autoregressive coefficients $\rho_Z \in \{\rho_O, \rho_{\mathcal{A}}, \rho_N, \rho_C, \rho_{I^H}, \rho_{I^O}, \rho_{I^K}, \rho_X, \rho_M, \rho_B, \rho_H, \rho_S, \rho_{\varepsilon}\}$ and innovations $\varepsilon^Z_{i,t} \in \{\varepsilon^O_{i,t}, \varepsilon^{\mathcal{A}}_{i,t}, \varepsilon^N_{i,t}, \varepsilon^C_{i,t}, \varepsilon^{I^H}_{i,t}, \varepsilon^{I^O}_{i,t}, \varepsilon^{I^K}_{i,t}, \varepsilon^X_{i,t}, \varepsilon^M_{i,t}, \varepsilon^B_{i,t}, \varepsilon^H_{i,t}, \varepsilon^S_{i,t}, \varepsilon^{\varepsilon}_{i,t}\}$ having variances $\sigma^2_Z \in \{\sigma^2_O, \sigma^2_{\mathcal{A}}, \sigma^2_N, \sigma^2_C, \sigma^2_{I^H}, \sigma^2_{I^O}, \sigma^2_{I^K}, \sigma^2_X, \sigma^2_M, \sigma^2_B, \sigma^2_H, \sigma^2_S, \sigma^2_{\varepsilon}\}$. Furthermore, the non-oil output price markup $\ln\hat{\vartheta}^Y_{i,t}$, wage markup $\ln\hat{\vartheta}^L_{i,t}$, absorption price markup $\ln\hat{\vartheta}^D_{i,t}$, non-oil export price markup $\ln\hat{\vartheta}^X_{i,t}$, non-oil import price markup $\ln\hat{\vartheta}^M_{i,t}$, monetary policy $\hat{\nu}^{i^B}_{i,t}$ and cartel oil supply $\hat{\nu}^O_t$ shocks follow serially uncorrelated processes driven by normally distributed innovations

$$\hat{\nu}^Z_{i,t} = \varepsilon^Z_{i,t},\ \varepsilon^Z_{i,t} \sim \mathcal{N}(0,\sigma^2_Z), \tag{192}$$

where $\hat{\nu}^Z_{i,t} \in \{\ln\hat{\vartheta}^Y_{i,t}, \ln\hat{\vartheta}^L_{i,t}, \ln\hat{\vartheta}^C_{i,t}, \ln\hat{\vartheta}^X_{i,t}, \ln\hat{\vartheta}^M_{i,t}, \hat{\nu}^{i^B}_{i,t}, \hat{\nu}^O_t\}$, with corresponding innovations $\varepsilon^Z_{i,t} \in \{\varepsilon^{\vartheta^Y}_{i,t}, \varepsilon^{\vartheta^L}_{i,t}, \varepsilon^{\vartheta^C}_{i,t}, \varepsilon^{\vartheta^X}_{i,t}, \varepsilon^{\vartheta^M}_{i,t}, \varepsilon^{i^B}_{i,t}, \varepsilon^O_t\}$ having variances $\sigma^2_Z \in \{\sigma^2_{\vartheta^Y}, \sigma^2_{\vartheta^L}, \sigma^2_{\vartheta^C}, \sigma^2_{\vartheta^X}, \sigma^2_{\vartheta^M}, \sigma^2_{i^B}, \sigma^2_O\}$. Finally, the mortgage loan markup $\ln\hat{\vartheta}^H_{i,t}$, corporate loan markup $\ln\hat{\vartheta}^F_{i,t}$, mortgage loan default $\hat{\nu}^{\delta^H}_{i,t}$, corporate loan default $\hat{\nu}^{\delta^F}_{i,t}$, government consumption

[5] The inclusion of the global terms of trade shifter in these equations is not microfounded.

demand $\hat{\nu}_{i,t}^{G^C}$, government investment demand $\hat{\nu}_{i,t}^{G^I}$, corporate tax rate $\hat{\nu}_{i,t}^{\tau^K}$, labor income tax rate $\hat{\nu}_{i,t}^{\tau^L}$, consumption tax rate $\hat{\nu}_{i,t}^{\tau^C}$, non-oil import tariff rate $\hat{\nu}_{i,t}^{\tau^M}$, oil consumption subsidy rate $\hat{\nu}_{i,t}^{\tau^O}$, transfer payment $\hat{\nu}_{i,t}^{T}$ and bank capital requirement $\hat{\nu}_{i,t}^{\kappa^R}$ shocks follow stationary first order autoregressive processes driven by normally distributed innovations

$$\hat{\nu}_{i,t}^{Z} = \rho_Z \hat{\nu}_{i,t-1}^{Z} + \varepsilon_{i,t}^{Z},\ \varepsilon_{i,t}^{Z} \sim \mathcal{N}(0,\sigma_Z^2), \tag{193}$$

where $\hat{\nu}_{i,t}^{Z} \in \{\ln \hat{g}_{i,t}^{H}, \ln \hat{g}_{i,t}^{F}, \hat{\nu}_{i,t}^{\delta^H}, \hat{\nu}_{i,t}^{\delta^F}, \hat{\nu}_{i,t}^{G^C}, \hat{\nu}_{i,t}^{G^I}, \hat{\nu}_{i,t}^{\tau^K}, \hat{\nu}_{i,t}^{\tau^L}, \hat{\nu}_{i,t}^{\tau^C}, \hat{\nu}_{i,t}^{\tau^M}, \hat{\nu}_{i,t}^{\tau^O}, \hat{\nu}_{i,t}^{T}, \hat{\nu}_{i,t}^{\kappa^R}\}$, with corresponding autoregressive coefficients $\rho_Z \in \{\rho_{g^H}, \rho_{g^F}, \rho_{\delta^H}, \rho_{\delta^F}, \rho_{G^C}, \rho_{G^I}, \rho_{\tau^K}, \rho_{\tau^L}, \rho_{\tau^C}, \rho_{\tau^M}, \rho_{\tau^O}, \rho_T, \rho_{\kappa^R}\}$ and innovations $\varepsilon_{i,t}^{Z} \in \{\varepsilon_{i,t}^{g^H}, \varepsilon_{i,t}^{g^F}, \varepsilon_{i,t}^{\delta^H}, \varepsilon_{i,t}^{\delta^F}, \varepsilon_{i,t}^{G^C}, \varepsilon_{i,t}^{G^I}, \varepsilon_{i,t}^{\tau^K}, \varepsilon_{i,t}^{\tau^L}, \varepsilon_{i,t}^{\tau^C}, \varepsilon_{i,t}^{\tau^M}, \varepsilon_{i,t}^{\tau^O}, \varepsilon_{i,t}^{T}, \varepsilon_{i,t}^{\kappa^R}\}$ having variances $\sigma_Z^2 \in \{\sigma_{g^H}^2, \sigma_{g^F}^2, \sigma_{\delta^H}^2, \sigma_{\delta^F}^2, \sigma_{G^C}^2, \sigma_{G^I}^2, \sigma_{\tau^K}^2, \sigma_{\tau^L}^2, \sigma_{\tau^C}^2, \sigma_{\tau^M}^2, \sigma_{\tau^O}^2, \sigma_T^2, \sigma_{\kappa^R}^2\}$. As an identifying restriction, all innovations are assumed to be contemporaneously uncorrelated.

4. Calibration

We calibrate our linearized global DSGE model to capture the diverse structural characteristics of the economies under consideration, informed by observed data and the empirical literature. In doing so, we allow most steady state equilibrium great ratios and some policy rule parameters to vary across economies, while restricting all behavioral and other steady state equilibrium parameters to coincide across them. The GCC countries excluding Bahrain, Qatar and the United Arab Emirates are members of the international oil cartel, as is the rest of the world, whereas the other systemic economies are not. The GCC countries have sovereign wealth funds, whereas the systemic economies do not. The conduct of monetary policy is represented by a flexible inflation targeting regime in the Euro Area and the United States, by a managed exchange rate regime in China and the rest of the world, and by a fixed exchange rate regime in the GCC countries, consistent with the de facto classification in IMF (2024). The dominant economy is represented by the United States.

The calibrated values of steady state equilibrium great ratios are reported in Table 3 of the Appendix. Some reported steady state equilibrium great ratios are set to their average observed values for each economy over the period 2015 through 2024, based on data extracted from the WEO database compiled by the IMF.[6] These are the steady state equilibrium oil output ratio $\frac{Y_i^O}{Y_i}$, government consumption ratio $\frac{G_i^C}{Y_i}$, government investment ratio $\frac{G_i^I}{Y_i}$, import ratio $\frac{M_i}{Y_i}$, government debt ratio $\frac{D_i^G}{P_i^Y Y_i}$, and world output share w_i^Y. Consistent with the specification of our DSGE model, the steady state equilibrium oil output ratio is set to the average observed oil export ratio, except for the rest of the world where it is residually determined to ensure clearing of the global oil market. Also consistent with the specification of our DSGE model, the steady state equilibrium import ratio is set to the average of observed export and import ratios, adjusted for observed re-exports and re-imports, the latter based on data extracted from the Comtrade database compiled by the UN. Some other reported steady state equilibrium great ratios are set proportional to the steady state equilibrium oil output ratio, or the implied steady state equilibrium non-oil output ratio $\frac{Y_i^N}{Y_i}$, given data availability constraints. These are the steady state equilibrium residential investment ratio $\frac{I_i^H}{Y_i}$ which satisfies $\frac{I_i^H}{Y_i} = 0.05\frac{Y_i^N}{Y_i}$, extraction investment ratio $\frac{I_i^O}{Y_i}$ which satisfies $\frac{I_i^O}{Y_i} = 0.15\frac{Y_i^O}{Y_i}$, business investment ratio $\frac{I_i^K}{Y_i}$ which satisfies $\frac{I_i^K}{Y_i} = 0.15\frac{Y_i^N}{Y_i}$, and labor income share $\frac{W_i L_i}{P_i^Y Y_i}$ which satisfies $\frac{W_i L_i}{P_i^Y Y_i} = \frac{2}{3}\frac{Y_i^N}{Y_i}$. The remaining reported steady state equilibrium great ratios are set to proxy values,

[6] In the WEO database, oil sector gross domestic product encompasses crude oil and natural gas production for all GCC countries.

given the specification of our DSGE model and data availability constraints. These are the steady state equilibrium oil import ratio $\frac{M_i^O}{Y_i}$, which is proxied by the average observed refined petroleum products consumption ratio based on data obtained from the US Energy Information Administration, and the steady state equilibrium world capital market capitalization share w_i^A, which is proxied by the average observed world government debt share based on data extracted from the WEO database compiled by the IMF. All unreported steady state equilibrium great ratios are derived from reported parameters given steady state equilibrium relationships.

The calibrated values of behavioral parameters lie within the range of estimates from the empirical literature where available, as reported in Table 4 of the Appendix. For example, the habit persistence in consumption α^C and labor supply α^L parameters are both set to 0.80. The intertemporal elasticity of substitution in consumption σ and intratemporal elasticity of substitution in labor supply η are both set to 1.00. In addition, the share of credit constrained households ϕ^C is set to 20 percent. The adjustment cost parameters for residential χ^H, extraction χ^O and business χ^K investment are all set to 3.00. The partial indexation parameters for output price γ^Y, wage γ^L, export price γ^X and import price γ^M determination are all set to 0.80, while that for absorption price γ^D determination is set to 0.40. Furthermore, the nominal rigidity parameters for output price ω^Y, wage ω^L, export price ω^X and import price ω^M determination are all set to imply average reoptimization intervals of 6 quarters, while that for absorption price ω^D determination is set to imply an average reoptimization interval of 4 quarters. The financial friction parameter for nominal mortgage and corporate loan rate determination ω^B is set to imply an average adjustment interval of 4 quarters. The intratemporal elasticity of substitution in import demand ψ^M is set to 1.50, while the intratemporal elasticity of substitution in oil demand ψ^O is set to 0.10. Finally, the international bond and stock market contagion coefficients λ^B and λ^S are set to 40 and 50 percent, respectively.

The calibrated values of other steady state equilibrium parameters also lie within the range of estimates from the empirical literature where available, as reported in Table 4 of the Appendix. For example, the subjective discount factor β is set to imply a steady state equilibrium annualized interbank rate of 2.0 percent. In addition, the steady state equilibrium annualized depreciation rate of the housing stock δ_H is set to 5.0 percent, of the extraction capital stock δ_O is set to 10.0 percent, of the private physical capital stock δ_K is set to 10.0 percent, and of the public capital stock δ_G is set to 7.5 percent. In steady state equilibrium, the annualized mortgage i^H and corporate i^F loan rates are set to 4.0 and 6.0 percent, while the annualized mortgage α^H and corporate α^F loan amortization rates are set to 2.5 and 5.0 percent, and the annualized mortgage δ^H and corporate δ^F loan default rates are set to 1.0 and 2.0 percent, respectively. Furthermore, the steady state equilibrium mortgage ϕ^H and corporate ϕ^F loan to value ratio limits are both set to 80 percent. Finally, the output ϑ^Y, export ϑ^X, import ϑ^M and absorption ϑ^D price markup parameters are all set to imply steady state equilibrium price markups of 15 percent. All unreported other steady state equilibrium parameters are derived from reported parameters given steady state equilibrium relationships.

The calibrated values of policy rule parameters also lie within the range of estimates from the empirical literature where available, as reported in Table 3 and Table 4 of the Appendix. In the monetary policy rule under a flexible inflation targeting regime, the response coefficient on consumption price inflation ξ^{π^C} is set to 2.0, while that on the non-oil output gap ξ^{Y^G} is set to 0.5 at the annual frequency. In addition, in the monetary policy rules under managed and fixed exchange rate regimes, the response coefficients on nominal effective $\xi^{\Delta\mathcal{E}}$ and bilateral $\xi^{\Delta\mathcal{E}_{i^*}}$ currency depreciation are set to 0.1 and 2.0, respectively. In the fiscal expenditure rules, the response coefficients for public consumption ξ^{G^C} and investment ξ^{G^I} with respect to the non-oil output gap are both set

to 1.0, while that for the oil consumption subsidy rate $\xi_i^{\tau^O}$ with respect to the relative quotation currency denominated global oil price is set to characterize the prevailing energy subsidy structure in each economy. In particular, $\xi_i^{\tau^O}$ is set to 1.0 for the GCC countries excluding Oman and the United Arab Emirates to eliminate substitution effects on oil consumption demand in equilibrium. Furthermore, in the fiscal revenue rules, the steady state equilibrium corporate τ^K, labor income τ^L and consumption τ^C tax rates are set to characterize the prevailing tax structure in each economy. In the countercyclical capital buffer rule, the steady state equilibrium bank capital ratio κ^R is set to 5.0 percent, while the response coefficient on real bank credit ξ^{κ^R} is set to 0.1. Finally, in the cartel oil supply rule, the response coefficient on the relative quotation currency denominated global oil price ξ^{O^Q} is set to 0.20.

All bilateral non-oil trade weights are calibrated to match their average observed values over the period 2015 through 2024, based on data for non-oil goods imports extracted from the Comtrade database compiled by the UN. To ensure multilateral consistency, they are normalized to sum to one across economies, with trade flows to and from the rest of the world residually determined. Finally, all innovation standard deviation parameters are normalized to one.

5. Scenario Analysis

To illustrate the simulation properties of our calibrated global DSGE model, we analyze global risk and domestic reform scenarios for the GCC countries. This scenario analysis indicates the extensive policy, risk and spillover analysis capabilities of the model.

A. Global Risk Scenario

The GCC economies are heavily exposed to global risks that could impact the oil market. To quantitatively assess the potential economic impacts of this exposure, we consider a global downside risk scenario under which key risks materialize simultaneously over one year across the systemic economies, generating a temporary but persistent global growth slowdown associated with tighter global financial conditions and lower oil prices. In China, the housing market correction intensifies, with housing risk premium shocks reducing the relative price of housing by 5 percent, and equity risk premium shocks lowering the relative price of equity by 10 percent. In the Euro Area, sovereign stress recurs, with duration risk premium shocks raising the term premium by 100 basis points, and equity risk premium shocks lowering the relative price of equity by 10 percent. In the United States, trade restrictions get augmented, with the non-oil import tariff rate rising by 10 percentage points, equity risk premium shocks lowering the relative price of equity by 10 percent, and foreign currency liquidity shocks depreciating the dollar by 10 percent in nominal effective terms. Finally, the rest of the world generates an oil supply glut, with oil supply shocks reducing the relative price of oil by 10 percent.

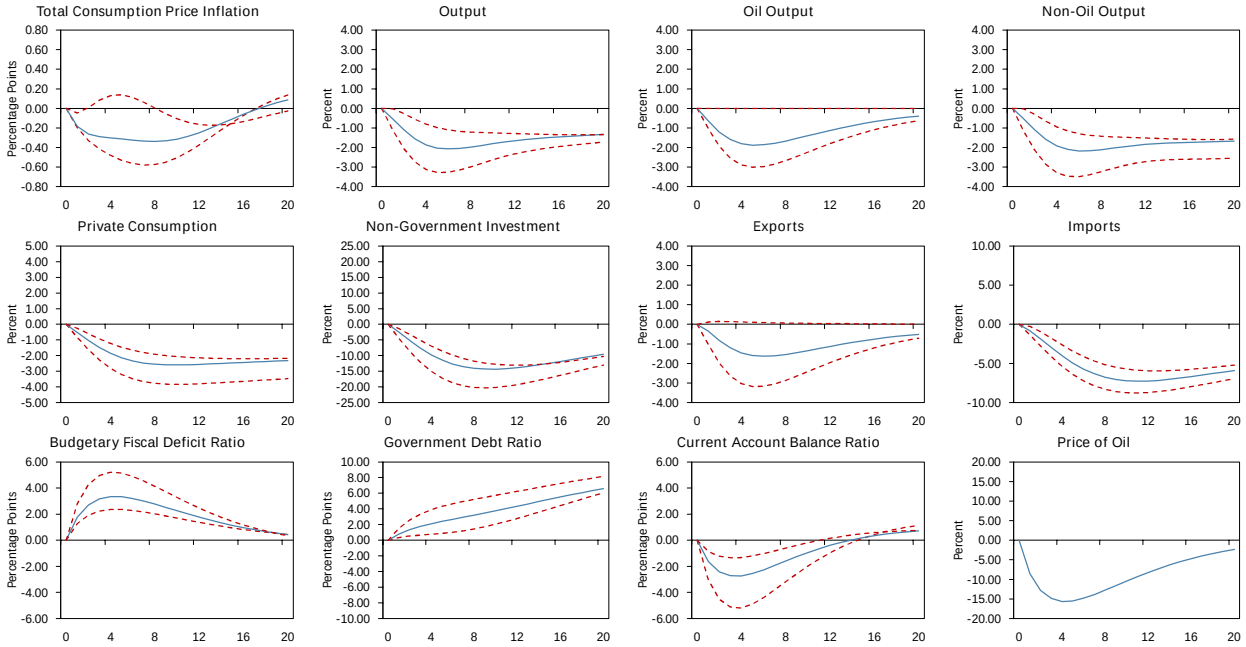
Table 1. Global Risk Scenario Assumptions

Layer 1: Housing market correction in China		
Relative price of housing, housing risk premium shocks	−5 percent	Temporary
Relative price of equity, equity risk premium shocks	−10 percent	Temporary
Layer 2: Sovereign stress in the Euro Area		
Term premium, duration risk premium shocks	+100 basis points	Temporary
Relative price of equity, equity risk premium shocks	−10 percent	Temporary
Layer 3: Trade restrictions in the United States		
Non-oil import tariff rate, non-oil import tariff rate shocks	+10 percentage points	Temporary
Relative price of equity, equity risk premium shocks	−10 percent	Temporary
Nominal effective exchange rate, foreign currency liquidity shocks	+10 percent	Permanent
Layer 4: Oil supply glut from the rest of the world		
Relative price of oil, oil supply shocks	−10 percent	Temporary

Note: All scenario assumptions are expressed as deviations from baseline at the annual frequency. All layers are phased in over one year.

Simulation results under this global risk scenario are depicted for the GCC economies in Figure 1. The global growth slowdown generated by the housing market correction in China, sovereign stress in the Euro Area and trade restrictions in the United States reduces global oil demand, while the oil supply glut from the rest of the world raises global oil supply. As a result, the price of oil falls by 15.6 percent peak after 4 quarters, in response to which the international oil cartel cuts its production quotas by 3.0 percent peak after 5 quarters. These reductions in oil prices and production adversely impact the GCC economies, which experience output losses ranging from 1.3 to 3.3 percent peak after 5 to 18 quarters. These output losses are generally concentrated in the oil sector and are distributed to widely varying degrees across demand components, with the non-oil sector exhibiting greater resilience to lower oil prices and production due to consumption smoothing by credit unconstrained households. They are accordingly concentrated among more oil-dependent economies—with Kuwait experiencing the largest impact and the United Arab Emirates the smallest—supporting economic diversification as a risk mitigation strategy. Lower oil revenues also deteriorate the fiscal and external balances, raising the budgetary fiscal deficit ratio by 2.4 to 5.2 percentage points peak after 4 quarters, and reducing the current account balance ratio by 1.3 to 5.2 percentage points peak after 3 to 4 quarters. Government debt accumulation to finance these higher budgetary fiscal deficits is mitigated by sovereign wealth fund drawdowns across the GCC economies, which serve as a partial substitute. Given their fixed exchange rate regimes, the assumed depreciation of the dollar in nominal effective terms mitigates the impact of this global risk scenario on the GCC economies via expenditure switching from imports to non-oil exports, while offsetting downwards pressure on consumption price inflation through exchange rate pass-through to import price inflation. The result that expenditure switching reinforces consumption smoothing as a contributor to the resilience of the non-oil sector under this global risk scenario hinges on the assumption that the dollar depreciates, and does not hold in general.

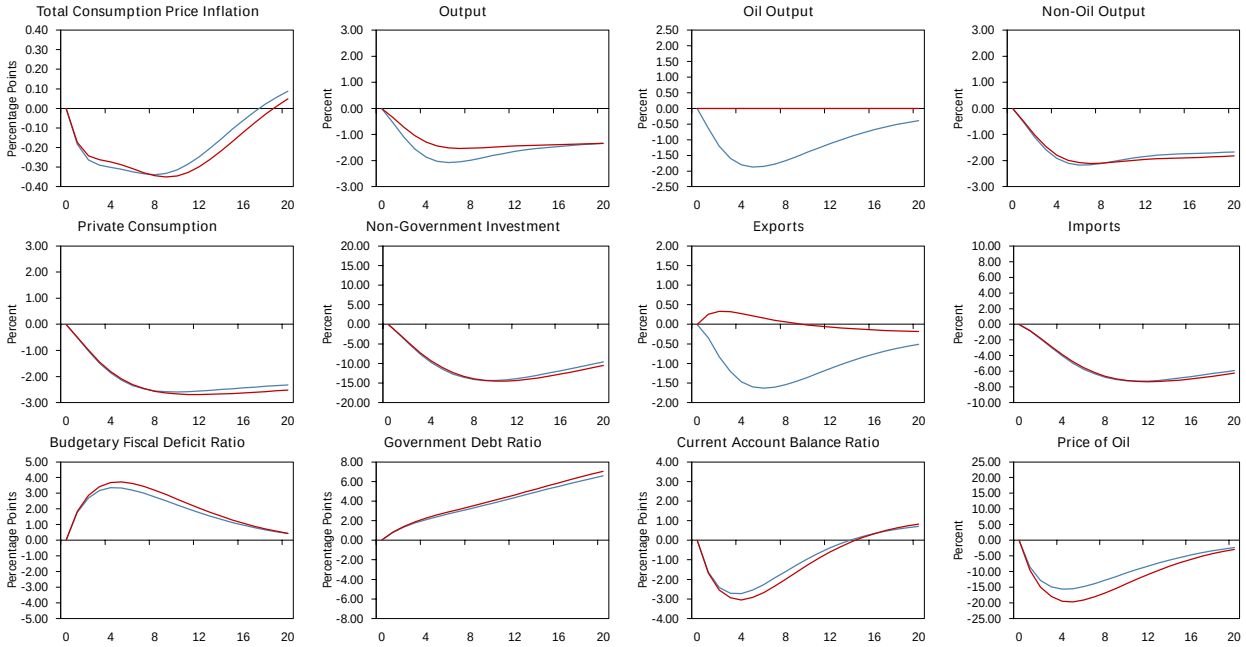
Figure 1. Global Risk Scenario Simulation Results for the GCC



Note: Depicts scenario simulation results for the output-weighted average aggregate GCC economy ■ bounded by economy-specific minima and maxima ■.

A key issue for the GCC economies under any scenario that significantly impacts the price of oil is how the OPEC+ cartel responds. We investigate this issue with a policy counterfactual experiment comparing simulation results under alternative versions of our global risk scenario where the international oil cartel can and cannot adjust its production quotas. These simulation results are depicted for the aggregate GCC economy in Figure 2. Production quota cuts by the oil cartel materially dampen the fall in the price of oil, from 19.7 percent peak after 5 quarters to 15.6 percent peak after 4 quarters. While this smaller oil price impact significantly mitigates the deteriorations in the fiscal and external balances for the aggregate GCC economy, this comes at the cost of a substantially larger output loss from lower oil production, of 2.1 percent peak after 6 quarters as opposed to 1.5 percent peak after 7 quarters. This illustrates the policy trade-off faced by the oil cartel when it intervenes in the oil market by adjusting the production quotas of its members to manage the price of oil, whereby stabilizing their fiscal and external balances requires destabilizing their output. This trade-off arises in our model because the budgetary fiscal deficit and current account balance ratios are more sensitive to changes in the price of oil than to the opposing changes in oil production quotas, and vice versa for output. It reflects the calibration of our model, which matches a relatively low oil demand elasticity of −0.1 and short run oil supply elasticity of 0.1, consistent with the empirical evidence reported in Caldara, Cavallo and Iacoviello (2019).

Figure 2. Unrestricted versus Restricted Global Risk Scenario Simulation Results for the GCC



Note: Depicts scenario simulation results for the output-weighted average aggregate GCC economy with ■ and without ■ international oil cartel responses.

B. Domestic Reform Scenario

The GCC countries are at varying stages of implementing economic diversification programs consisting of fiscal reforms to reinforce intergenerational equity combined with structural reforms to enhance competitiveness supported by public investment boosts to develop infrastructure. To quantitatively assess the economic effects of these diversification programs, we consider an illustrative domestic reform scenario under which the government implements a permanent front-loaded fiscal consolidation package over five years that rationalizes current spending and mobilizes non-oil revenue while boosting public investment, alongside structural reforms that raise productivity. We also allow for illustrative confidence effects that boost private domestic demand. The design of this domestic reform scenario is generic, to show how the simulation properties of the model vary across the GCC economies with their structural characteristics in response to a common set of shocks. At the current juncture, the economic reforms under consideration are not all relevant or applicable to all of the GCC economies. Current spending is rationalized by managing the public sector wage bill and reforming energy subsidies, both of which reduce the non-oil primary fiscal deficit by 2.5 percent of GDP. Non-oil revenue is mobilized by raising the corporate and consumption tax rates, both by 2.5 percentage points. Public investment is boosted by 2.5 percent of GDP, mitigating the fiscal consolidation while shifting the composition of government spending. Finally, structural reforms are assumed to raise productivity by 2.5 percent.

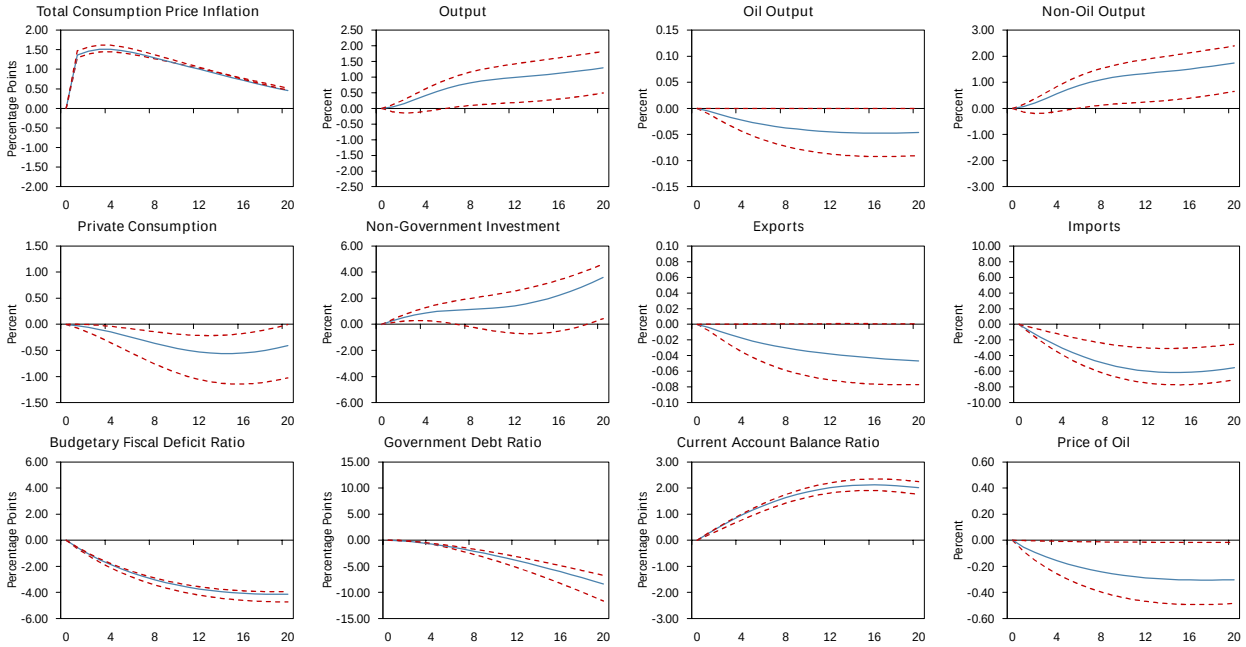
Table 2. Domestic Reform Scenario Assumptions

Layer 1: Public sector wage bill rationalization		
Non-oil primary fiscal deficit ratio, government consumption demand shocks	−2.5 percentage points	Permanent
Layer 2: Energy subsidy reform		
Non-oil primary fiscal deficit ratio, oil consumption subsidy rate shocks	−2.5 percentage points	Permanent
Layer 3: Non-oil revenue mobilization		
Corporate tax rate, corporate tax rate shocks	+2.5 percentage points	Permanent
Consumption tax rate, consumption tax rate shocks	+2.5 percentage points	Permanent
Layer 4: Public investment boost		
Non-oil primary fiscal deficit ratio, government investment demand shocks	+2.5 percentage points	Permanent
Layer 5: Productivity-enhancing structural reforms		
Productivity, productivity shocks	+2.5 percent	Permanent
Layer 6: Confidence effects		
Private consumption, private consumption demand shocks	+1.0 percent	Permanent
Residential investment, residential investment demand shocks	+4.0 percent	Permanent
Business investment, business investment demand shocks	+4.0 percent	Permanent

Note: All scenario assumptions are expressed as deviations from baseline at the annual frequency. All layers are phased in over two years.

Simulation results under this domestic reform scenario are depicted for the GCC economies in Figure 3. This economic diversification program is moderately expansionary, raising output by 0.5 to 1.8 percent at the end of the 5-year simulation horizon, with the public investment boost and productivity-enhancing structural reforms collectively dominating the fiscal consolidation. These medium-term output gains are concentrated in the non-oil sector, with the oil sector being insulated. They are decreasing in trade openness—with Saudi Arabia experiencing the largest output gain and Bahrain the smallest—reflecting demand spillovers. This economic diversification program is also moderately inflationary, raising total consumption price inflation by 1.4 to 1.6 percentage points peak after 3 to 4 quarters, primarily reflecting the reduction in the oil consumption subsidy rate and increase in the consumption tax rate. The fiscal and external balances improve, with the budgetary fiscal deficit ratio falling by 3.9 to 4.7 percentage points at the end of the 5-year simulation horizon, and the current account balance ratio rising by 1.8 to 2.2 percentage points. The savings from lower budgetary fiscal deficits are used to reduce government debt and accumulate sovereign wealth fund assets, reinforcing intergenerational equity by increasing the rate of transformation of exhaustible oil wealth into return-generating public financial wealth.

Figure 3. Domestic Reform Scenario Simulation Results for the GCC



Note: Depicts scenario simulation results for the output-weighted average aggregate GCC economy ■ bounded by economy-specific minima and maxima ■.

6. Conclusion

Oil-dependent economies tend to exhibit distinct structural characteristics that shape their responses to shocks. Typically, a state-owned enterprise extracts oil using a capital-intensive technology subject to an oil production quota set by the OPEC+ cartel, and remits profits to the government. The government in turn subsidizes domestic oil consumption and invests oil profits not needed to finance its budget predominantly abroad via a sovereign wealth fund, often under a fixed exchange rate regime.

This paper develops a global DSGE model for oil-dependent economies to support quantitative scenario-based policy, risk and spillover analysis. Our model is the first to integrate the full range of structural characteristics that distinguish these economies, exemplified by the members of the GCC. Simulating global risk and domestic reform scenarios for these economies using the calibrated model supports several conventional views while elucidating the mechanisms at work. First, more oil-dependent members of the GCC experience larger output losses under our global downside risk scenario—supporting economic diversification as a risk mitigation strategy for them—as the non-oil sector dampens the economic impact of lower oil prices and production through consumption smoothing. Second, a key determinant of the impact of this global risk scenario on the GCC economies is the response of the international oil cartel, which faces a policy trade-off between stabilizing the fiscal and external balances of its members versus destabilizing their output when adjusting its production quotas to manage the price of oil. Third, sovereign wealth fund transactions help stabilize government debt dynamics across the GCC economies under our global risk and domestic reform scenarios, by serving as an alternative source of net financing for the budget.

The application of our global DSGE model to scenario analysis for the members of the GCC is subject to important caveats, as it simplifies structural characteristics that set them apart from other oil-dependent economies. Notably, the GCC economies exhibit labor market duality to varying degrees, with expatriates disproportionately employed in the private sector and citizens in the public sector, amid large public sector wage premiums and generally tighter credit constraints for expatriates. In addition, the GCC economies tend to have multiple sovereign wealth funds of different types, including fiscal stabilization and intergenerational savings funds which primarily invest oil profits not needed to finance the budget abroad, as well as economic development funds which finance off-budget public investment projects that have expanded as economic diversification programs have progressed. Extending our global DSGE model for oil-dependent economies to more fully encompass these distinctive structural characteristics of the GCC economies remains an objective for future research.

Appendix. Calibrated Parameter Values

Table 3. Calibrated Heterogeneous Parameters

	BHR	CHN	EUR	KWT	OMN	QAT	ROW	SAU	UAE	USA
Y_i^O / Y_i	0.241	0.002	0.002	0.427	0.305	0.383	0.045	0.231	0.151	0.008
I_i^H / Y_i	0.038	0.050	0.050	0.029	0.035	0.031	0.048	0.038	0.042	0.050
I_i^O / Y_i	0.036	0.000	0.000	0.064	0.046	0.057	0.007	0.035	0.023	0.001
I_i^K / Y_i	0.114	0.150	0.150	0.086	0.104	0.093	0.143	0.115	0.127	0.149
G_i^C / Y_i	0.154	0.168	0.212	0.248	0.224	0.174	0.167	0.245	0.130	0.140
G_i^I / Y_i	0.037	0.155	0.035	0.066	0.106	0.075	0.070	0.056	0.115	0.035
M_i / Y_i	0.566	0.177	0.396	0.392	0.364	0.383	0.219	0.266	0.465	0.137
M_i^O / Y_i	0.035	0.023	0.025	0.059	0.046	0.031	0.027	0.041	0.022	0.025
$W_i L_i / P_i^Y Y_i$	0.506	0.665	0.665	0.382	0.463	0.411	0.637	0.513	0.566	0.661
$\xi_i^{\tau^O}$	1.000	0.000	0.000	1.000	1.000	1.000	0.200	1.000	0.000	0.000
τ_i^K	0.150	0.150	0.150	0.150	0.150	0.150	0.150	0.150	0.150	0.150
τ_i^L	0.000	0.150	0.150	0.000	0.000	0.000	0.150	0.000	0.000	0.150
τ_i^C	0.100	0.100	0.150	0.000	0.050	0.000	0.100	0.150	0.050	0.050
$D_i^G / P_i^Y Y_i$	$1.029S$	$0.644S$	$0.893S$	$0.085S$	$0.425S$	$0.506S$	$0.933S$	$0.214S$	$0.280S$	$1.148S$
w_i^Y	0.000	0.166	0.153	0.002	0.001	0.002	0.413	0.009	0.005	0.248
w_i^A	0.000	0.116	0.150	0.000	0.000	0.001	0.419	0.002	0.001	0.309

Note: The calibration is a function of the seasonal frequency S, evaluated at $S = 4$.

Table 4. Calibrated Homogeneous Parameters

α^C	0.800	i^F	$0.060 / S$	ρ^{G^I}	0.800	$\rho_{\mathcal{A}}$	0.800
α^L	0.800	δ^H	$0.010 f(\alpha^H, i^H)$	ρ^{i^K}	0.800	ρ_N	0.800
α^H	0.025	δ^F	$0.020 f(\alpha^F, i^F)$	ρ^{i^L}	0.800	ρ_C	0.400
α^F	0.050	ω^Y	$(6-1)/6$	ρ^{i^C}	0.800	ρ_{i^H}	0.600
β	$1/(1+0.020/S$	ω^L	$(6-1)/6$	ρ^{i^M}	0.800	ρ_{i^O}	0.600
χ^H	3.000	ω^X	$(6-1)/6$	ρ^{κ^R}	0.800	ρ_{i^K}	0.600
χ^O	3.000	ω^M	$(6-1)/6$	ρ^{O^Q}	0.400	ρ_X	0.800
χ^K	3.000	ω^D	$(4-1)/4$	σ	1.000	ρ_M	0.800
χ^B	0.500	ω^B	$(4-1)/4$	ς	1.000	ρ_B	0.800
δ_H	$0.050 / S$	ω^G	0.950	ϑ^Y	1.150	ρ_H	0.800
δ_O	$0.100 / S$	ϕ^A	0.950	ϑ^X	1.150	ρ_S	0.800
δ_K	$0.100 / S$	ϕ^C	0.200	ϑ^M	1.150	$\rho_{\mathcal{E}}$	0.400
δ_G	$0.075 / S$	ϕ_H^C	0.300	ϑ^D	1.150	ρ_{q^H}	0.600
η	1.000	ϕ_B^C	0.300	ξ^{π^C}	2.000	ρ_{q^F}	0.600
η^K	1.000	ϕ_S^C	0.050	ξ^{Y^G}	$0.500 / S$	ρ_{δ^H}	0.400
η^B	25.000	ϕ^H	0.800	$\xi^{\Delta\mathcal{E}}$	0.100	ρ_{δ^F}	0.400
γ^Y	0.800	ϕ^F	0.800	$\xi^{\Delta\mathcal{E}_{i^*}}$	2.000	ρ_{ζ^C}	0.400
γ^L	0.800	ϕ_B^F	0.750	ξ^{G^C}	1.000	ρ_{ζ^I}	0.400
γ^X	0.800	ϕ_H^F	0.200	ξ^{G^I}	1.000	ρ_{τ^K}	0.400
γ^M	0.800	ϕ_F^F	0.050	ξ^{T^N}	$0.000 / S$	ρ_{τ^L}	0.400
γ^D	0.400	ϕ^O	0.950	ξ^{T^D}	$0.040 / S$	ρ_{τ^C}	0.400
κ^R	0.050	ψ^M	1.500	ξ^{κ^R}	0.100	ρ_{τ^M}	0.400
λ^A	0.500	ψ^O	0.100	ξ^{O^Q}	0.200	ρ_{τ^O}	0.800
λ^B	0.400	ρ^{A^T}	0.950	ζ^{δ^H}	$0.067 f(\alpha^H, i^H)$	ρ_T	0.800
λ^S	0.500	ρ^{δ^H}	0.800	ζ^{δ^F}	$0.133 f(\alpha^F, i^F)$	ρ_{κ^R}	0.400
μ	1.000	ρ^{δ^F}	0.800	ζ^F	0.250		
μ^L	0.050	ρ^{i^B}	0.800	$\zeta^{\mathcal{E}}$	$0.040 / S^2$		
i^H	$0.040 / S$	ρ^{G^C}	0.800	ρ_O	0.800		

Note: The calibration is a function of the seasonal frequency S, evaluated at $S = 4$. Also, $f(\alpha^Z, i^Z) = \frac{1+i^Z}{\alpha^Z + i^Z}\frac{1}{S}$ for $Z \in \{H, F\}$.

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